The most entangled states of four and seven qubits

Antony Lewis

Department of Physics & Astronomy, University of Sussex, Brighton BN1 9QH, UK

Abstract

Pure states of four and seven qubits cannot be maximally entangled across every balanced bipartition; we determine how entangled they can be on average. For four qubits we prove the conjecture of Higuchi and Sudbery: the mean von Neumann entropy of the three 2|2 cuts is at most $1 + \frac{1}{2}\log_2 3$ bits, with equality exactly on the local-unitary orbits of their state M_4 and its complex conjugate. For seven qubits the mean purity of the 35 balanced 3|4 cuts is at least $121/917$ and their mean entropy is at most $2.965356029\ldots$ bits, each with a unique optimiser up to local unitaries and qubit permutations. The two optimisers are distinct members of one icosahedral family, so minimum purity and maximum entropy select different states.

The proofs bound moments of the marginal spectra by positive identities on several copies of the state, some of them spin flipped, and bound the entropy by a polynomial in those moments. The four-qubit maximum has an analytic proof; the seven-qubit proofs, and some of the four-qubit refinements below, are computer-assisted, with exact arithmetic and rigorous error bounds. We also prove the Gour–Wallach conjecture for Rényi entropies of all orders, with the maximiser changing from M_4 to cluster states at $q_* \approx 1.588$, and obtain sharp Tsallis bounds, mixed-state bounds and quantitative stability.

The results and text were all produced by AI agents under human supervision.

1 Introduction

Multipartite entanglement requires a compromise: a state maximally entangled across one division of its qubits need not be maximally entangled across another. A state that achieves maximal entanglement across every bipartition is called *absolutely maximally entangled* (AME). Such states are closely related to quantum error-correcting codes [29] and "perfect" tensors [31]; Ref. [4] reviews them. They exist for two, three, five and six qubits, but not for four [1, 3], seven [16], or eight or more [29]. Four and seven qubits are therefore the first two sizes at which no state can maximise entanglement across every cut simultaneously; at most four of the six two-qubit marginals, and at most 32 of the 35 three-qubit marginals, can be maximally mixed [1, 16, 5]. Balanced cuts expose this obstruction: if all balanced marginals are maximally mixed, their smaller reductions are too. Conversely, states with all one- and two-qubit marginals maximally mixed exist for every $n \geq 5$ [29], so small marginals alone do not reveal the seven-qubit obstruction. We determine the greatest mean entropy across balanced cuts at both sizes; minimisers of the mean balanced purity are the maximally multipartite entangled states of Facchi et al. [22], and highly entangled states have been sought numerically since Refs. [23, 24]. At four qubits this is the question posed by Higuchi and Sudbery in "How entangled can two couples get?" [1].

Label the qubits $1, 2, 3, 4$. There are three ways to divide them into two groups of two:

$$12|34, \qquad 13|24, \qquad 14|23.$$

A *bipartition*, or *cut*, is balanced when its two groups are as close in size as possible: 2|2 here, and 3|4 for seven qubits. For a pure state ψ, the reduced density matrix $\rho_{12} = \mathrm{tr}_{34}\,|\psi\rangle\langle\psi|$ describes qubits $1, 2$ alone; define ρ_{13} and ρ_{14} similarly. Its von Neumann entropy

$$S(\rho) = -\,\mathrm{tr}\,\rho \log_2 \rho$$

measures the entanglement across the cut. Operationally, it is the number of Bell pairs per copy that the two sides can distil from many copies of the pure state, or need to prepare those copies, using local operations and classical communication [2]. The complementary pair has the same entropy, so each cut is counted once. A pair of qubits has a four-dimensional state space, so $S \le 2$ bits, with equality when ρ is maximally mixed, $I_4/4$, with four equal eigenvalues $1/4$. Entropies are in bits throughout, except in the Tsallis corollary and in proofs that use ln, which say so.

We give the three cuts equal weight and study the average

$$E(\psi) = \frac{1}{3}[S(\rho_{12}) + S(\rho_{13}) + S(\rho_{14})] = \frac{1}{3}\sum_{j=2}^{4} S(\rho_{1j}).$$

It is the entanglement, in Bell pairs, available across a balanced cut chosen at random, and it is unchanged by relabelling the qubits. Reaching $E = 2$ would require all three reduced states to be maximally mixed, hence an AME state. Their compatibility—all three come from one wavefunction—prevents this [1]: three maximally mixed cuts would give total purity $P = \sum_j \operatorname{tr}\rho_{1j}^2 = 3\cdot\frac{1}{4}$, whereas every pure four-qubit state has $P \ge 1$ (Lemma 2). The best average is therefore strictly below 2.

The two main results. Higuchi and Sudbery (HS) exhibited the state

$$M_4 = \tfrac{1}{\sqrt{6}}(|0011\rangle + |1100\rangle + \omega(|0101\rangle + |1010\rangle) + \omega^2(|0110\rangle + |1001\rangle)), \qquad \omega = e^{2\pi i/3},$$

whose amplitudes sit on the six strings of Hamming weight two, grouped into complementary pairs, one for each pairing of the qubits, with phases $1, \omega, \omega^2$. An even permutation of the qubits permutes the three pairings cyclically (or fixes each of them) and only multiplies M_4 by a phase (Remark 5), so the three cuts are identical: each reduced state has eigenvalues $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$. Every cut then has the same entropy, and $E(M_4) = 1 + \frac{1}{2}\log_2 3 \approx 1.792$. They conjectured that this is the maximum [1]. Brierley and Higuchi proved that M_4 is a local maximum [7]. Derksen et al. proved that M_4 maximises a different measure, the geometric measure of entanglement (distance to the nearest product state) [6]. Gour and Wallach proved the corresponding Tsallis bound among *critical* states, those whose one-qubit marginals are all maximally mixed, and conjectured the Rényi envelope [8]; the precise scope of these results is at the end of this section. Independent single-qubit unitaries are called *local unitaries* (LU). Two states are *LU-equivalent* if local unitaries map one to the other up to an overall phase, and the states equivalent to a given one form its *LU orbit*. Since these operations preserve cut entropies, maximisers occur in whole LU orbits.

Theorem 1 (four qubits). For every pure state of four qubits, $E(\psi) \le 1 + \frac{1}{2}\log_2 3$. Equality holds if and only if ψ is equivalent to M_4 or to its complex conjugate $\overline{M_4}$ by independent single-qubit unitaries.

At seven qubits a balanced cut puts three qubits on one side, and there are $\binom{7}{3} = 35$ such cuts, one for each set S of three qubits with reduced state ρ_S. Write $\pi_3 = \frac{1}{35}\sum_{|S|=3}\operatorname{tr}\rho_S^2$ and $E_3 = \frac{1}{35}\sum_{|S|=3} S(\rho_S)$ for the mean purity and mean entropy of the 35 cuts. An AME state would have $\pi_3 = 1/8$ and $E_3 = 3$ bits.

Theorems 11 and 12 (seven qubits). Every pure seven-qubit state has $\pi_3 \ge 121/917$, with equality exactly on the local-unitary and permutation orbit of an explicit icosahedral state Ψ_{ico}. Every pure seven-qubit state has $E_3 \le E_3(\psi_{x_*}) = 2.965356029\ldots$ bits, with equality exactly on the local-unitary and permutation orbit of a different member ψ_{x_*} of the same icosahedral family (Section 5). With qubit labels fixed, each equality set consists of 42 LU orbits.

For four qubits, many states tie for the minimum mean purity, but the HS cut spectra uniquely maximise the entropy. At seven qubits even the purity minimiser is globally unique (up to local unitaries and qubit permutations), yet it is not the entropy maximiser. Both optimisers lie in one explicit one-parameter icosahedral family, and our proofs separate them globally with five-copy certificates. "Most entangled" thus depends on the measure, even when only balanced cuts are compared; at four qubits the Rényi entropies show the same effect, with the maximiser changing at q_*.

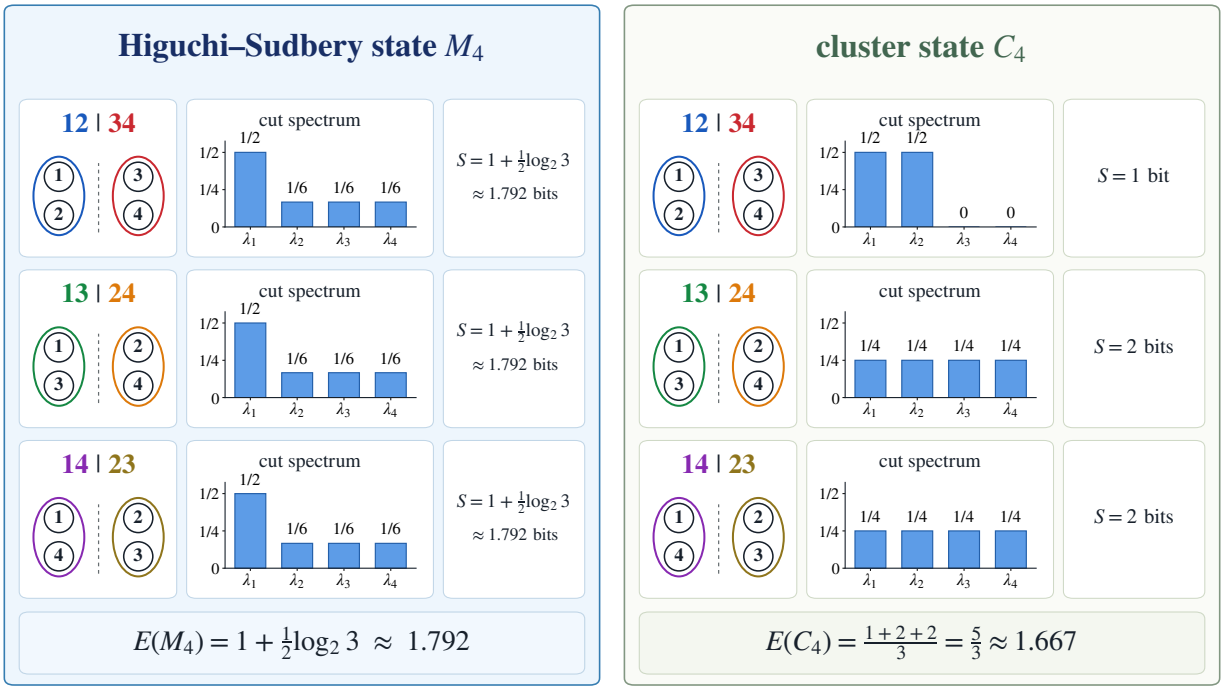


Figure 1: Four eigenvalues of each reduced density matrix for the three balanced 2|2 cuts (same vertical scale throughout). For M_4 each cut has spectrum $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ and entropy $1 + \frac{1}{2}\log_2 3$ bits. For the cluster state C_4, the 12|34 cut has spectrum $(\frac{1}{2}, \frac{1}{2}, 0, 0)$ and entropy 1 bit, while the other two cuts are maximally mixed and have entropy 2 bits, so $E(C_4) = (1+2+2)/3 = 5/3 < E(M_4)$: balancing the three cuts beats perfecting two at the expense of one.

Four qubits: compromise between cuts. To see why a compromise is needed, compare the four-qubit linear cluster state $C_4 = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle)$. Its entropy is 1 bit for 12|34 and 2 bits for each of the other two cuts, so $E(C_4) = 5/3 \approx 1.667 < E(M_4)$. Making two cuts perfect costs more on the third than is gained. Figure 1 compares the reduced-state *spectra*, or lists of eigenvalues, of the two states.

The LU orbits of M_4 and $\overline{M_4}$ are distinct [12, Prop. 7] (Appendix A gives an elementary separating invariant), but an odd permutation of the qubits exchanges them (Remark 5), so the maximiser is unique up to local unitaries and relabelling the qubits.

Purity alone cannot identify the entropy maximiser: the cluster state and M_4 both attain the minimum summed purity $P = 1$ but have different entropies, so a proof must also control higher moments of the cut spectra. The entanglement of a cut depends only on the spectrum of its marginal ρ_S, and spectra are what copies of a state measure: $\operatorname{tr}\rho_S^k$ is the expectation value of the operator that cyclically permutes the subsystems S of k copies of ψ. More generally, any homogeneous polynomial of degree m in $|\psi\rangle\langle\psi|$ is an m-copy expectation value. A bound is proved by writing the gap between the bound and the objective as the m-copy expectation value of a sum of manifestly positive operators $A^\dagger A$; the optimisers are the states on which every term vanishes.

The useful positive operators also act on copies replaced by the spin flip $\tilde{\psi} = \sigma_y^{\otimes n}\bar{\psi}$, time reversal ($\sigma_y$ composed with complex conjugation) applied to every spin-$\frac{1}{2}$. A Pauli string is a tensor product of Pauli matrices and identities; its weight is the number of nonidentity factors. On density matrices, $\tilde{\rho} = \sigma_y^{\otimes n}\rho^T\sigma_y^{\otimes n}$ negates the Pauli-string components of odd weight and keeps those of even weight; on one qubit it reverses the Bloch vector. For a product state $\tilde{\psi}$ is orthogonal to ψ, while for a Bell pair it equals ψ up to phase (for two qubits $|\langle\psi|\tilde{\psi}\rangle|$ is Wootters' concurrence [33]). Since $\epsilon\bar{u}\epsilon^{-1} = u$ for $u \in SU(2)$ and $\epsilon = i\sigma_y$, a flipped copy transforms like an ordinary one, so these operators remain local-unitary covariant.

The number of copies is set by two things. First, the entropy is not a polynomial. The

proof bounds $-t\log_2 t$ by a *majorant*, a polynomial lying above it on $[0,1]$, that touches it at the optimiser's eigenvalues, and the degree needed grows with the number of touching points (Section 2.3). The cuts of M_4 have spectrum $(\frac{1}{2},\frac{1}{6},\frac{1}{6},\frac{1}{6})$, with two distinct eigenvalues, so a cubic suffices: two copies give $P \geq 1$ and three bound the cubic moments. Second, a few copies may not be enough even for a polynomial objective. Allowing $m = 2, 3, \ldots$ copies gives a *hierarchy* of ever stronger bounds, but a few copies constrain only low-order correlations between copies, and they admit moment data that pass every positivity test at that level yet come from no pure state. At seven qubits both the purity and the entropy need five copies; Section 5 explains why and compares the two sizes in Table 4.

The four-qubit proof. Write P and T for the sums of the second and third moments of the three two-qubit marginals, and $F = 17P - 12T - 12$. The two-copy spin flip [8, 11] gives $P \geq 1$ (Lemma 2); a three-copy sum of squared tensor norms, the sextic identity, gives $F \geq 0$, that is $T \leq 5/12 + (17/12)(P-1)$ (Theorem 3 and Appendix A). A cubic majorant of $-x\ln x$, its Hermite interpolant (the cubic matching its value and slope at $1/6$ and $1/2$, which lies above it), turns these moment bounds into the entropy bound. With $\Delta_H \geq 0$ the total gap of the majorant over the twelve cut eigenvalues, the entropy gap is

$$3\ln 2\,\big[E(M_4) - E(\psi)\big] = (\tfrac{15}{2} - 6\ln 3)(P-1) + \tfrac{3}{2}(\ln 3 - 1)F + \Delta_H,$$

with all three terms nonnegative (Section 2.3, equation (EG)). Equality forces $P = 1$ and every cut eigenvalue to be $1/6$ or $1/2$, hence the M_4 spectrum on each cut; the critical normal form identifies the state from just one cut (Proposition 4). The classification of all zeros of the moment inequality, also given in Appendix A, is not needed for equality.

Relation to earlier work. Gour and Wallach studied the analogous optimisation for Tsallis and Rényi entropies [8], including a conjecture on Rényi entropies that we prove (Corollary 9).[1] Their Theorem 10(a) proves the Tsallis-entropy bound for exponents $0 < q < 2$ among the critical states (the normal form of [34]), with the bound attained by M_4. It does not cover every pure state. Their equality clause names only the class of M_4; with labelled qubits M_4 and its conjugate are distinct LU classes, exchanged by an odd permutation (Remark 5). For $q > 2$ their Theorem 9(a) identifies the unrestricted Tsallis maximiser, a state other than M_4 (we prove only that the cluster state beats M_4 there), and their Theorem 12 identifies the cluster-state maximisers of the Rényi entropies for $q \geq 2$ [8]; Corollary 23 gives a self-contained elementary proof of that high-order ceiling and its equality cut spectra. For the von Neumann entropy itself, this paper supplies a short, self-contained analytic proof of the bound for all pure states, the classification of the equality case, and unrestricted corollaries for other spectral entropies.[2] At seven qubits, earlier numerical searches found mean three-qubit purities of 0.13387 [28] and 0.131952 [10]; the latter agrees with our minimum $121/917$ to six digits. *Stabilizer states*, joint eigenstates of n commuting, independent Pauli strings (graph states and the cluster state C_4 are examples), do worse: the best, including that of Zha et al., have purity $19/140 \approx 0.1357$ [27, 16]. Before this work, the best explicit universal lower bound known to us was the unattainable AME value $1/8$; Rico et al. obtained a numerical semidefinite estimate of 0.127(8) [30], without a certificate. It agrees with the exact three-copy bound $\pi_3 \geq 125/978 \approx 0.12781$ proved here (Table 8), the three-copy level of our hierarchy.

Proofs and verification. The four-qubit von Neumann bound and its equality case have a fully analytic proof here. Some extensions, particularly at seven qubits, use exact-rational or

[1] It is distinct from the statement on four-qubit states maximising the absolute value of the hyperdeterminant that is usually called the Gour–Wallach conjecture, proved by Chen and Đoković [9].

[2] A proof of the unrestricted entropy bound and its attainment at M_4, with a Lean verification, appeared independently while this work was in preparation [39]; it does not address uniqueness of the maximiser or the equality case, and we make no priority claim for the moment inequality as such.

rigorously bounded numerical certificates. Table 1 is a roadmap of the paper: for each result it gives the optimiser, how it is proved and where the complete proof is. The supplement uses S-prefixed numbering; the companion repository [40] contains both documents, the scripts and their captured outputs, and the Lean formalisation. Appendix I lists the computer checks behind each result, and `verifiers/MAP.md` maps them to scripts. The analytic arguments do not depend on Lean. Every theorem, proposition, lemma and corollary is proved, analytically or with the exact and computer-assisted checks named in its proof; numerical observations are labelled as such. Table 5 in Section 6 collects what is known at other qubit counts.

Question	Optimiser	How proved	Where
Four qubits: maximum mean entropy	M_4, $\overline{M_4}$	Analytic (two- and three-copy identities, cubic majorant)	Sections 2.1–2.4; Appendices A, B
Four qubits: Tsallis $0 < q < 2$	M_4	Analytic (same majorant)	Section 3; Appendix E
Four qubits: Rényi $0 < q < 1$	M_4	Analytic (from the Tsallis bound)	Section 3
Four qubits: Rényi $1 < q < 2$	M_4 for $q < q_*$, cluster for $q > q_*$	Analytic with exact finite checks	Appendix F; supplement, Section S6
Four qubits: Rényi $q \geq 2$	cluster	Analytic (elementary purity bound)	Appendix F.1
Four qubits: orbit stability	–	Analytic with exact finite checks; some constants computer-assisted with rigorous error bounds	Section 4; Appendix D; supplement, Sections S7, S8
Seven qubits: minimum mean purity	Ψ_{ico}	Computer-assisted with rigorous error bounds (five-copy identity)	Appendices G, H; supplement, Section S11
Seven qubits: maximum mean entropy	$\psi_{x_*} \neq \Psi_{\text{ico}}$	Computer-assisted with rigorous error bounds (five-copy identity, joint majorant)	Supplement, Section S12; uses Theorem 11

Table 1: Optimisers, methods and the location of complete proofs. Optimisers are listed up to local unitaries and qubit permutations; both classes attain the Rényi bound at $q = q_*$. Supplementary sections have an S prefix. Exact finite checks use exact arithmetic; computer-assisted proofs evaluate explicit identities in floating point with rigorous error bounds. Neither uses a numerical search as evidence. Appendix I lists the checks behind each result.

2 The four-qubit maximum

Throughout this section, ψ is a unit four-qubit state, $E(\psi) = \frac{1}{3}\sum_{j=2}^{4} S(\rho_{1j})$ is its average $2|2$ entropy in bits, and M_4 is the Higuchi–Sudbery state, all as defined in Section 1. Write the one-qubit marginals as $\rho_i = (I + \mathbf{r}_i \cdot \boldsymbol{\sigma})/2$, where $\boldsymbol{\sigma}$ is the vector of Pauli matrices and $\mathbf{r}_i$ is the Bloch vector.

Theorem 1 (maximum and equality)**.** *For every unit $\psi \in (\mathbb{C}^2)^{\otimes 4}$,*

$$E(\psi) \leq 1 + \tfrac{1}{2}\log_2 3,$$

with equality if and only if ψ is local-unitarily equivalent to M_4 or to $\overline{M_4}$.

The bound is proved in Sections 2.1–2.3 and the equality case in Section 2.4; the two orbits are related by a qubit permutation in Remark 5.

Symbol (unit states)	Meaning
$P = \sum_{j=2}^{4} \operatorname{tr} \rho_{1j}^2$	Sum of the cut purities
$T = \sum_{j=2}^{4} \operatorname{tr} \rho_{1j}^3$	Sum of the third moments
$F = 17P - 12T - 12$	Nonnegative defect in the third-moment bound
$W_1 = \sum_{i=1}^{4} \lVert \mathbf{r}_i \rVert^2$	Total squared one-qubit Bloch-vector length
$\tau = \lvert \psi^T \sigma_y^{\otimes 4} \psi \rvert^2$	Squared spin-flip overlap (four-tangle)

Table 2: Notation for the four-qubit proof. A state is critical when all $\mathbf{r}_i = 0$, equivalently when all one-qubit marginals are maximally mixed. Homogeneous formulas below use $N = \lVert z \rVert^2$ and unnormalised marginals; unit states have $N = 1$.

2.1 Purity identity

The first ingredient bounds the sum of the three cut purities. Its equality condition also identifies the locus on which an entropy maximiser must lie.

Lemma 2 (purity identity)**.** *For a unit state, $P = \sum_{j=2}^{4} \operatorname{tr} \rho_{1j}^2 = 1 + \frac{1}{2}\Delta$ with $\Delta = \sum_{i=1}^{4} \lVert \mathbf{r}_i \rVert^2 + \tau$, where $\mathbf{r}_i$ are the one-qubit Bloch vectors and $\tau = \lvert \psi^T \sigma_y^{\otimes 4} \psi \rvert^2$. In particular $P \geq 1$, with equality iff all $\mathbf{r}_i = 0$ (critical) and $\tau = 0$.*

Here τ is the four-tangle of Wong and Christensen [11], and the lemma is Gour and Wallach's identity $\tau_2 = (4\tau_1 - \tau_{ABCD})/3$ [8, Theorem 6], with τ_k the averaged linear entropies $2(1 - \operatorname{tr} \rho^2)$ of the k-qubit marginals. We include the short spin-flip proof to keep the argument self-contained.

Proof. For a 2×2 matrix A, spin flip is $\sigma_y A^T \sigma_y = (\operatorname{tr} A) I - A$. Tensoring this identity over the four qubits, a product operator $A_1 \otimes \cdots \otimes A_4$ is sent to $\bigotimes_i ((\operatorname{tr} A_i) I - A_i)$. By linearity the spin flip $\tilde{\rho} = \sigma_y^{\otimes 4} \rho^T \sigma_y^{\otimes 4}$ of a four-qubit density matrix is $\tilde{\rho} = \sum_{S \subseteq \{1,\ldots,4\}} (-1)^{|S|} \rho_S \otimes I_{S^c}$, where ρ_S is the marginal on the sites in S and $\rho_\varnothing = 1$. Since $\operatorname{tr}[\rho(\rho_S \otimes I)] = \operatorname{tr} \rho_S^2$,

$$\operatorname{tr}(\rho \tilde{\rho}) = \sum_S (-1)^{|S|} \operatorname{tr} \rho_S^2.$$

For $\rho = |\psi\rangle\langle\psi|$ we have $\rho^T = |\bar{\psi}\rangle\langle\bar{\psi}|$, hence $\tilde{\rho} = |\tilde{\psi}\rangle\langle\tilde{\psi}|$ with $\tilde{\psi} = \sigma_y^{\otimes 4} \bar{\psi}$, and

$$\operatorname{tr}(\rho \tilde{\rho}) = |\langle \psi | \tilde{\psi} \rangle|^2 = |\psi^\dagger \sigma_y^{\otimes 4} \bar{\psi}|^2 = |\psi^T \sigma_y^{\otimes 4} \psi|^2 = \tau,$$

because $\psi^\dagger \sigma_y^{\otimes 4} \bar{\psi}$ is the complex conjugate of $\psi^T \sigma_y^{\otimes 4} \psi$ (as $(\sigma_y^{\otimes 4})^* = \sigma_y^{\otimes 4}$). For a pure four-qubit state, complementary purities agree ($\operatorname{tr} \rho_S^2 = \operatorname{tr} \rho_{S^c}^2$), so the sum has the two terms $S = \varnothing, \{1,2,3,4\}$ equal to 1, the eight terms with $|S| = 1, 3$ adding to $-2 \sum_i \operatorname{tr} \rho_i^2$, and the six terms with $|S| = 2$ adding to $2P$: $\tau = 2 - 2 \sum_i \operatorname{tr} \rho_i^2 + 2P$. Now $\operatorname{tr} \rho_i^2 = (1 + \lVert \mathbf{r}_i \rVert^2)/2$ gives the result. □

Every term of $P - 1 = \frac{1}{2}\Delta$ is quadratic in $|\psi\rangle\langle\psi|$ and hence an expectation value in two copies of the state. The term τ is the squared overlap of one copy with the spin flip of the other. This two-copy positivity is the first level of the hierarchy used at seven qubits (defined in Section 5 and Appendix G).

2.2 The sharp moment inequality

The purity bound alone does not give the sharp entropy bound. We also need to constrain the third moments of the same marginals. We state this bound homogeneously so that its sum-of-squares identity applies to unnormalised vectors.

For $z \in \mathbb{C}^{16}$, write $N = \lVert z \rVert^2$ and let R_{1j} be its unnormalised reduced matrix on qubits $1, j$. Put

$$P = \sum_{j=2}^{4} \operatorname{tr} R_{1j}^2, \qquad T = \sum_{j=2}^{4} \operatorname{tr} R_{1j}^3, \qquad F = 17NP - 12T - 12N^3.$$

For unnormalised z the letters P, T denote these homogeneous moments; at $N = 1$ they reduce to the unit-state P, T of Table 2, and $F = 17P - 12T - 12$.

Theorem 3 (moment inequality with polarisation remainder)**.** *Let* $W_1 = \sum_i \sum_{a=x,y,z} \langle z|\sigma_a^{(i)}|z\rangle^2$. *Then*

$$F(z) \geq \frac{3}{4} N W_1 \geq 0,$$

and the constant $3/4$ *is attained by every product state. For unit states this implies* $T \leq 5/12 + (17/12)(P-1)$. *The state* M_4 *has* $(P,T) = (1, 5/12)$ *and a product of two Bell pairs has* $(3/2, 9/8)$*; both satisfy* $F = 0$*, so both lie on the line* $T = 5/12 + (17/12)(P-1)$. *Consequently* $17/12$ *is the least* κ *for which* $T \leq 5/12 + \kappa(P-1)$ *holds for all unit states (the Bell point forces* $\kappa \geq 17/12$*). Every zero* $z \neq 0$ *of* F *is critical: all its one-qubit marginals are* $NI/2$.

Proof. The key identity expresses the moment defect as a sum of three kinds of nonnegative terms: one-qubit polarisation, the squared norms of contractions of two copies of the state with one spin-flipped copy, and the disagreement between the actions of the one-qubit marginals on the state. Let $\epsilon = i\sigma_y$, $\tilde{z} = \epsilon^{\otimes 4}\bar{z} = \sigma_y^{\otimes 4}\bar{z}$ and $w = z \otimes z \otimes \tilde{z}$, so that each qubit carries three slots x, y, t, one from each copy. At one qubit define maps from three slots to one by $L|xyt\rangle = \epsilon_{yt}|x\rangle$ and $M|xyt\rangle = \epsilon_{xt}|y\rangle$. Let S_i be the orthogonal projector onto the symmetric subspace of the three slots of qubit i, and contract the other three qubits:

$$U_i = S_i \prod_{j \neq i} L_j w, \qquad V_i = S_i \prod_{j \neq i} (L_j + M_j) w,$$

vectors on six qubits. Write R_i for the unnormalised one-qubit marginal of z and $R_i^{(i)}$ for its action on qubit i of $\mathbb{C}^{16}$. Appendix A proves the identity

$$\boxed{8F = 6NW_1 + 3\sum_i \|V_i - 8U_i\|^2 + 12\sum_{i<j} \|(R_i^{(i)} - R_j^{(j)})z\|^2.} \tag{S}$$

Identity (S) is a polynomial identity of degree six in the amplitudes of z, that is, cubic in $|z\rangle\langle z|$ (a three-copy identity); we call it the *sextic identity*. The derivation uses contractions with the two-dimensional antisymmetric tensor, the vanishing of three-vector antisymmetrisation in dimension two, and elementary swap traces. Nonnegativity and criticality follow at once. At a product state of norm N every $R_i^{(i)} z = Nz$ and $F = 3N^3 = \frac{3}{4} N W_1$. □

2.3 Entropy bound

We now combine the two moment bounds with a scalar interpolation inequality. The resulting gap formula makes both the bound and its equality conditions explicit.

Let $f(x) = -x \ln x$, and let h be its cubic Hermite interpolant at the nodes $\xi_1 = \frac{1}{6}$ and $\xi_2 = \frac{1}{2}$: it agrees with f and f' at both nodes. Its cubic and quadratic coefficients are

$$a = 18(\ln 3 - 1) > 0, \qquad b = 18 - \tfrac{39}{2} \ln 3.$$

These coefficients are fixed by matching f and f' at both nodes. Since $f^{(4)}(x) = -2/x^3 < 0$, the Hermite remainder has the sign

$$h(x) - f(x) = -\frac{f^{(4)}(\zeta_x)}{4!}(x - \tfrac{1}{6})^2(x - \tfrac{1}{2})^2 \geq 0 \qquad (x > 0),$$

for some $\zeta_x > 0$. Equality occurs only at the two nodes; at the endpoint $x = 0$, $h(0) - f(0) = \frac{1}{2} - \frac{3}{8} \ln 3 > 0$. Thus h majorises the entropy function on the entire spectral interval (Figure 2). In general a polynomial majorant of $-t \log_2 t$ on $[0, 1]$ touching it at k interior points has degree at least $2k - 1$, attained by the Hermite interpolant (Proposition S81 of the supplement). Two

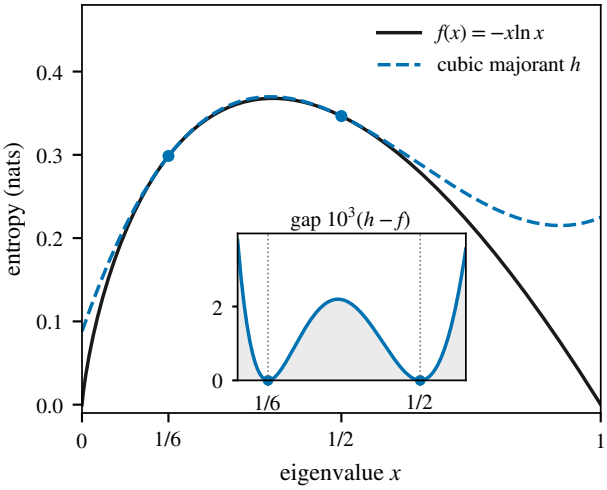


Figure 2: The single-eigenvalue entropy contribution $f(x) = -x \ln x$ (in nats) and its cubic majorant h, tangent at the eigenvalues $1/6$ and $1/2$ of every cut of M_4. The inset magnifies the gap $h - f$ by 10^3 on $[0.1, 0.6]$: it is nonnegative, with double zeros at the nodes. Summing h over the cut spectra gives an affine upper bound on the entropy in terms of P and T; equality in this scalar bound requires every eigenvalue to be at a node.

tangencies thus need a cubic; at seven qubits this degree count leads to a two-variable majorant (Section 5).

Return now to unit states. Each cut has four eigenvalues λ with $\sum\lambda = 1$, $\sum\lambda^2 = \mathrm{tr}\rho_{1j}^2$, $\sum\lambda^3 = \mathrm{tr}\rho_{1j}^3$. Summing h over the 12 eigenvalues of the three cuts,

$$3 \ln 2 \; E(\psi) = \sum_{j,\lambda} f(\lambda) = \kappa_0 + bP + aT - \Delta_H, \qquad \Delta_H = \sum_{j,\lambda} (h - f)(\lambda) \geq 0,$$

with a constant κ_0 independent of ψ. At M_4 every cut has spectrum $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$, so $\Delta_H = 0$, $P = 1$, $T = \frac{5}{12}$ and $E(M_4) = \frac{1}{2} + \frac{1}{2}\log_2 6 = 1 + \frac{1}{2}\log_2 3$. Subtracting, and using $T = (17P - 12 - F)/12$ (the definition of F at $N = 1$),

$$\begin{aligned} 3 \ln 2 \, (E(M_4) - E(\psi)) &= -b(P - 1) - a(T - \tfrac{5}{12}) + \Delta_H \\ &= (\tfrac{15}{2} - 6 \ln 3)(P - 1) + \tfrac{3}{2}(\ln 3 - 1)F + \Delta_H. \end{aligned} \tag{EG}$$

Here $-b - \frac{17}{12}a = \frac{15}{2} - 6 \ln 3$ and $a/12 = \frac{3}{2}(\ln 3 - 1)$. All three terms are nonnegative: $\frac{15}{2} - 6 \ln 3 > 0$ (since $1 < \ln 3 < 5/4$), $P \geq 1$ (Lemma 2), $F \geq 0$ (Theorem 3), $\Delta_H \geq 0$. This proves $E(\psi) \leq E(M_4) = 1 + \frac{1}{2}\log_2 3$.

Why the bound is tight. The entropy bound uses only the nonnegativity of F, rather than its stronger polarisation remainder. Its unit zeros are exactly the two M_4 LU orbits and the three LU orbits of products of two Bell pairs, one for each pairing of the qubits (Proposition 14 of Appendix A). Only the two HS orbits saturate the entropy bound, since products of two Bell pairs have $P = \frac{3}{2}$. The cubic majorant touches the entropy function exactly at the eigenvalues of M_4, and its coefficient signs make both moment slacks, $P - 1$ and F, enter (EG) with nonnegative weights. The same criterion yields the Tsallis bound for $0 < q < 2$ and the Rényi bound for $0 < q < 1$ (Proposition 6). At Tsallis order $q = 2$ all critical states with $\tau = 0$ tie. Appendix C gives a correlation-volume interpretation of the moment inequality. The sextic identity also controls the spin flip: for unit states $F \geq \frac{3}{4}W_1 + \tau - \tau^{3/2}$ (Theorem 15), with equality on product states and, for every $\tau \in [0, 1]$, on a family of critical states. With the gap formula (EG) this gives the Tsallis and von Neumann stability estimates in W_1 and τ of Appendix E.

Being cubic in $|\psi\rangle\langle\psi|$, the sextic identity is a certificate at the three-copy level of the same hierarchy, for the objective F instead of a purity (Appendix G).

2.4 Equality and one-cut rigidity

Proposition 4 (one-cut rigidity)**.** *If $P = 1$ and a $2|2$ cut has spectrum $(1/2, 1/6, 1/6, 1/6)$, then ψ is LU-equivalent to M_4 or $\overline{M_4}$.*

Proof. Relabel the chosen cut as $12|34$. Lemma 2 gives criticality and $\tau = 0$. Write two-qubit states in the Bell basis with the phases of the magic basis,

$$m_1 = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad m_2 = \tfrac{i}{\sqrt{2}}(|01\rangle + |10\rangle), \quad m_3 = \tfrac{1}{\sqrt{2}}(|01\rangle - |10\rangle), \quad m_4 = \tfrac{i}{\sqrt{2}}(|00\rangle - |11\rangle),$$

and a four-qubit state as $\psi = \sum_{j,k} Z_{jk}\, m_j \otimes m_k$, with qubits $1, 2$ indexing the rows of Z and qubits $3, 4$ the columns. In this basis single-qubit unitaries on each pair act, up to phase, as real four-dimensional rotations, so local unitaries act by $Z \mapsto O_1 Z O_2^T$ with $O_1, O_2 \in SO(4)$. By the critical normal form (Theorem 16 of Appendix B), ψ is LU-equivalent to $G(v) = \sum_k v_k\, m_k \otimes m_k$, that is $Z = \mathrm{diag}(v)$, with $v \in \mathbb{C}^4$ a unit vector. The cut spectrum and τ are LU invariant (the latter by (τ_G)). On $G(v)$ the spectrum of the cut $12|34$ is $\{|v_k|^2\}$ (the m_k are orthonormal), and $\tau = |\sum_k v_k^2|^2$ by (τ_G). Fix the large coordinate to $1/\sqrt{2}$ by a common phase. Then $\tau = 0$ says that the other three squares have modulus $1/6$ and sum to $-1/2$. Their sum has the largest possible modulus, $3(1/6) = 1/2$, so equality in the triangle inequality forces all three squares to equal $-1/6$. Thus, up to position of the large coordinate,

$$v = (\sqrt{3}, \pm i, \pm i, \pm i)/\sqrt{6}.$$

The rotations $Z \mapsto O_1 Z O_2^T$ reduce these sign choices to two. Applying the same determinant-one signed permutation on both sides of $G(v)$ moves the large entry to the first position without changing the other entries except for their order. A diagonal rotation on one side can flip any pair of small signs. Thus only their parity matters, giving the representatives $(\sqrt{3}, i, i, i)/\sqrt{6}$ and $(\sqrt{3}, -i, -i, -i)/\sqrt{6}$. Expanding the m_k shows that

$$M_4 = G((1, 1, i\sqrt{3}, 1)/\sqrt{6}).$$

Multiplying by $-i$ and moving its third coordinate to the first gives the second representative. Complex conjugation gives the first, since each $m_k \otimes m_k$ is a real vector, so $\overline{G(v)} = G(\bar{v})$. Both representatives therefore belong to the stated pair of LU orbits. Relabelling preserves the pair of orbits, since it permutes the coefficient triple $(1, \omega, \omega^2)$ up to common phase and conjugation. □

Proof of equality in Theorem 1. Suppose $E(\psi) = E(M_4)$. The positive coefficients in (EG) force $P = 1$ and $\Delta_H = 0$. The strict Hermite remainder above then forces every cut eigenvalue to be $\frac{1}{6}$ or $\frac{1}{2}$. Four such eigenvalues with sum 1 are $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$, so Proposition 4 applied to any one cut shows that ψ is LU-equivalent to M_4 or $\overline{M_4}$. This identifies the equality states without classifying all zeros of F or separately excluding Bell-pair products. Conversely, M_4 and its conjugate have this spectrum on every cut and attain the bound. This proves Theorem 1, including equality.

Remark 5 (a single orbit). The transposition of qubits 2 and 3 exchanges $|0011\rangle \leftrightarrow |0101\rangle$ and $|1100\rangle \leftrightarrow |1010\rangle$ and fixes $|0110\rangle$ and $|1001\rangle$. It therefore sends the coefficient triple $(1, \omega, \omega^2)$ of M_4 to $(\omega, 1, \omega^2) = \omega\,(1, \omega^2, \omega)$, that is,

$$M_4 \longmapsto \omega\, \overline{M_4}.$$

Since E is invariant under qubit permutations and ω is a global phase, the two local-unitary orbits of Theorem 1 are exchanged by a permutation of the qubits: the set of maximisers is a single orbit under local unitaries together with qubit permutations. This is a special case of the observation of Lyons, Arnold and Swogger [13]: even permutations of the qubits fix M_4 up to phase, and odd ones map it to $\overline{M_4}$.

3 Rényi, Tsallis and mixed-state entropies

Different spectral entropies give different weights to the marginal eigenvalues. For $q > 0$, $q \neq 1$, the Rényi [35] and Tsallis [36] entropies

$$H_q(\rho) = \frac{\log_2 \operatorname{tr} \rho^q}{1-q}, \qquad S_q(\rho) = \frac{1 - \operatorname{tr} \rho^q}{q-1}$$

tend as $q \to 1$ to $S(\rho)$ in bits and $(\ln 2)S(\rho)$ in nats, respectively. As $q \to 0$, H_q tends to the base-two logarithm of the rank of ρ, which is the Schmidt rank across a pure-state cut; $H_2 = -\log_2 \operatorname{tr} \rho^2$ is fixed by the purity, the quantity minimised at seven qubits and the one that randomised-measurement experiments estimate directly [20]; as $q \to \infty$, H_q depends only on the largest eigenvalue. Each is Schur-concave: averaging any pair of eigenvalues cannot lower it. For pure states it therefore does not increase under deterministic local operations and classical communication [19]: every member measures how entangled a cut is. Asking which state maximises them all tests whether "most entangled" is a robust notion. The answer depends on the order: M_4 maximises the mean Rényi entropy for $q < q_* \approx 1.588$, and the cluster states maximise it for $q > q_*$. For the Tsallis entropies M_4 is the maximiser for $0 < q < 2$; the cluster states tie with it at $q = 2$ and beat it for every $q > 2$. The maximiser for $q > 2$ is identified by Gour and Wallach [8, Theorem 9(a)]; we do not reprove that result. The cut spectra explain the change. Those of the cluster state are flat, $(\frac{1}{2}, \frac{1}{2})$ on one cut and uniform $\frac{1}{4}$ on the other two, so its mean Rényi entropy is $5/3$ for every q; the lopsided spectrum $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ of M_4 gives a mean H_q falling from 2 bits as $q \to 0$ to 1 bit as $q \to \infty$. Small q favours full rank and large q a small largest eigenvalue, hence the crossing at q_*. The Tsallis tie at $q = 2$ holds because $1 - \operatorname{tr} \rho^2$ depends only on the cut purities, and both states have $P = 1$.

The cubic-majorant argument applies to a wider class of spectral functionals. We state that criterion and apply it to Tsallis entropies and Rényi orders below one; orders above one need additional estimates (Appendix F). We then transfer the pure-state bounds to mixed-state convex roofs. We retain the notation P, T, F, τ from the maximum proof; proofs not given here are in Appendix E.

Proposition 6 (Cubic spectral majorants)**.** *For continuous $f : [0,1] \to \mathbb{R}$, set*

$$\mathcal{E}_f(\psi) = \frac{1}{3} \sum_{j=2}^{4} \operatorname{tr} f(\rho_{1j}).$$

Suppose $h(x) = ax^3 + bx^2 + cx + d$ majorises f on $[0,1]$, agrees with f at $1/6$ and $1/2$, and satisfies $a \geq 0$, $b + 17a/12 \leq 0$. Then $\mathcal{E}_f(\psi) \leq \mathcal{E}_f(M_4)$ for every pure four-qubit state. If both coefficient inequalities are strict, equality holds precisely on the two HS LU orbits.

Proof. Put $\alpha = -b - 17a/12$ and $\Delta_f = \sum_{j,\lambda \in \operatorname{spec} \rho_{1j}} (h - f)(\lambda) \geq 0$. Summing the cubic and subtracting its value at M_4 gives

$$3(\mathcal{E}_f(M_4) - \mathcal{E}_f(\psi)) = \alpha(P-1) + \frac{a}{12} F + \Delta_f. \tag{SF}$$

All three terms are nonnegative, since $P \geq 1$ (Lemma 2) and $F \geq 0$ (Theorem 3). The same two nonnegative moment defects therefore control every functional satisfying the coefficient conditions. If $a, \alpha > 0$, equality requires $P = 1$ and $F = 0$, so Proposition 14 gives the HS orbits; the other zeros of F, the products of two Bell pairs, have $P = 3/2$ and are excluded. No strict scalar majorisation is required. For differentiable f, a candidate touching cubic is its Hermite interpolant at the two nodes. □

Corollary 7 (Tsallis entropies)**.** *For every $0 < q < 2$, $q \neq 1$, the mean Tsallis-q entropy of the three cuts satisfies*

$$\boxed{\frac{1}{3} \sum_{j=2}^{4} \frac{1 - \operatorname{tr} \rho_{1j}^q}{q-1} \leq \frac{1 - 2^{-q} - 3\,6^{-q}}{q-1}}$$

with equality iff ψ is LU-equivalent to M_4 or $\overline{M_4}$.

The range is sharp: at $q = 2$ equality is no longer unique, and for every $q > 2$ the cluster state beats M_4 (proved in Appendix E, which also gives the corresponding quadratic deficit bounds).

Rényi entropies. With H_q as above, write

$$\mathcal{E}_q(\psi) = \frac{1}{3}\sum_{j=2}^{4} H_q(\rho_{1j}), \qquad h_q = \mathcal{E}_q(M_4) = \frac{\log_2(2^{-q} + 3\,6^{-q})}{1-q}.$$

At $q = 1$ these have their continuous von Neumann values in bits.

Corollary 8 (Rényi entropies below one)**.** *For $0 < q < 1$, concavity of* log *and the Tsallis bound imply*

$$\frac{1}{3}\sum_{j=2}^{4} \frac{\log_2 \operatorname{tr} \rho_{1j}^{q}}{1-q} \leq \frac{\log_2(2^{-q} + 3\,6^{-q})}{1-q},$$

again with exactly the same LU equality classes. The argument does not assert the analogous Rényi bound for $q > 1$.

Equality needs equality in the Tsallis bound, hence the HS orbits, and in Jensen's inequality for the logarithm over the three cuts, which holds there because the three cuts of M_4 have identical spectra.

Orders above one require additional estimates. The full result is as follows, with q_* the unique order for which $h_{q_*} = 5/3$ (Proposition 24).

Corollary 9 (Rényi envelope of Gour and Wallach)**.** *For every pure four-qubit state and every $q > 0$, $\mathcal{E}_q(\psi) \leq \max\{h_q, 5/3\}$. For $q < q_*$ the maximisers are exactly the two HS LU orbits; for $q > q_*$ they are the cluster states up to local unitaries and qubit permutations (every permutation of the three cuts is induced by a permutation of the qubits). At $q = q_*$ both classes, and only those classes, attain the bound.*

Below one the bound follows from the preceding corollary, and at $q = 1$ from Theorem 1. For $q > 1$ it is proved in Appendix F, which opens with a plan of the argument; for $1 < q < 2$ the proof is analytic apart from exact finite checks, namely an exact six-copy certificate bounding the variance of the three cut purities and exact polynomial sign checks. Figure 3 shows the change of maximiser.

3.1 Mixed-state entanglement

For a mixed state, the entropy of a marginal also counts classical mixing, which is not entanglement: the maximally mixed four-qubit state has maximally mixed marginals but no entanglement. We therefore use the *convex roof*: the least average of a pure-state spectral functional over all ensembles representing the state, as in the entanglement of formation [37]. For a general f obeying the hypotheses below, this construction is an entanglement monotone when the induced function of the marginal is concave and vanishes on pure marginals, as for the von Neumann and Tsallis entropies; the ceiling below needs neither property. In the cut notation below, $\widehat{1j}$ denotes the complementary pair.

Corollary 10 (Convex roofs)**.** *Let f satisfy the hypotheses of Proposition 6. For each cut define the convex roof*

$$E_f^{1j|\widehat{1j}}(\rho) = \inf_{\rho=\sum_\ell p_\ell |\psi_\ell\rangle\langle\psi_\ell|} \sum_\ell p_\ell \operatorname{tr} f(\rho_{1j}^{(\ell)}).$$

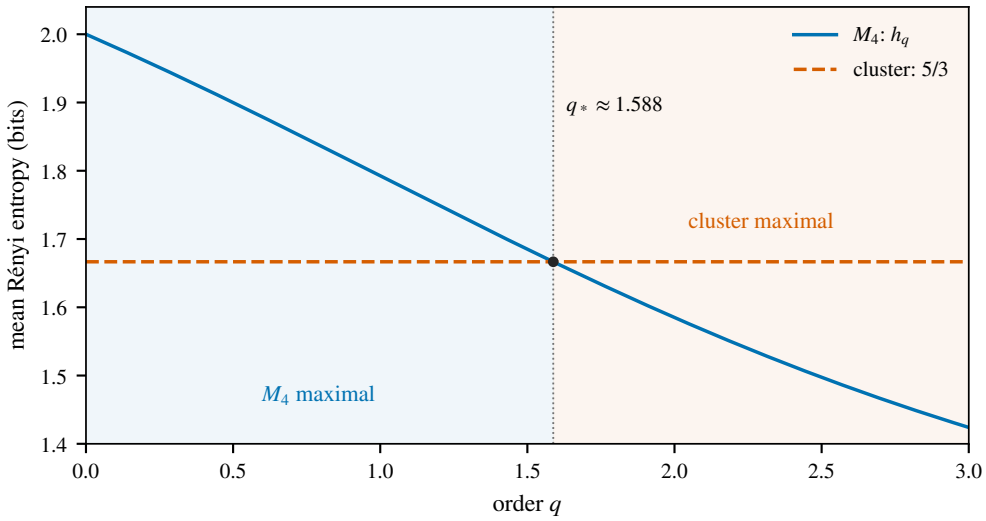


Figure 3: Change of Rényi-entropy maximiser with the order q. The mean entropy of M_4 (h_q, solid) and that of the cluster states (5/3, dashed) cross once, at $q_* \approx 1.588$ (Proposition 24). The larger of the two, $\max\{h_q, 5/3\}$, is the maximum over all pure four-qubit states (Corollary 9): M_4 attains it below q_* and the cluster states above q_* (shaded regions). Values at $q = 0$ and $q = 1$ are continuous limits.

Then every mixed four-qubit state satisfies

$$\frac{1}{3}\sum_{j=2}^{4} E_f^{1j|\widehat{1j}}(\rho) \le \mathcal{E}_f(M_4).$$

If both coefficient inequalities of Proposition 6 are strict, equality holds exactly for pure states in the two M_4 LU orbits. For $f = -x\log_2 x$ these are the three bipartite entanglements of formation. The same HS ceiling and pure-only equality classification hold for the Tsallis entropies with $0 < q < 2$ and the Rényi entropies with $0 < q < q_*$, including their von Neumann limits at $q = 1$.

To prove the ceiling, fix any pure-state ensemble $\rho = \sum_\ell p_\ell |\psi_\ell\rangle\langle\psi_\ell|$. Each cut's roof is at most that cut's ensemble average, so the mean of the three roofs is at most $\sum_\ell p_\ell \mathcal{E}_f(\psi_\ell) \le \mathcal{E}_f(M_4)$ by Proposition 6 (in the Rényi case, with $\mathcal{E}_q$ in place of $\mathcal{E}_f$, by Corollary 9). Appendix E excludes mixed maximisers: every mixed-state support contains a vector of deficient Schmidt rank (fewer than four nonzero Schmidt coefficients across a given 2|2 cut), whereas every pure maximiser has full rank there.

4 Stability of the four-qubit maximum

A nearly maximal four-qubit entropy forces the state close to a maximising orbit. Let $d(\psi)$ be the minimum Hilbert-space distance from a unit state ψ to the LU orbits of M_4 and $\overline{M_4}$, allowing a global phase, and $\delta(\psi) = 1 + \frac{1}{2}\log_2 3 - E(\psi)$ the entropy deficit. Then $\delta \ge d^2/20$ for every pure state (Theorem 19): an entropy within ε of the maximum guarantees distance at most $\sqrt{20\varepsilon}$ from a maximiser. Near the orbits the bound improves to $\delta \ge \frac{3}{10}d^2$ when $d \le 1/20$, and the power two is optimal. On the critical locus, where all one-qubit marginals are maximally mixed, the optimal bound is $\delta \ge C_* d^2$ with $C_* \approx 0.19058$, attained by explicit states (Theorem 21), so no universal coefficient can exceed C_*. The coefficient $\frac{3}{10}$ holds only for $d \le 1/20$, whereas the states attaining C_* lie at distance $d \approx 0.55$. The proof compares the infidelity to the orbits with a sextic moment defect, with optimal coefficient one (Theorem 18). The same estimates transfer to the Tsallis entropies of Section 3 (Corollary 22). Appendix D states these results with their constants and proof strategy.

5 Seven qubits: compatible moments and entropy

The four-qubit proof bounds compatible marginal moments and then converts those bounds into an entropy bound. We follow the same strategy for seven qubits, where AME states again do not exist [16]. Five-copy positivity determines the purity minimum and its equality orbit. For entropy, a majorant involving each marginal and its spin flip gives a second five-copy certificate. The two optima occur at different members of one icosahedral family.

The two optimisation problems. A balanced cut divides the seven qubits into groups of three and four. It suffices to consider the three-qubit side S, whose reduced state ρ_S has eight eigenvalues. There are $\binom{7}{3} = 35$ such subsets, and we average over all of them:

$$E_3(\psi) = \frac{1}{35} \sum_{|S|=3} S(\rho_S), \qquad \pi_3(\psi) = \frac{1}{35} \sum_{|S|=3} \operatorname{tr} \rho_S^2.$$

Every cut has entropy at most 3 bits and purity at least 1/8, and reaching either bound on every cut would require an AME state.

The mean purity has a direct interpretation in terms of correlations. Expanding each marginal in Pauli strings writes π_3 as 1/8 plus a positively weighted sum of the squared expectation values of all Pauli strings acting on one, two or three qubits (Appendix G). An AME state would make all these low-weight correlations vanish. Minimising π_3 asks how small they can be when compatibility prevents them all from vanishing. Every seven-qubit stabilizer state has $\pi_3 \geq 19/140$ (Proposition 43), strictly above the optimum 121/917 of Theorem 11 below; hence the purity minimiser is not a stabilizer state.

The two optima. Both optimisers are members of an explicit one-parameter family of seven-qubit states ψ_x, defined below. Each has one distinguished qubit s; the other six qubits are labelled by the six axes of an icosahedron. A single mixing weight $x \in [0, 1/2]$ parametrises the family. The purity minimiser is the *icosahedral state* $\Psi_{\text{ico}} = \psi_{37/524}$, and the entropy maximiser is ψ_{x_*}, where $x_* = 0.0698\ldots$ is the root of equation (2) below.

Theorem 11 (the seven-qubit purity minimum)**.** *Every pure seven-qubit state satisfies* $\pi_3(\psi) \geq 121/917$. *Equality holds if and only if* ψ *is equivalent to* Ψ_{ico} *under local unitaries and permutations of the qubits.*

The proof is computer-assisted: a five-copy identity verified in floating point with rigorous error bounds, followed by an exact equality analysis. It is outlined below and given in full in Section S11.6 of the supplement.

Theorem 12 (seven-qubit mean-entropy maximum and equality)**.** *Every pure seven-qubit state satisfies*

$$E_3(\psi) \leq E_3(\psi_{x_*}) = 2.965356029477675899\ldots \text{ bits.}$$

Equality holds if and only if ψ *is local-unitarily and qubit-permutation equivalent to* ψ_{x_*}. *With fixed qubit labels, the equality set consists of exactly* 42 *LU orbits.*

The proof is computer-assisted: a five-copy identity verified in floating point with rigorous error bounds, an exact certificate for the scalar majorant, Theorem 11, and an exact equality analysis. It is outlined below and given in full in Section S12 of the supplement.

At four qubits M_4 both minimises the summed purity $P = 3\pi_2$ and maximises entropy. At seven qubits these objectives select different states (Table 3).

	ideal AME value (unattainable)	optimum	optimiser
$\min \pi_3$	0.125	$\frac{121}{917} \approx 0.131952$	Ψ_{ico}, unique up to local unitaries and permutations
$\max E_3$ (bits)	3	$2.96535602947\ldots$	ψ_{x_*}, unique up to local unitaries and permutations

Table 3: Mean three-qubit purity and entropy of seven-qubit pure states. The AME values are not attained. The states Ψ_{ico} and ψ_{x_*} both lie in the icosahedral family ψ_x defined below.

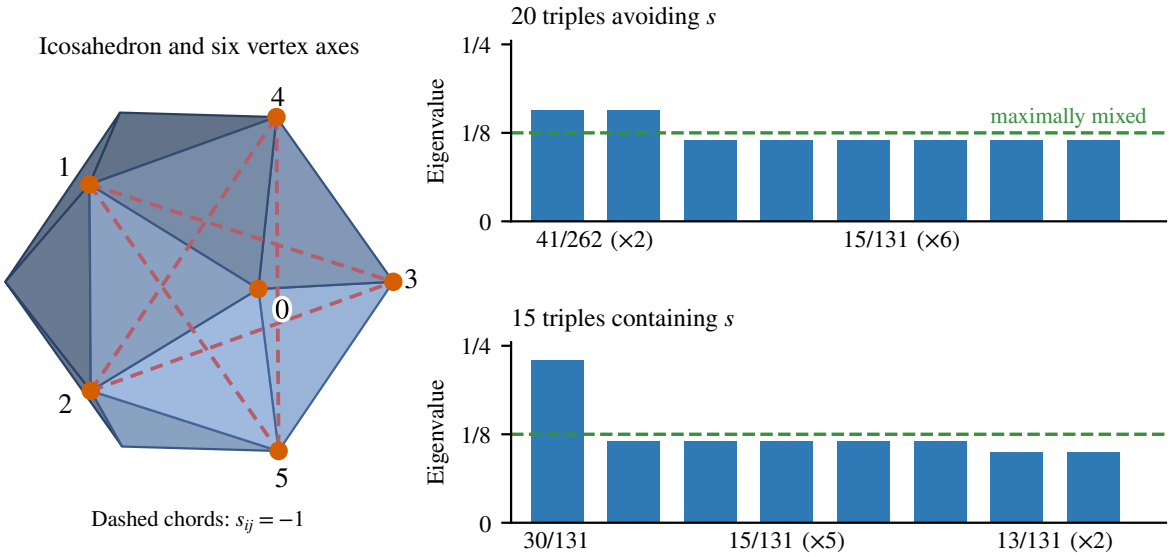


Figure 4: Icosahedral geometry and marginal spectra of $\Psi_{\text{ico}} = \psi_{37/524}$. Left: six oriented representatives of the axes through opposite icosahedral vertices label qubits $0, \ldots, 5$; the distinguished seventh qubit s is separate. Dashed chords mark pairs whose amplitude sign s_{ij} is -1. Right: spectra of the 20 three-qubit marginals avoiding s and the 15 containing it; multiplicities in parentheses (the values $(4u)^6, (3u+x)^2$ and $8u, (4u)^5, (u+x)^2$ above at $u = 15/524$); every marginal in a class has the spectrum shown. The dashed horizontal line marks the maximally mixed eigenvalue $1/8$. This state minimises the mean purity over all pure states. The entropy maximiser ψ_{x_*}, $x_* = 0.0698\ldots$, is a different member of the same family; its spectra have the same multiplicities, and each eigenvalue differs from the one shown by less than 10^{-3}, too little to see at this scale.

An explicit state. Label the qubits $s, 0, 1, \ldots, 5$, with $0, \ldots, 5$ the axes through opposite vertices of the icosahedron. Let e_i denote the six-bit string with a single 1 at site i. Set $s_{ij} = -1$ for $\{i, j\}$ an edge of the pentagon $1-3-2-4-5-1$, and $s_{ij} = +1$ otherwise. For $0 \le x \le 1/2$ put $\alpha = \sqrt{(1-2x)/30}$ and (equation (14) of Appendix H)

$$\psi_x = |0\rangle_s \Big(\alpha \sum_{i<j} s_{ij} |e_i + e_j\rangle + \sqrt{x}\, |111111\rangle \Big) + |1\rangle_s \Big(\sqrt{x}\, |000000\rangle - \alpha \sum_{i<j} s_{ij} |111111 - e_i - e_j\rangle \Big).$$

Every member has maximally mixed one-qubit marginals, and its three-qubit marginals fall into two spectral classes (Theorem 41). Writing $u = (1-2x)/30$, the 20 marginals avoiding s have eigenvalues $4u$ six times and $3u + x$ twice; the 15 containing s have eigenvalues $8u$ once, $4u$ five times and $u + x$ twice. Consequently the mean purity $\pi_3(\psi_x)$ is an explicit quadratic in x, with minimum $121/917$ at $x = 37/524$: the icosahedral state Ψ_{ico} (Figure 4).

The icosahedron enters through the signs. The matrix C with zero diagonal and entries s_{ij} off it is a symmetric *conference matrix*: $C^2 = 5I$. Its eigenvalues are $\pm\sqrt{5}$, each three times, so $I + C/\sqrt{5}$

is positive semidefinite of rank three with unit diagonal. It is therefore the Gram matrix of six unit vectors in $\mathbb{R}^3$ with $v_i \cdot v_j = s_{ij}/\sqrt{5}$: six equiangular lines [38] at angle $\arccos(1/\sqrt{5})$, which are the axes through opposite vertices of an icosahedron. The same identity block-diagonalises every two- and three-qubit marginal of ψ_x in the computational basis, which is how Theorem 41 obtains the spectra.

The family is the fixed set of a symmetry group generated by local unitaries, qubit permutations and complex conjugation (up to signs and phase). Hence a member stationary within the family is stationary on the whole sphere for the purity, the entropy or any other differentiable functional invariant under these symmetries (Lemma S72 of the supplement, which identifies the group).

Why an icosahedron. An exact purity identity shows why minimising purity selects this geometry. Widen the family: replace the pair amplitudes αs_{ij} of ψ_x by complex numbers t_{ij} in the $|0\rangle_s$ branch and by their conjugates in the $|1\rangle_s$ branch. They form a complex symmetric 6×6 matrix $Q = (t_{ij})$ with zero diagonal; put $p_i = \sum_j |t_{ij}|^2$ and $\sigma = \sum_{i<j} |t_{ij}|^2$, so normalisation is $2\sigma + 2x = 1$. For every such state the weight-one and weight-three Pauli correlations on the six ordinary qubits cancel between the two branches (Theorem S51 of the supplement), and (Theorem S53 of the supplement)

$$\begin{aligned} \pi_3 - \frac{121}{917} = \frac{(524x-37)^2}{137550} + \frac{4}{35}\sum_i \Big(p_i - \frac{\sigma}{3}\Big)^2 \\ + \frac{48}{35}\sum_{i<j}\Big(|t_{ij}|^2 - \frac{\sigma}{15}\Big)^2 + \frac{24}{35}\sum_{i<j}|(QQ^\dagger)_{ij}|^2. \end{aligned} \tag{1}$$

At equality $x = 37/524$, all $|t_{ij}|$ are equal and the rows of Q are orthogonal: Q is a multiple of a complex conference matrix of order six, which is unique up to diagonal phases and permutations (Proposition S54 of the supplement) and is the icosahedral sign pattern above. Minimum purity thus forces the six rows of pair amplitudes to be orthogonal with equal moduli. The identity is exact only within this family and does not bound arbitrary states; the global bound is the five-copy certificate of Theorem 11, whose equality analysis reduces an arbitrary minimiser to this family and this identity.

Entropy selects a different state. Since the marginal eigenvalues of ψ_x are affine in x, its mean entropy is explicit: with $f(t) = -t\log_2 t$ and $u = (1-2x)/30$,

$$E_3(\psi_x) = \tfrac{1}{35}\big[195f(4u) + 40f(3u+x) + 15f(8u) + 30f(u+x)\big]$$

(Appendix H). It is strictly concave in x, and its unique maximum occurs at the root $x_* \in (0, 1/2)$ of

$$2^{32}(1-2x)^{15} = 3^8(1+8x)^8(1+28x)^7 \tag{2}$$

(Proposition S73 of the supplement), giving $E_3(\psi_{x_*}) = 2.965356029\ldots$ bits. This root lies strictly below $37/524$ (Corollary 42), so Ψ_{ico} is not stationary for the entropy, even within the family. By Theorem 11 no purity minimiser maximises the mean entropy: unlike at four qubits, the two problems have different solutions. Their objective values are nevertheless very close: $E_3(\Psi_{\text{ico}})$ lies only $7.1597\ldots \cdot 10^{-6}$ bits below the maximum, and $\pi_3(\psi_{x_*})$ exceeds $121/917$ by $1.2204\ldots \cdot 10^{-6}$ (rigorous interval enclosures).

The marginal moments give a heuristic explanation for this difference; the exact derivative in Corollary 42 supplies the proof. In the Taylor expansion of f about the mean eigenvalue $1/8$, the second-moment contribution favours small π_3, while the positive cubic coefficient favours a large $b_3 = \frac{1}{35}\sum_{|S|=3} \operatorname{tr}\rho_S^3$. At four qubits M_4 does both: it minimises P and maximises the cubic moment T on the slice $P = 1$ (Theorem 3). Along ψ_x, instead, $b_3(\psi_x) = (35008x^3 + 17328x^2 - 3084x + 709)/31500$ has $b_3'(37/524) = -432/120127 < 0$ (Appendix H): moving x below $37/524$ raises b_3 at first order and costs π_3 only at second order.

The marginal spectra and pair purities of the minimiser agree, to the six digits published, with those of a seven-qubit state found numerically by Di Martino, Facchi and Florio [10]. This suggests that the two states are LU-equivalent (Appendix H); Theorem 41 gives an explicit closed-form state.

Copies of the state. A purity is an expectation value in two copies, and a homogeneous polynomial of degree m in $|\psi\rangle\langle\psi|$ is an expectation value in m copies. Averaging the projector onto $\psi^{\otimes m}$ over local unitaries, qubit permutations and complex conjugation gives the *twirled m-copy moment*. Replacing some copies by the spin-flipped state $\tilde{\psi} = \epsilon^{\otimes n}\bar{\psi}$ (which differs from $\sigma_y^{\otimes n}\bar{\psi}$ only by the phase i^n), $\epsilon = i\sigma_y$, gives further positive operators, all linear images of it. Minimising π_3 over all moment operators subject to these positivity conditions gives lower bounds that increase with m to the true minimum, a bosonic-extension hierarchy in the spirit of [17], and close to the copy hierarchy of Yu et al. for the pure-state marginal problem [18] (Appendix G and Theorem S44 of the supplement). A certificate at level m is an identity

$$\text{(objective)} - \text{(bound)} = \text{a sum of squared norms of linear images of } m \text{ copies of } \psi, \text{ some spin flipped.}$$

At four qubits $P \geq 1$ follows at the two-copy level from Rains' shadow inequalities [21], and the sextic identity (S) is a three-copy certificate for the different objective F (Appendix G). At seven qubits the levels separate: two copies give only the AME value $1/8$, three copies provably cannot reach $121/917$, four copies provably fall short (numerically by about $7.5 \cdot 10^{-5}$), and five copies are the first level that attains the sharp purity bound (Table 8).

Proof outline for the purity minimum. The proof has three steps; the entropy proof below repeats them at a different point of the family.

1. *Bound.* A five-copy identity writes $\pi_3 - 121/917$ as a sum of twelve terms, each the expectation value of a positive operator in five copies of ψ, some of them spin flipped, averaged over local unitaries and qubit permutations. Every term is therefore nonnegative, which gives the bound. Unlike at four qubits, the identity needs computer assistance: floating-point linear algebra with rigorous error bounds establishes that suitable positive definite coefficient matrices exist.

2. *Equality moments.* At equality all twelve terms vanish. Their vanishing fixes the averaged five-copy moment of ψ to that of Ψ_{ico} (Theorem S66 of the supplement): every conjugation-even, permutation-invariant five-copy invariant of ψ takes its icosahedral value.

3. *Equality state.* A reconstruction theorem (Theorem S64 of the supplement) then shows that ψ differs from Ψ_{ico} only by local unitaries and a qubit permutation. Some of these invariants put ψ in the form of the widened family above, and the identity (1) makes the amplitude matrix Q a conference matrix. With qubit labels fixed, the orbit of Ψ_{ico} consists of 42 LU orbits: 7 choices of the distinguished qubit times 6 inequivalent sign patterns of the conference matrix.

Appendix G defines the terms and records the tangent-line step, the invariants used and the sizes of the computations; Section S11 of the supplement gives the full proof.

Proof outline for the entropy maximum. The proof follows the four-qubit pattern: a polynomial majorant touching the entropy at the spectra of the maximiser, then a moment certificate. Across the 35 cuts of ψ_{x_*}, four distinct eigenvalues occur. A single one-variable majorant of $-t \log_2 t$ sharp on every cut would therefore touch at four points and need degree at least seven, and hence seven copies (Proposition S81 of the supplement). We instead use each marginal ρ_S together with its spin flip $\widetilde{\rho}_S = \sigma_y^{\otimes 3}\rho_S^T\sigma_y^{\otimes 3}$, which is also the marginal of $\tilde{\psi}$ on S; it has the spectrum of ρ_S but

in general different eigenvectors. At ψ_{x_*} the two commute, and their joint eigenvalue pairs (λ, μ) take only five values. With $u = (1 - 2x_*)/30$, the cut eigenvalues of ψ_{x_*} are $\nu_1 = 3u + x_*$, $\nu_2 = 8u$, $\nu_3 = 4u$ and $\nu_4 = u + x_*$ (Theorem 41), and the pairs are (ν_3, ν_3), (ν_1, ν_1), (ν_4, ν_4), (ν_2, ν_3) and (ν_3, ν_2). A scalar inequality bounds the average entropy of two eigenvalues by a degree-five polynomial: there is a symmetric polynomial $H(\lambda, \mu)$ of degree five majorising $\frac{1}{2}[-\lambda \log_2 \lambda - \mu \log_2 \mu]$, with first-order contact at those five pairs (Section S12.6 of the supplement proves it, with exact one-variable sign checks). For arbitrary marginals the two eigenbases need not coincide. Their squared overlaps nevertheless form a doubly stochastic matrix, so averaging the scalar inequality with these weights gives, with h_{kl} the coefficients of H,

$$S(\rho_S) \leq \sum_{k,l} h_{kl} \operatorname{tr}(\rho_S^k \widetilde{\rho}_S^l),$$

even when ρ_S and $\widetilde{\rho}_S$ do not commute. The right side is a five-copy moment.

It suffices to bound this polynomial for states attaining or exceeding $E_3(\psi_{x_*})$, so the certificate is only needed on a compact domain containing them, cut out by two technical restrictions (Proposition S79 and Corollary S78 of the supplement). The purity minimum and the entropy–purity slope for eight eigenvalues give the *purity band* $121/917 \leq \pi_3 < 8693/64000$, and spin-flip compatibility between overlapping cuts caps every marginal eigenvalue at $497/1000$. On this domain a positive five-copy identity bounds the polynomial by the candidate value $E_3(\psi_{x_*})$ (Theorem S85 of the supplement; Appendix G lists its terms). At equality all its terms vanish, and the equality-moment argument and reconstruction theorem used for purity identify the orbit (Theorems S66 and S64 of the supplement).

Stability. Both optima are quadratically stable: each optimiser is a strict local optimum modulo local unitaries and phase with an explicit quadratic basin, and compactness extends this to a global lower bound on the objective deficit by a constant times the squared distance to the orbit under local unitaries and qubit permutations. We do not make the constant explicit (Theorems S69, S71, S88 and S89 and Corollary S90 of the supplement).

Comparison with four qubits. The two sizes differ in the equality set. At four qubits the purity minimum $P = 1$ is attained on a positive-dimensional set, containing M_4 and the cluster state, and the entropy needs the cut spectrum to single out M_4 (Proposition 4). At seven qubits the purity alone selects a single orbit under LU and qubit permutations, and the two-copy level of the hierarchy is no longer sharp (Section 6 compares other sizes).

Table 4 summarises the common method and the additional constraints needed at seven qubits. The degree rule of Section 2.3 explains both majorants: $k = 2$ gives the cubic of Section 2.3, and the joint majorant evades $k = 4$ by adding the spin-flip variable. It also shows that degree five is the least possible for a joint majorant. A joint majorant sharp at ψ_{x_*} must touch at the three diagonal pairs (λ, λ) among its contacts. On the diagonal it is therefore a majorant of $-t \log_2 t$ touching at three distinct points, which needs degree at least five (Corollary S83 of the supplement). Within this majorant method five copies are therefore necessary for the entropy; that says nothing about other certificates.

6 Discussion and open problems

Intermediate Rényi orders. Gour and Wallach conjectured that M_4 maximises the mean Rényi-q entropy for $q < \alpha_0 \approx 1.59$ and the cluster states for $q > \alpha_0$, and proved the cluster-state bound for $q \geq 2$ [8, Fig. 2 and Theorem 12]; their α_0 is the crossing q_* of Proposition 24, and Corollary 9 proves the conjecture. For $1 < q < 2$ a single-crossing property of individual spectra (Lemma 25) transfers two anchor bounds to all intermediate orders, and Appendix F gives the argument and the finite checks it uses.

	four qubits, 2|2 cuts	seven qubits, 3|4 cuts
mean purity minimum	$\pi_2 = 1/3$ (two copies)	$\pi_3 = 121/917$ (five copies)
purity equality set	positive-dimensional, contains M_4	one orbit, Ψ_{ico}
entropy maximiser	M_4	$\psi_{x_*} \neq \Psi_{\text{ico}}$
majorant	cubic in λ, contacts $1/6, 1/2$	degree five in (λ, μ), five contact pairs
moment certificate	sextic identity, three copies	certified identity, five copies
extra constraints	none	purity band, cap $\lambda \leq 497/1000$
entropy equality	M_4 orbit (Theorem 1)	ψ_{x_*} orbit

Table 4: The copy-hierarchy method at four and seven qubits. Here $\pi_2 = P/3$ and π_3 are the mean balanced cut purities. Orbits include local unitaries and qubit permutations; μ is an eigenvalue of the spin-flipped marginal.

Quantitative stability. The universal coefficient $1/20$ in Theorem 19 is not claimed to be sharp, and whether the critical optimum C_* of Section 4 holds for all states is open. Numerical tests of locally filtered critical states have found no smaller δ/d^2 (Remark S42 of the supplement). Local filtering does not preserve the entropy deficit, even though the overlap bound of Theorem 18 holds without a criticality assumption.

Seven-qubit questions. The five-copy proofs give sharp purity and entropy bounds, but an analytic certificate comparable to the four-qubit sextic identity would better explain the global role of the icosahedral geometry. The entropy ceiling also holds for mixed-state convex roofs, with equality only for pure states (Proposition S92 of the supplement); where the four-qubit argument finds a support vector of deficient Schmidt rank, the seven-qubit one uses that the optimiser's local-unitary tangent space contains no complex line.

The method. At both sizes the proof starts from the spectra of the candidate optimiser, builds a polynomial majorant touching the entropy at those spectra, bounds the required moments by positivity of multicopy operators with some copies spin flipped, and reads off the equality states from the vanishing terms. The spin-flip constraints rely on the pseudoreality $\epsilon\bar{u}\epsilon^{-1} = u$ of $SU(2)$. For $SU(d)$, $d \geq 3$, the defining representation is inequivalent to its conjugate, so this form of the multicopy argument does not extend directly to qudits.

Purity and entropy at other sizes. Table 5 collects what is known. Four qubits is the smallest size at which the question has content. At five qubits AME states exist, so the mean entropy reaches its elementary upper bound, exactly at the AME states; their LU classes form a continuum, each LU-equivalent to a point of the unique $((5, 2, 3))$ code (five qubits, a two-dimensional code space and distance three), with points identified under a group of order 24 [15]. The minimum purity is the value of the Rains linear programme [21, 25], the two-copy level of the hierarchy, attained by a graph state, at every $n \leq 18$ except 7 and 13 (Theorems S101, S103, S104 and S105 of the supplement [40]). A graph state is the stabilizer state obtained from $|+\rangle^{\otimes n}$ by a controlled-Z gate along each edge of a graph on the qubits; every stabilizer state is locally equivalent to one [32, 26], and C_4 is locally equivalent to the graph state of the four-vertex path. Seven qubits is the first size at which the two-copy level is not sharp: the linear-programming value would require the weight distribution of a pure $((7, 1, 4))_2$ code, which does not exist [16]. At thirteen qubits it is not sharp either: the three-copy level gives $\min \pi_6 \geq 0.018171249$, above the linear-programming value $25/1408$, and an explicit stabilizer state gives $\min \pi_6 \leq 85/4576$, but the exact minimum is open (Proposition S116 of the supplement); here the linear-programming value would require a pure $((13, 1, 6))_2$ code, which the three-copy bound excludes.

The maximum entropy is known only at the AME sizes and at four and seven qubits. Elsewhere each purity minimum, or at thirteen qubits its lower bound, gives an explicit entropy ceiling through

a sharp linear inequality: every d-dimensional density matrix, $d \geq 3$, satisfies $\log_2 d - S(\rho) \geq c_d(d \operatorname{tr} \rho^2 - 1)$ with $c_d = (d-1)\log_2(d-1)/(d(d-2))$ (Lemma S77 of the supplement). Being linear in the purity, this inequality averages over the balanced cuts, $E_k \leq k - c_{2^k}(2^k \pi_k - 1)$, whereas the largest entropy at fixed purity is not concave in the purity and does not average this way. At thirteen qubits an explicit state also gives $\max E_6 > 5.8336$ (Corollary S119 of the supplement). The ceilings are not attained by the graph-state purity minimisers. A stabilizer state's marginals are normalised projectors, so unless $n \in \{2, 3, 5, 6\}$ some balanced marginal is rank-deficient; such a marginal can always be perturbed to raise the entropy at order $t^2|\ln t|$. Hence no stabilizer state is even a local maximum of the mean balanced entropy (Corollary S94 of the supplement); in particular neither seven-qubit optimiser is a stabilizer state.

n	ideal purity 2^{-k}	$\min \pi_k$	$\max E_k$ (bits)
2, 3	1/2	1/2	1 (AME)
4	1/4	1/3	$1 + \frac{1}{2}\log_2 3$ (Theorem 1)
5	1/4	1/4	2 (AME)
6	1/8	1/8	3 (AME)
7	1/8	121/917	2.96535602947… (Theorem 12)
8	1/16	3/35	< 3.9029
9	1/16	1/14	< 3.9627
10	1/32	1/21	< 4.9163
11	1/32	1/28	< 4.9772
12	1/64	5/224	< 5.9594
13	1/64	[0.018171249, 85/4576]	(5.8336, 5.9846)
14	1/128	331/27456	< 6.9702
15	1/128	1/104	< 6.9874
16	1/256	1/156	< 7.9800
17	1/256	1/208	< 7.9928
18	1/512	7/2288	< 8.9901

Table 5: Mean balanced purity and entropy of n-qubit pure states. With $k = \lfloor n/2 \rfloor$, π_k and E_k are the mean purity and mean von Neumann entropy of the k-qubit marginals over all $\binom{n}{k}$ subsets of k qubits. The ideal purity 2^{-k} and entropy k are attained exactly at the AME sizes $n = 2, 3, 5, 6$. The known purity minima other than at $n = 7$ are Rains linear-programming values attained by graph states (Table S6 and Theorem S105 of the supplement); at $n = 13$ the minimum is open, and the entry is the interval of Proposition S116 of the supplement. Single entropy entries without $<$ are exact maxima; entries with $<$ are upper bounds on unknown maxima, the ceilings $k - c_{2^k}(2^k \min \pi_k - 1)$ rounded up, with the lower end of the purity interval in place of $\min \pi_k$ at $n = 13$. There the maximum is open; the upper end of its interval is that ceiling, and the lower end is exceeded by an explicit state (Corollary S119 of the supplement).

Purity and time reversal. A large overlap with the spin-flipped state costs purity: at $n = 4$, $\pi_2 = \frac{1}{3} + \frac{1}{6}(W_1 + \tau)$ (Lemma 2), and Rains' shadow inequalities give sharp analogues at larger sizes (Section S17 of the supplement [40]).

Formal verification. The four-qubit maximum, its attainment by M_4 and its equality case are formalised in Lean 4 with Mathlib [40], together with the moment inequality, the general and critical envelopes, most of the four-qubit spectral corollaries, the overlap bound with the global orbit-stability estimates, the sharp quadratic variance bound with the Rényi envelope above order one, and the exact computations for the icosahedral family. Finite checks of two certificates, for the overlap bound and the quadratic variance bound, use compiled code; the results depending on them inherit that trust assumption, and everything else uses the Lean kernel alone. The global seven-qubit theorems are not formalised. Section S19 of the supplement lists the scope, declarations and trust assumptions.

Use of AI

All results, documents and scripts were generated by AI agents (mainly Claude Opus and Sonnet 5.5, and OpenAI's GPT 6.1 sol) with human supervision and many cross-checks, including AI-generated Lean verification of most of the four-qubit results.

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A The sextic from elementary swap contractions

The proof of Theorem 3 has two parts. We first express three tensor contractions as marginal moments in (C). We then eliminate their mixed moment Π_2 using norms of one-qubit marginals acting on the state. The derivation uses two-dimensional antisymmetry, partial traces and matrix algebra; the eleven-type table evaluates the longest contraction. An independent coefficient-by-coefficient check of (S) in the sixteen complex amplitudes is also available (SI-1, SI-2, Appendix I); it does not use that table.

Write $R_S = \mathrm{tr}_{S^c}\,|z\rangle\langle z|$, $N = \|z\|^2$, $P = \sum_{j=2}^4 \mathrm{tr}\,R_{1j}^2$, $T = \sum_{j=2}^4 \mathrm{tr}\,R_{1j}^3$, $\Pi_1 = \sum_i \mathrm{tr}\,R_i^2$ and $\Pi_2 = \sum_{i<j} \mathrm{tr}[R_{ij}(R_i \otimes R_j)]$. The unnormalised Bloch vectors $\mathbf{r}_i = (r_i^a)$ have components $r_i^a = \langle z|\sigma_a^{(i)}|z\rangle$; put $W_1 = \sum_i \|\mathbf{r}_i\|^2 = 2\Pi_1 - 4N^2$ and $F = 17NP - 12T - 12N^3$.

A.1 One residual tensor and the local moment map

Let $\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_y$ and $w = z_x \otimes z_y \otimes \tilde{z}_t$, $\tilde{z} = \epsilon^{\otimes 4}\bar{z}$. At one physical site define maps from three qubits to one by

$$L|xyt\rangle = \epsilon_{yt}|x\rangle, \qquad M|xyt\rangle = \epsilon_{xt}|y\rangle.$$

For each site i, let S_i symmetrise its three slots (the orthogonal projector onto the symmetric subspace). On the other three sites contract with L, or with $L + M$, respectively:

$$U_i = S_i \prod_{j\neq i} L_j w, \qquad V_i = S_i \prod_{j\neq i} (L_j + M_j) w.$$

These are ordinary vectors on six qubits (three slots at site i, one at each other site), with the inherited Hilbert-space norm. Write $R_i^{(i)}$ for the one-qubit marginal R_i acting on qubit i of $\mathbb{C}^{16}$. The exact identity is

$$\boxed{8F = 6NW_1 + 3\sum_i \|V_i - 8U_i\|^2 + 12\sum_{i<j} \|(R_i^{(i)} - R_j^{(j)})z\|^2.} \tag{S}$$

It immediately proves the sharp bound $F \geq \frac{3}{4}NW_1 \geq 0$. For a product state $z = \sqrt{N}\, a_1 \otimes \cdots \otimes a_4$ every $R_i^{(i)} z = Nz$, so the pair norms vanish, and $F = 3N^3 = \frac{3}{4}NW_1$; the constant $3/4$ cannot be improved. Proposition S95 of the supplement writes $32F$ as a sum of site terms alone; (S) keeps the separate W_1 term, which gives this bound and the envelope below.

How the terms are chosen. All norms and overlaps of U_i, V_i are expectation values in $z^{\otimes 3}$ of slot permutations (Section A.2), so they are polynomials in the marginal moments. Besides N^3, NP, T, they involve Π_1 (equivalently W_1) and the mixed moment Π_2, which F does not contain. The pair norms contain exactly this mixed moment. Since $(R_i^{(i)})^2$ acts on qubit i only, and a 2×2 matrix of trace N has $\mathrm{tr}\,R^3 = \frac{1}{2}(3N\,\mathrm{tr}\,R^2 - N^3)$,

$$\|(R_i^{(i)} - R_j^{(j)})z\|^2 = \mathrm{tr}\,R_i^3 + \mathrm{tr}\,R_j^3 - 2\,\mathrm{tr}[R_{ij}(R_i \otimes R_j)], \qquad \sum_{i<j} \|(R_i^{(i)} - R_j^{(j)})z\|^2 = \tfrac{9}{2}N\Pi_1 - 6N^3 - 2\Pi_2. \tag{Pr}$$

In $\sum_i \|V_i - tU_i\|^2$ the coefficient of N^3 is $-\frac{4}{3}(t-8)(t+4)$. At the root $t = 8$ the N^3 terms cancel, and the Π_2 that remains is removed by the pair norms, leaving the nonnegative multiple $6NW_1 = 18\|U\|^2$; at the other root $t = -4$ the same elimination leaves a negative multiple of NW_1.

The first tensor is just local polarisation. Contracting the three epsilon pairs gives $U_i = -S_i(z_x \otimes (R_i\epsilon^T)_{yt})$. Diagonalise R_i with eigenvalues λ, μ, and write $z = \sqrt{\lambda}\,|0\rangle|\varphi_0\rangle + \sqrt{\mu}\,|1\rangle|\varphi_1\rangle$ with orthonormal φ_a on the remaining sites. Then $R_i\epsilon^T$ has entries $-\lambda$ at $(y,t) = (0,1)$ and μ at $(1,0)$, the two Schmidt terms of U_i are orthogonal, and symmetrising $|0\rangle(\mu|10\rangle - \lambda|01\rangle)$ and $|1\rangle(\mu|10\rangle - \lambda|01\rangle)$ gives $(\mu - \lambda)$ times $\frac{1}{3}(|001\rangle + |010\rangle + |100\rangle)$ and $\frac{1}{3}(|011\rangle + |101\rangle + |110\rangle)$, each of squared norm $1/3$; the two Schmidt terms carry the weights λ and μ. Hence

$$\|U_i\|^2 = \frac{(\lambda+\mu)(\lambda-\mu)^2}{3} = \frac{N}{3}\|\mathbf{r}_i\|^2. \tag{U}$$

Thus the first term in (S) is $18\|U\|^2$, an ordinary one-qubit marginal defect. For nonzero z, equality $F = 0$ forces every $R_i = NI/2$, hence $U_i = 0$; it then requires $V_i = 0$ for all i. Section A.4 classifies the remaining zeros.

A.2 The three contractions

Write $\|U\|^2 = \sum_i \|U_i\|^2$, $\|V\|^2 = \sum_i \|V_i\|^2$ and $\langle U, V\rangle = \sum_i \langle U_i, V_i\rangle$. We will show

$$\begin{aligned} \|U\|^2 &= (2N\Pi_1 - 4N^3)/3, \\ \mathrm{Re}\langle U, V\rangle &= (-8N^3 - 2N\Pi_1 + 8\Pi_2)/3, \\ \|V\|^2 &= (128N^3 - 226N\Pi_1 + 136NP + 152\Pi_2)/3 - 32T. \end{aligned} \tag{C}$$

These contraction identities imply (S) by direct substitution: $\sum_i \|V_i - 8U_i\|^2 = \|V\|^2 - 16\,\mathrm{Re}\langle U, V\rangle + 64\|U\|^2$, and (C) turns this into

$$\sum_i \|V_i - 8U_i\|^2 = \tfrac{136}{3}NP - 32T - 22N\Pi_1 + 8\Pi_2,$$

in which N^3 has cancelled. Adding twelve times (Pr) and $6NW_1 = 12N\Pi_1 - 24N^3$ to three times this gives $136NP - 96T - 96N^3 = 8F$, as claimed. The rest of this subsection proves (C). Every step uses only the invariance of $z^{\otimes 3}$ under simultaneous permutations of the copies, the vanishing of the one-site antisymmetriser and averaging over sites, so (C) holds as a linear identity in the twirled three-copy moment with one flipped copy.

Slot permutations. In this subsection $p = (12)$, $q = (13)$, $r = (23)$ are the transpositions of the three slots (x, y, t) of one site, acting by permuting the three tensor factors, and c is the cycle $c|a_1a_2a_3\rangle = |a_2a_3a_1\rangle$, with inverse $c^{-1}|a_1a_2a_3\rangle = |a_3a_1a_2\rangle$. (Here the letters p, q, r denote only these transpositions; in particular q is not the Tsallis exponent.) Products are composed as operators, right factor first; then

$$cr = q, \qquad pr = c^{-1}, \qquad qr = c, \qquad pq = c, \qquad qp = c^{-1}.$$

Antisymmetrising three vectors in dimension two gives zero, so

$$I - p - q - r + c + c^{-1} = 0, \qquad S = \tfrac{1}{6}(I + p + q + r + c + c^{-1}) = (p + q + r)/3, \tag{A}$$

where S is the symmetriser of the three slots of one site (a projector).

The kernels. Let Γ_3 denote the partial transpose on slot 3, and for an operator A on the three slots of a site put $\mathfrak{t}(A) = \big[(I\otimes I\otimes\epsilon)^T A (I\otimes I\otimes\epsilon)\big]^{\Gamma_3}$. Since $\tilde z = \epsilon^{\otimes 4}\bar z$, moving ϵ from the vector $\tilde z$ to the operator and transposing slot 3 (which turns $\bar z$ into z) gives, for an operator $A = \bigotimes_{s=1}^4 A_s$ with A_s acting on the three slots of site s,

$$\langle w, Aw\rangle = \langle z^{\otimes 3}, \bigotimes\nolimits_s \mathfrak{t}(A_s)\, z^{\otimes 3}\rangle.$$

The matrix elements (output $x'y't'$, input xyt) of the four needed operators are listed below, with $\epsilon_{01} = 1 = -\epsilon_{10}$; for instance $L^\dagger L = I\otimes|\epsilon\rangle\langle\epsilon| = I - r$, where $|\epsilon\rangle = |01\rangle - |10\rangle$. Conjugation by ϵ on slot 3 acts on the two ϵ's that carry the index t, t' as $\sum_s \epsilon_{as}\epsilon_{sb} = (\epsilon^2)_{ab} = -\delta_{ab}$; the two signs cancel, and each such ϵ-pair becomes a Kronecker δ. Then Γ_3 exchanges t and t'. Hence

$$\begin{aligned}
L^\dagger L = \delta_{xx'}\epsilon_{y't'}\epsilon_{yt} \;&\to\; \delta_{xx'}\delta_{y't}\delta_{yt'}: \quad |xyt\rangle \mapsto |xty\rangle = r,\\
M^\dagger M = \delta_{yy'}\epsilon_{x't'}\epsilon_{xt} \;&\to\; \delta_{yy'}\delta_{x't}\delta_{xt'}: \quad |xyt\rangle \mapsto |tyx\rangle = q,\\
L^\dagger M = \delta_{x'y}\epsilon_{y't'}\epsilon_{xt} \;&\to\; \delta_{x'y}\delta_{y't}\delta_{xt'}: \quad |xyt\rangle \mapsto |ytx\rangle = c,\\
M^\dagger L = \delta_{xy'}\epsilon_{x't'}\epsilon_{yt} \;&\to\; \delta_{xy'}\delta_{x't}\delta_{yt'}: \quad |xyt\rangle \mapsto |txy\rangle = c^{-1}.
\end{aligned}$$

On permutation operators, $\mathfrak{t}(I) = I$ and $\mathfrak{t}(p) = p$ (slot 3 is not involved), while for $q = \frac12(I + \sum_a \sigma_a\otimes I\otimes\sigma_a)$ and likewise r, the identity $\epsilon^T\sigma_a\epsilon = \sigma_y\sigma_a\sigma_y = -\sigma_a^T$ gives $\mathfrak{t}(q) = I - q$, $\mathfrak{t}(r) = I - r$. Hence

$$\begin{aligned}
\mathfrak{t}(S) &= (2I + p - q - r)/3 =: S',\\
\mathfrak{t}((L+M)^\dagger(L+M)) &= r + q + c + c^{-1} = -I + p + 2q + 2r =: K, \qquad \text{(K)}\\
\mathfrak{t}(L^\dagger(L+M)) &= r + c, \qquad \mathfrak{t}(L^\dagger L) = r,
\end{aligned}$$

using (A) for $c + c^{-1} = p + q + r - I$. Since $S_i^\dagger S_i = S_i$,

$$\|V\|^2 = \sum_i \langle z^{\otimes 3}, S_i' \textstyle\prod_{j\neq i} K_j\, z^{\otimes 3}\rangle.$$

The vector $z^{\otimes 3}$ is fixed by every simultaneous slot permutation (the same permutation at all four sites); these can be multiplied on either side of any word inside its expectation, or used to conjugate it.

Words and the table. Expanding $S_i'\prod_{j\neq i}K_j$ gives a combination of *words* $(\pi_1,\dots,\pi_4)$, $\pi_s \in \{I,p,q,r\}$, with π_s acting on the slots of site s. The coefficient of S' on I,p,q,r is $(2,1,-1,-1)/3$ and that of K is $(-1,1,2,2)$; their ratios are $(-\frac23, \frac13, -\frac16, -\frac16)$. So a word with a letters I and $\beta_p, \beta_q, \beta_r$ letters p, q, r has total coefficient (summed over the focused site i)

$$(-1)^a 2^{\beta_q+\beta_r}\big[-2a + \beta_p - (\beta_q + \beta_r)/2\big]/3.$$

Conjugation by a global slot permutation (the same permutation on all four sites) permutes the transposition labels p, q, r in all six ways and does not change expectations, so the expectation of a word depends only on its *type* $(a; b, c, d)$: a is the number of identity letters and $b \geq c \geq d$ are the multiplicities of the three transpositions, sorted (together with the placement on the four sites). Let # be the number of words of a type (these numbers sum to 4^4), C the sum of their coefficients, and J the mean expectation of the type over its placements and labellings; then $\|V\|^2 = \sum CJ$. Multiplying every letter on the left by one global transposition g (written $\times g$) does not change the expectation, and changes $pq \to c$, $pr \to c^{-1}$. The table gives, for each type, a representative word, its reduction, J and C.

$$J_* := \tfrac12(N^3 + NP + \Pi_2) - \tfrac34 N\Pi_1 - \tfrac13 T.$$

Table 6: The eleven contraction types in $\|V\|^2$. # counts the words of each type, J is their mean expectation over site placements and transposition labels, and C is the sum of their coefficients, so $\|V\|^2 = \sum CJ$. A word acts on the three copies at each of the four sites; I is the identity and p, q, r are slot transpositions. Types are grouped by the number of identity letters.

type	#	word	expectation of the word	J	C
$(4;0,0,0)$	1	IIII	N^3	N^3	$-8/3$
$(3;1,0,0)$	12	$pIII$	$N \operatorname{tr} R_1^2$	$N\Pi_1/4$	$124/3$
$(2;2,0,0)$	18	$ppII$	$N \operatorname{tr} R_{12}^2$	$NP/3$	-84
$(2;1,1,0)$	36	$pqII$	$\operatorname{tr}[R_{12}(R_1 \otimes R_2)]$	$\Pi_2/6$	-136
$(1;3,0,0)$	12	$pppI$	$N \operatorname{tr} R_{123}^2 = N \operatorname{tr} R_4^2$	$N\Pi_1/4$	$220/3$
$(1;2,1,0)$	72	$ppqI \overset{\times p}{=} IIcp$	$\operatorname{tr}[R_{34}^2(R_3 \otimes I)]$	$\frac{NP+\Pi_2}{6} - \frac{N\Pi_1}{8}$	296
$(1;1,1,1)$	24	$pqrI$	see (c)	J_*	64
$(0;4,0,0)$	3	$pppp \overset{\times p}{=} IIII$	N^3	N^3	-20
$(0;3,1,0)$	24	$pppq \overset{\times p}{=} IIIc$	$\operatorname{tr} R_4^3$	$\frac{3N\Pi_1}{8} - \frac{N^3}{2}$	$-248/3$
$(0;2,2,0)$	18	$ppqq \overset{\times p}{=} IIcc$	$\operatorname{tr} R_{34}^3$	$T/3$	-48
$(0;2,1,1)$	36	$ppqr \overset{\times p}{=} IIcc^{-1}$	$\operatorname{tr}[(R_{34}^\Gamma)^3]$	J_*	-16

Here $\operatorname{tr} R_S^k$ is the expectation of the k-cycle on the sites of S (and $\operatorname{tr} R_S^2$ that of a transposition), as for the usual replica trick, and R^Γ is a partial transpose of a two-qubit matrix on one qubit (see (e)); the entries come from following the index lines of $z^{\otimes 3}$. The means are obtained as follows. To make the nontrivial partial-transpose entry explicit, consider the representative $ppqr$. Multiplication on the left by the global p gives $IIcc^{-1}$. Trace out sites $1, 2$ and write $R = R_{34}$ with entries $R_{ab,a'b'}$. For partial transpose on the second qubit, $(R^\Gamma)_{ab,a'b'} = R_{ab',a'b}$, and the opposite cycles give

$$\begin{aligned}\langle z^{\otimes 3}, (IIcc^{-1}) z^{\otimes 3}\rangle &= \sum_{a,a';a'',b,b',b''} R_{ab',a'b} R_{a'b'',a''b'} R_{a''b,ab''} \\ &= \sum_{a,a';a'',b,b',b''} (R^\Gamma)_{ab,a'b'} (R^\Gamma)_{a'b',a''b''} (R^\Gamma)_{a''b'',ab} \\ &= \operatorname{tr}[(R^\Gamma)^3].\end{aligned}$$

All six indices run over $\{0, 1\}$. This explains why opposite local cycles produce a partial-transpose moment, whereas equal cycles give $\operatorname{tr} R^3$; part (e) below evaluates its mean in the marginal invariants.

(a) Types $(4;0,0,0)$, $(0;4,0,0)$: N^3 on every word. Types $(3;1,0,0)$, $(1;3,0,0)$, $(0;3,1,0)$: average over the four sites of $N \operatorname{tr} R_l^2$ or $\operatorname{tr} R_l^3$, using $\operatorname{tr} R_{123}^2 = \operatorname{tr} R_4^2$ for a pure state and $\operatorname{tr} R^3 = (3N/2) \operatorname{tr} R^2 - N^3/2$ for a 2×2 matrix of trace N.

(b) Types $(2;2,0,0)$, $(0;2,2,0)$: complementary pairs have equal spectra, so over the six pairs $\sum_{i<j} \operatorname{tr} R_{ij}^2 = 2P$ and $\sum_{i<j} \operatorname{tr} R_{ij}^3 = 2T$, giving the means $NP/3$ and $T/3$. Type $(2;1,1,0)$: the mean of $\operatorname{tr}[R_{ij}(R_i \otimes R_j)]$ over the six pairs is $\Pi_2/6$.

(c) Type $(1;1,1,1)$: replace the identity letter by $p + q + r - c - c^{-1}$ (equation (A)). The three transpositions give words of type $(0;2,1,1)$. In the words with c or c^{-1}, multiplying all letters by c^{-1} or c turns the cycle into I and permutes the three transpositions, so the type is $(1;1,1,1)$ again. Thus $J_{(1;1,1,1)} = 3J_{(0;2,1,1)} - 2J_{(1;1,1,1)}$: the two types have the same mean.

To evaluate the remaining means, we use the Pauli expansion of a two-qubit matrix, derived for $N = 1$ in Appendix C; the homogeneous forms below follow by scaling. For the unnormalised

$R_{ij} = \frac{1}{4}(NI + \mathbf{r}_i\cdot\boldsymbol{\sigma}_i + \mathbf{r}_j\cdot\boldsymbol{\sigma}_j + \sum_{a,b} C_{ij}^{ab}\sigma_a\otimes\sigma_b)$, with $\mathbf{r}_i, C_{ij}$ the (unnormalised) expectations and $u_{ij} = \|\mathbf{r}_i\|^2 + \|\mathbf{r}_j\|^2 + \|C_{ij}\|_F^2$,

$$\operatorname{tr} R_{ij}^2 = \tfrac{1}{4}(N^2 + u_{ij}), \qquad \operatorname{tr}[R_{ij}(R_i\otimes R_j)] = \tfrac{1}{4}(N^3 + N\|\mathbf{r}_i\|^2 + N\|\mathbf{r}_j\|^2 + \mathbf{r}_i^T C_{ij}\mathbf{r}_j),$$
$$\operatorname{tr} R_{ij}^3 = \tfrac{1}{16}(N^3 + 3Nu_{ij} + 6\,\mathbf{r}_i^T C_{ij}\mathbf{r}_j - 6\det C_{ij}), \qquad \operatorname{tr}[R_{ij}^2(\mathbf{r}_i\cdot\boldsymbol{\sigma}_i\otimes I)] = \tfrac{1}{2}(N\|\mathbf{r}_i\|^2 + \mathbf{r}_i^T C_{ij}\mathbf{r}_j).$$

(d) Type $(1;2,1,0)$: writing $R_i\otimes I = \frac{N}{2}I + \frac{1}{2}\mathbf{r}_i\cdot\boldsymbol{\sigma}_i\otimes I$, the last formula gives the two-qubit identity

$$\operatorname{tr}[R_{ij}^2(R_i\otimes I)] = \frac{N}{2}\operatorname{tr} R_{ij}^2 + \operatorname{tr}[R_{ij}(R_i\otimes R_j)] - \frac{N}{2}\operatorname{tr} R_j^2,$$

since both sides exceed $\frac{N}{2}\operatorname{tr} R_{ij}^2$ by $\frac{1}{4}(N\|\mathbf{r}_i\|^2 + \mathbf{r}_i^T C_{ij}\mathbf{r}_j)$. Averaging over the twelve ordered pairs (i,j), whose sums are $4P$, $2\Pi_2$, $3\Pi_1$ for the three terms, gives $\frac{1}{12}(2NP + 2\Pi_2 - \frac{3}{2}N\Pi_1)$, the stated J.

(e) Type $(0;2,1,1)$: partial transposition of the first qubit (or the second; they differ by a full transpose, which does not change the spectrum) reverses the sign of σ_y there, so with $D = \operatorname{diag}(1,-1,1)$ the data become $\mathbf{r}_i \to D\mathbf{r}_i$, $C_{ij} \to DC_{ij}$ and $\mathbf{r}_j$ is unchanged. Hence $\mathbf{r}_i^T C_{ij}\mathbf{r}_j$ and u_{ij} are unchanged and $\det C_{ij} \to -\det C_{ij}$, and the cubic formula above gives $\operatorname{tr}[(R_{ij}^\Gamma)^3] = \operatorname{tr} R_{ij}^3 + \frac{3}{4}\det C_{ij}$. Solving the formulas above for $\det C_{ij}$ and summing over the six pairs, with $\sum_{i<j} u_{ij} = 8P - 6N^2$, $\sum_{i<j}(\|\mathbf{r}_i\|^2 + \|\mathbf{r}_j\|^2) = 3W_1$ and $\sum_{i<j}\operatorname{tr} R_{ij}^3 = 2T$, gives $\sum_{i<j}\det C_{ij} = 4N^3 + 4NP + 4\Pi_2 - 6N\Pi_1 - \frac{16}{3}T$; hence $\sum_{i<j}\operatorname{tr}[(R_{ij}^\Gamma)^3] = 3(N^3 + NP + \Pi_2) - \frac{9}{2}N\Pi_1 - 2T$, whose mean over six pairs is J_*.

Summing CJ over the table proves the last line of (C).

The other two contractions. By (K), $\langle U_i, V_i\rangle$ is the expectation of $S_i'\prod_{j\neq i}(r+c)_j$. Multiply on the right by the global r: the factors at $j\neq i$ become $(r+c)r = I + q$, and at i we get $S_i' r_i = \frac{1}{3}(2r_i - I) + \frac{1}{3}(c_i^{-1} - c_i)$. Conjugation by the global q fixes q and exchanges c and c^{-1}, so the expectation of the last term equals its own negative and vanishes. Hence

$$\operatorname{Re}\langle U, V\rangle = \tfrac{1}{3}\sum_i \Big\langle (2r_i - I)\prod_{j\neq i}(I + q_j)\Big\rangle = \tfrac{1}{3}\sum_i \sum_{m=0}^{3} \sum_{A\subseteq\{j\neq i\},\,|A|=m} \langle (2r_i - I)\, q_A\rangle,$$

where q_A puts q on the sites of A. There are $4\binom{3}{m}$ pairs (i,A) for each m, and the terms $r_i q_A$ have the types $(3;1,0,0)$, $(2;1,1,0)$, $(1;2,1,0)$, $(0;3,1,0)$ for $m = 0,1,2,3$, and the terms $I_i q_A$ the types $(4;0,0,0)$, $(3;1,0,0)$, $(2;2,0,0)$, $(1;3,0,0)$, each placement occurring equally often. With $J_{(a;b,c,d)}$ the means of the table,

$$\operatorname{Re}\langle U, V\rangle = \tfrac{4}{3}\Big[2\big(J_{(3;1,0,0)} + 3J_{(2;1,1,0)} + 3J_{(1;2,1,0)} + J_{(0;3,1,0)}\big) - \big(J_{(4;0,0,0)} + 3J_{(3;1,0,0)} + 3J_{(2;2,0,0)} + J_{(1;3,0,0)}\big)\Big].$$

The first bracket is $\frac{1}{2}NP + \Pi_2 + \frac{1}{4}N\Pi_1 - \frac{1}{2}N^3$ and the second is $NP + N\Pi_1 + N^3$, which gives $(-8N^3 - 2N\Pi_1 + 8\Pi_2)/3$. Only the first six rows and the row $(0;3,1,0)$ of the table are used. The same argument for $\|U\|^2$, with $L^\dagger L \mapsto r$ on the other sites, leaves $\frac{1}{3}\sum_i\langle 2r_i - I\rangle = \frac{1}{3}\sum_i(2N\operatorname{tr} R_i^2 - N^3) = (2N\Pi_1 - 4N^3)/3$, agreeing with (U). This completes the derivation of (C).

A.3 The critical envelope

On the critical locus $W_1 = 0$, and the moment bound can be sharpened by the spin-flip invariant. For a critical normal vector v let $\phi = \sum_k v_k m_k \in \mathbb{C}^4$ be the auxiliary two-qubit vector (Appendix B) and $\Theta_{\alpha\beta} = \langle\phi|\sigma_\alpha\otimes\sigma_\beta|\phi\rangle$ its Bloch correlation matrix.

Proposition 13 (critical envelope)**.** *Every critical unit four-qubit state satisfies*

$$\boxed{F = \frac{1}{2} + \frac{7}{4}\tau + \frac{9}{4}\operatorname{perm}\Theta \;\geq\; \tau - \tau^{3/2}.} \tag{CE}$$

For $0 < \tau \leq 1$, with $\kappa = \sqrt{\tau}$, equality holds exactly when $\Theta = O(\kappa I + (1-\kappa)ss^T)$ with $s \in \{\pm 1\}^3/\sqrt{3}$ and O one of the twelve signed cyclic permutation matrices whose three signs have product -1; such states exist for every τ. Since $P = 1 + \tau/2$ on this locus, the cubic moment obeys $T \leq \frac{5}{12} + \frac{17}{12}(P-1) - \frac{1}{12}(x - x^{3/2})$, $x = 2(P-1)$, with equality attained for every $P \in [1, \frac{3}{2}]$: this is the exact upper boundary of the critical moment region, and it meets the supporting line of Theorem 3 only at its two ends.

Proof. F and τ are LU invariant, so take $\psi = G(v)$ with $\|v\| = 1$ (Theorem 16). *The formula.* The twelve magic Bell rays have signed-permutation correlation matrices (Lemmas S36(1) and S33 of the supplement), so the cut $k \in \mathbb{Z}/3$ has the four eigenvalues $(1 + \sigma \cdot g^{(k)})/4$, where $g^{(k)}_\beta = \Theta_{\beta+k,\beta}$ is a cyclic diagonal of Θ and σ runs over the four sign triples with $\sigma_1\sigma_2\sigma_3 = -1$. Over these triples $\sum(\sigma \cdot g) = 0$, $\sum(\sigma \cdot g)^2 = 4|g|^2$ and $\sum(\sigma \cdot g)^3 = -24 g_1 g_2 g_3$. The three cyclic diagonals partition the entries of Θ, and the sum of their products is the even half $e = (\operatorname{perm}\Theta + \det\Theta)/2$ of the permanent. Hence $P = (3 + \|\Theta\|^2)/4$, $T = (3 + 3\|\Theta\|^2 - 6e)/16$ and $F = -\frac{3}{2} + 2\|\Theta\|^2 + \frac{9}{2}e$. In Schmidt form $\phi = (U \otimes V)(\cos\varphi|00\rangle + \sin\varphi|11\rangle)$, so $\Theta = R_a \operatorname{diag}(\kappa, -\kappa, 1) R_b^T$ with $R_a, R_b \in SO(3)$ the rotations induced by U, V and $\kappa = \sin 2\varphi$ the concurrence of ϕ, which in the magic basis is $|\sum_k v_k^2| = \sqrt{\tau}$ by (τ_G). Thus Θ has singular values $(1, \kappa, \kappa)$ and $\det\Theta = -\kappa^2$. So $\|\Theta\|^2 = 1 + 2\tau$, which gives the formula in (CE) and $P = 1 + \tau/2$ (checks CE-1, CE-2).

The permanent bound. Write the polar decomposition $\Theta = OD$, with $O \in O(3)$, $\det O = -1$, $D = (\Theta^T\Theta)^{1/2} = \kappa I + (1-\kappa)ss^T$ and $ODO^T = \kappa I + (1-\kappa)rr^T$; here r, s are unit top singular vectors of Θ, the local Bloch directions of ϕ. For every 3×3 matrix X, $\operatorname{perm} X = \langle \omega, X^{\otimes 3}\omega\rangle$ with the unit vector $\omega = 6^{-1/2}\sum_\pi e_{\pi 1} \otimes e_{\pi 2} \otimes e_{\pi 3}$. Cauchy–Schwarz gives

$$|\operatorname{perm}\Theta| = |\langle (D^{1/2}O^T)^{\otimes 3}\omega, (D^{1/2})^{\otimes 3}\omega\rangle| \leq \big(\operatorname{perm}(ODO^T)\operatorname{perm} D\big)^{1/2}.$$

For a unit vector u and $y_a = u_a^2$, expanding the permanent by rows,

$$\operatorname{perm}(\kappa I + (1-\kappa)uu^T) = \kappa^3 + \kappa^2(1-\kappa) + 2\kappa(1-\kappa)^2 e_2(y) + 6(1-\kappa)^3 e_3(y) \leq \frac{2 + 3\kappa^2 + 4\kappa^3}{9},$$

by Maclaurin's inequalities $e_2(y) \leq \frac{1}{3}$, $e_3(y) \leq \frac{1}{27}$ on the simplex (check CE-3); for $\kappa < 1$ equality needs $y = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Thus $-\operatorname{perm}\Theta \leq (2 + 3\tau + 4\tau^{3/2})/9$, and $F \geq \frac{1}{2} + \frac{7}{4}\tau - \frac{1}{4}(2 + 3\tau + 4\tau^{3/2}) = \tau - \tau^{3/2}$.

Equality. Let $\kappa > 0$, so D is invertible. Equality forces $r, s \in \{\pm 1\}^3/\sqrt{3}$ when $\kappa < 1$, and equality with a negative sign in Cauchy–Schwarz, $(O^T)^{\otimes 3}\omega = -\omega$. As ω is the symmetric tensor of $x_1x_2x_3$, this says $x_1x_2x_3$ changes sign under O^T; then O^T preserves the union of the coordinate planes, so O is a signed permutation, with sign product -1; then $\det O = -\operatorname{sgn}\pi = -1$ makes the permutation π even, that is, cyclic (check CE-4). Conversely, for such O and s, $\operatorname{perm}\Theta = \langle (O^T)^{\otimes 3}\omega, D^{\otimes 3}\omega\rangle = -\operatorname{perm} D$ attains the bound. Every matrix with singular values $(1, \kappa, \kappa)$ and determinant $-\kappa^2$ has the form $R_a \operatorname{diag}(\kappa, -\kappa, 1) R_b^T$ with $R_a, R_b \in SO(3)$, so it is the correlation matrix of a suitable ϕ in Schmidt form; and every unit ϕ is $\sum_k v_k m_k$ for a critical $G(v)$. The moment statement is (CE) rewritten with $\tau = 2(P-1)$ and $F = 17P - 12T - 12$. □

On critical states $W_1 = 0$, so for unit states (CE) is the critical case of $F \geq \frac{3}{4}W_1 + \tau - \tau^{3/2}$, which Theorem 15 proves for all states.

A.4 All zeros and why the moment bound is tight

The critical envelope classifies the zeros of F. This classification supplements the moment inequality; it is not used in the proof of Theorem 1.

Proposition 14 (complete zero set of the moment inequality)**.** *For unit four-qubit states, $F = 0$ holds exactly on the two M_4 LU orbits and the three LU orbits of products of Bell pairs, one for each pairing of the labelled qubits.*

Proof. A unit zero has $W_1 = 0$ by Theorem 3, so it is critical, and since $\tau - \tau^{3/2} > 0$ for $0 < \tau < 1$, Proposition 13 leaves $\tau \in \{0, 1\}$. At $\tau = 1$, $F = 0$ is equality in (CE), so Θ is one of the twelve signed cyclic permutation matrices O; one cyclic diagonal then has entries ± 1 with product -1, and that cut has spectrum $(1, 0, 0, 0)$. The pure cut splits the state into two two-qubit pure states with maximally mixed one-qubit marginals, that is, two Bell pairs. At $\tau = 0$, $\Theta = rs^T$ with unit r, s, and the formula in (CE) reads $F = \frac{1}{2} + \frac{27}{2}\prod_a r_a s_a$. Its zero $\prod_a r_a s_a = -\frac{1}{27}$ forces $|r_a| = |s_a| = 3^{-1/2}$ by the AM–GM inequality, so every cyclic diagonal $g^{(k)}$ has entries $\pm\frac{1}{3}$ with product $-\frac{1}{27}$ and every cut has spectrum $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$; Proposition 4 then gives the M_4 orbits. Conversely M_4, $\overline{M_4}$ and the Bell-pair products have $F = 0$. Section S5 of the supplement gives an independent proof by a weighted-variance identity on the critical normal form. □

There is also an elementary way to distinguish the two classes. Let D_{1j} be the determinant of the $1j|\widehat{1j}$ flattening, with qubit indices in increasing order. Every invertible local transformation multiplies all three determinants by the same factor $\prod_i (\det A_i)^2$; their ratios are therefore invariant, even under invertible local filters. Directly,

$$\frac{D_{13}(M_4)}{D_{12}(M_4)} = \frac{1 - i\sqrt{3}}{2},$$

whereas conjugation conjugates this nonreal ratio. Thus the two classes are distinct even under such filters. LU-inequivalence alone was first proved in [12, Prop. 7].

By Proposition 14, the supporting line $17P - 12T - 12 = 0$ meets the pure-state moment region at exactly $(1, 5/12)$ and $(3/2, 9/8)$; its open line segment is unattainable.

A.5 The general envelope

Theorem 15 (general envelope and complete equality; exact certificate, FI, FG)**.** *Every nonzero four-qubit vector satisfies, with $s = z^T \sigma_y^{\otimes 4} z$,*

$$F - \tfrac{3}{4} N W_1 - N|s|^2 + |s|^3 \;\geq\; \frac{7|s|^2}{8(N + |s|)}\, W_1. \tag{GE}$$

For unit states, with $\tau = |s|^2$,

$$\boxed{F \;\geq\; \Big(\frac{3}{4} + \frac{7\tau}{8(1 + \sqrt{\tau})}\Big) W_1 + \tau - \tau^{3/2}.}$$

Equality in $F \geq \frac{3}{4} W_1 + \tau - \tau^{3/2}$ holds exactly on fully product states and on the critical equality family of Proposition 13, including its HS endpoint at $\tau = 0$ and Bell-pair-product endpoint at $\tau = 1$.

The coefficient $\frac{7}{8}$ of the added W_1 term is optimal. On $z = \alpha|0000\rangle + \beta|1111\rangle$ with real $\alpha, \beta \geq 0$, $\alpha^2 + \beta^2 = 1$, one has $s = 2\alpha\beta$, $\tau = 4\alpha^2\beta^2$, $W_1 = 4(1 - \tau)$ and $F = 3 + \frac{3}{2}\tau$, so the boxed bound with $\frac{7}{8}$ replaced by C leaves the slack $(\frac{7}{2} - 4C)\tau + (1 + 4C)\tau^{3/2}$, negative for small $\tau > 0$ when $C > \frac{7}{8}$ (exact check ESH). The proof (Section S3 of the supplement) follows $H = F - \frac{3}{4} N W_1$ along a path of unit states from a state with $s = 0$ to z, along which the spin-flip overlap grows from 0 to s; an exact identity for the derivative of H along this path (exact certificate; checks FI, FG), integrated, gives (GE), and the zeros of the remainder give the equality cases.

B Critical four-qubit states: an elementary normal form

The equality proof needs an LU normal form for *critical* states, whose one-qubit marginals are all maximally mixed. We derive it by elementary linear algebra, including all ranks and singular-value multiplicities. No orbit-closure or Kempf–Ness argument is needed. Exact checks are listed in Appendix I.

B.1 Setting

Let $m_1, \dots, m_4$ be the magic basis of two qubits,

$$m_1 = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad m_2 = \tfrac{i}{\sqrt{2}}(|01\rangle + |10\rangle), \quad m_3 = \tfrac{1}{\sqrt{2}}(|01\rangle - |10\rangle), \quad m_4 = \tfrac{i}{\sqrt{2}}(|00\rangle - |11\rangle).$$

A four-qubit state is $\psi = \sum_{j,k} Z_{jk}\, m_j \otimes m_k$ with $Z \in \mathbb{C}^{4\times 4}$ (qubits 1,2 index the rows, qubits 3,4 the columns), $\|Z\|_F = 1$. In this basis $u \otimes u' \in SU(2) \otimes SU(2)$ acts as a real rotation, $SU(2) \otimes SU(2) \to SO(4)$ onto, so the local unitaries act as $Z \mapsto O_1 Z O_2^T$, $O_1, O_2 \in SO(4)$. For $v \in \mathbb{C}^4$ put

$$G(v) = \sum_{k=1}^{4} v_k\, m_k \otimes m_k.$$

Expanding the m_k shows $G(v) = G_{abcd}$, the family $\frac{a+d}{2}(|0000\rangle + |1111\rangle) + \frac{a-d}{2}(|0011\rangle + |1100\rangle) + \frac{b+c}{2}(|0101\rangle + |1010\rangle) + \frac{b-c}{2}(|0110\rangle + |1001\rangle)$ with $(a, b, c, d) = (v_1, -v_2, v_3, -v_4)$ (CN-2).

Spin flip on the normal form. The invariant $\tau = |\psi^T \sigma_y^{\otimes 4} \psi|^2$ of Lemma 2 is LU invariant, because $u^T \sigma_y u = (\det u)\sigma_y$ for $u \in U(2)$: a product unitary multiplies $\psi^T \sigma_y^{\otimes 4} \psi$ by $\prod_i \det u_i$. Directly from the definitions, $\sigma_y^{\otimes 2} m_k = s_k m_k$ with $s = (-, +, -, +)$, and $m_k^T m_l = \eta_k \delta_{kl}$ with $\eta = (+, -, +, -)$ (the m_k are orthonormal and real up to the phases $1, i, 1, i$). Hence $m_k^T \sigma_y^{\otimes 2} m_l = s_l \eta_k \delta_{kl} = -\delta_{kl}$, and for $\psi = G(v)$

$$\psi^T \sigma_y^{\otimes 4} \psi = \sum_{k,l} v_k v_l \left(m_k^T \sigma_y^{\otimes 2} m_l\right)^2 = \sum_k v_k^2, \qquad \text{so} \qquad \tau = \Big|\sum_k v_k^2\Big|^2. \tag{τ_G}$$

Theorem 16 (Critical normal form)**.** *For a unit four-qubit state ψ with matrix Z the following are equivalent.*

1. *All four one-qubit marginals of ψ equal $I/2$ (ψ is* critical*).*
2. *$ZZ^\dagger$ and $Z^\dagger Z$ are real matrices. Equivalently, for $Z = X + iY$ with X, Y real, XY^T and $X^T Y$ are symmetric.*
3. *$Z = O_1 \operatorname{diag}(v)\, O_2^T$ with $O_1, O_2 \in SO(4)$ and $v \in \mathbb{C}^4$.*
4. *ψ is local-unitarily equivalent to $G(v)$ for some v with $\|v\| = 1$.*

Lemma 17 (Simultaneous real singular values)**.** *For any complex $m \times n$ matrix Z, the two Gram matrices $ZZ^\dagger$ and $Z^\dagger Z$ are real if and only if*

$$Z = O_1 D O_2^T, \qquad O_1 \in O(m), \quad O_2 \in O(n),$$

where D is a complex rectangular diagonal matrix. This includes all ranks and multiplicities.

Proof. Write $Z = X + iY$. The Gram conditions are precisely $XY^T = YX^T$ and $X^T Y = Y^T X$. Take a real SVD of X and apply its two orthogonal factors to Y too. On the nonzero diagonal of X, write $d_i > 0$. The two symmetry equations give

$$d_i Y_{ji} = d_j Y_{ij}, \qquad d_i Y_{ij} = d_j Y_{ji}, \qquad (d_i^2 - d_j^2) Y_{ij} = 0.$$

Thus different positive singular-value blocks do not mix, each equal-value block of Y is symmetric, and its entries joining a positive block to either nullspace vanish. Diagonalise each positive block of Y by the same real orthogonal transformation on its left and right; this leaves $X = dI$ fixed. On the remaining, possibly rectangular nullspace block, take a real SVD of Y. Both X and Y are now diagonal. The converse is immediate from the two Gram products. □

B.2 Proof of the critical normal form

Proof of Theorem 16. Marginals are moment maps.

In the magic basis the six operators $i\sigma_a \otimes 1$, $i\,1 \otimes \sigma_a$ $(a = x, y, z)$ have real antisymmetric matrices, and they are linearly independent, hence span $\mathfrak{so}(4)$ (CN-1: check of the explicit 4×4 matrices). The reduced state of qubits 1,2 is $\rho_{12} = ZZ^\dagger$ in the magic basis. For a real antisymmetric L and Hermitian $H = ZZ^\dagger$, $\operatorname{tr}(L\,\mathrm{Re}H) = 0$, so $\langle \sigma \otimes 1\rangle = \operatorname{tr}(H(-iL)) = \operatorname{tr}(L\,\mathrm{Im}H)$. The six expectations $\langle \sigma_a \otimes 1\rangle, \langle 1 \otimes \sigma_a\rangle$ are therefore the six independent components of the antisymmetric matrix $\mathrm{Im}(ZZ^\dagger)$ under an invertible linear change of coordinates, and they vanish iff $\mathrm{Im}(ZZ^\dagger) = 0$. Thus the marginals of qubits 1 and 2 are both $I/2$ iff $ZZ^\dagger$ is real. In the same way $\rho_{34} = (Z^\dagger Z)^T$, so qubits 3 and 4 are maximally mixed iff $Z^\dagger Z$ is real. This proves (1)⇔(2). (Writing $ZZ^\dagger = XX^T + YY^T + i(YX^T - XY^T)$ and $Z^\dagger Z = X^TX + Y^TY + i(X^TY - Y^TX)$ gives the symmetry form of (2).)

(3)⇒(2). $ZZ^\dagger = O_1 \operatorname{diag}(|v_k|^2) O_1^T$ and $Z^\dagger Z = O_2 \operatorname{diag}(|v_k|^2) O_2^T$ are real.

(2)⇒(3). Apply the lemma with $m = n = 4$. If an orthogonal factor has determinant -1, change its first column's sign and the first diagonal entry's sign; thus both factors can be taken in $SO(4)$.

(3)⇔(4). $(U_{12} \otimes U_{34})\, G(v)$ has matrix $O_1 \operatorname{diag}(v) O_2^T$ when U_{12}, U_{34} act as O_1, O_2 in the magic basis. □

B.3 Consequences

1. On the unitary slice $Z = W/2$, the lemma is the real-orthogonal KAK factorisation. Here $|v_k| = 1/2$ and the v_k^2 are the eigenvalues of ZZ^T.

2. The sextic zero classification and Proposition 4 use this theorem.

C Correlation geometry of the sextic

We use the notation F, P, T, τ of Sections 2.1 and 2.2 for a unit pure four-qubit state.

C.1 Correlation-volume identity

For a unit state, write $r_i^a = \langle \sigma_a^{(i)} \rangle$ and $C_{ij}^{ab} = \langle \sigma_a^{(i)} \sigma_b^{(j)} \rangle$, with C_{ij} a real 3×3 matrix. Expanding a two-qubit marginal in Pauli matrices, and putting $u_j = \|r_1\|^2 + \|r_j\|^2 + \|C_{1j}\|_F^2$, gives

$$\operatorname{tr} \rho_{1j}^2 = \frac{1 + u_j}{4}, \qquad \operatorname{tr} \rho_{1j}^3 = \frac{1 + 3u_j + 6 r_1^T C_{1j} r_j - 6 \det C_{1j}}{16}.$$

To see this, write $\rho_{1j} = \frac{1}{4}(I + X)$ with X the traceless part, so that $\operatorname{tr}\rho^3 = \frac{1}{64}(4 + 3\operatorname{tr}X^2 + \operatorname{tr}X^3)$ and $\operatorname{tr}X^2 = 4u_j$. In $\operatorname{tr}X^3$, the terms with one Pauli matrix on each of the two qubits in every factor give $\operatorname{tr}(\sigma_a\sigma_c\sigma_e)\operatorname{tr}(\sigma_b\sigma_d\sigma_f) C^{ab}C^{cd}C^{ef} = (2i)^2 \epsilon_{ace}\epsilon_{bdf} C^{ab}C^{cd}C^{ef} = -24 \det C_{1j}$, using $\operatorname{tr}(\sigma_a\sigma_c\sigma_e) = 2i\epsilon_{ace}$; the terms with one Bloch vector on each qubit and one factor C give $6 \cdot 4\, r_1^T C_{1j} r_j$ (six orderings); the terms with two Bloch vectors on one qubit, or three, and those with one Bloch vector and two factors C, are traceless (the last because CC^T is symmetric while ϵ_{ace} is antisymmetric). Hence $\operatorname{tr}X^3 = 24(r_1^T C_{1j} r_j - \det C_{1j})$, which gives the stated formulas. Lemma 2 implies $\sum_j u_j =$

$4P - 3 = 1 + 2(W_1 + \tau)$, where $W_1 = \sum_i \|r_i\|^2$. Substitution yields the compact identity

$$\boxed{2F = 1 + 8(W_1 + \tau) - 9\sum_{j=2}^{4}(r_1^T C_{1j} r_j - \det C_{1j}).} \tag{CV}$$

The sum singles out qubit 1 only to label the three balanced cuts $\{1,j\}|\{k,l\}$: by the formulas above each summand equals $(16\,\mathrm{tr}\,\rho_{1j}^3 - 12\,\mathrm{tr}\,\rho_{1j}^2 + 2)/6$, a function of the cut spectrum alone, so the right-hand side is permutation invariant. Therefore Theorem 3 is equivalent to an *oriented correlation-volume monogamy inequality* for three compatible two-qubit marginals. At M_4 the Bloch vectors and τ vanish and each $\det C_{1j} = -1/27$; at Bell $\otimes$ Bell, $\tau = 1$ and the three determinants are $(-1,0,0)$ up to ordering. This explains the two tight configurations: the spin-flip invariant compensates for concentration of the correlation volume in a single Bell pair.

Equation (CV) gives a reduced-state reformulation. Its nonnegativity still requires the moment inequality. Pairwise density-matrix positivity and known absolute steering-ellipsoid volume-monogamy bounds do not visibly supply the required *oriented* and τ-sensitive inequality (compare the volume monogamy relation for pure four-qubit states of Cheng, Milne, Hall and Wiseman [14], Eq. (28): in terms of normalised steering-ellipsoid volumes $v_{j|1} = |\det(C_{1j} - r_1 r_j^T)|/(1 - \|r_1\|^2)^2$ of the ellipsoids that qubit 1 steers on qubit j, it reads $\sum_j v_{j|1}^{2/3} \le 1$, which for $r_1 = 0$ is $\sum_j |\det C_{1j}|^{2/3} \le 1$ and is already strict at M_4, giving $1/3 < 1$). RG-1 and RG-2 (Appendix I) check the local Pauli formulas and (CV) exactly.

Replica witness. Let $p = (12)$ and $c = (123)$ permute the *three copies* of a qubit, and use a subscript i for the physical qubit on which they act. The Hermitian replica witness

$$K_{\mathrm{rep}} = 17\sum_{j=2}^{4} p_1 p_j - 6\sum_{j=2}^{4}(c_1 c_j + c_1^{-1} c_j^{-1}) - 12I$$

satisfies $F(z) = \langle z^{\otimes 3}, K_{\mathrm{rep}} z^{\otimes 3}\rangle$ for unit z (by the usual swap/cycle formulas for reduced moments). It is not positive on the copy-symmetric space. On the sector where all four local triples carry spin 1/2, let $M \simeq \mathbb{C}^2$ be the standard S_3 multiplicity space and take the three diagonal-S_3-fixed pairings $\delta_{12}\delta_{34}$, $\delta_{13}\delta_{24}$, $\delta_{14}\delta_{23}$ in $M^{\otimes 4}$. Their Gram matrix has diagonal entries 4 and off-diagonal entries 2; the matrix of K_{rep} on these pairings has diagonal entries -52 and off-diagonal entries 42. The generalised eigenvalues are $4, -47, -47$ (RG-3, exact). Thus K_{rep} is indefinite even on $\mathrm{Sym}^3((\mathbb{C}^2)^{\otimes 4})$. By uniqueness of polarisation of a bidegree-$(3,3)$ polynomial, F cannot be a sum of squared moduli of *holomorphic* cubic forms in z alone. This does not contradict $F(z) \ge 0$ on pure-state tensor powers.

D Quantitative stability of the maximising LU orbits

The entropy deficit $\delta(\psi) = E(M_4) - E(\psi)$ controls the squared Hilbert-space distance to the two HS orbits. The gap identity (EG) controls W_1 and the spin-flip overlap τ, but both also vanish at cluster states; the sextic moment defect is needed to distinguish the maximisers.

The defect Q below bounds HS infidelity with sharp coefficient one (Theorem 18). This gives a uniform, nonoptimal entropy-distance bound, while the optimal coefficient on critical states is known (Theorem 19; Appendix D.1). We also transfer the bounds to other spectral functionals (Appendix D.2). Complete proofs and further stability estimates are in Sections S7 and S8 of the supplement [40]. Write

$$\mathcal{M} = \{e^{i\theta}(U_1 \otimes \cdots \otimes U_4)\xi : U_i \in SU(2),\ \xi \in \{M_4, \overline{M_4}\}\}, \qquad d(\psi) = \min_{\chi\in\mathcal{M}} \|\psi - \chi\|.$$

Thus global phase is included, qubits are labelled, and $d^2 \le 2(1 - 1/\sqrt{6})$ for unit states (Lemma S27 of the supplement). Equivalently, if $\Phi(\psi) = \max_{\chi\in\mathcal{M}} |\langle\psi,\chi\rangle|^2$, then $d^2 = 2(1 - \sqrt{\Phi})$. Use N, P, T, F

from Theorem 3, and put

$$Q(z) = F(z) + N(P(z) - N^2) = 18NP - 12T - 13N^3, \qquad k = \frac{\ln 3 - 1}{2 \ln 2} > \frac{16}{225}.$$

This nonnegative homogeneous sextic vanishes on the unit sphere precisely at $\mathcal{M}$. On unit states also set $\mathcal{D} = F + 3(W_1 + \tau) = F + 6(P - 1) = Q + 5(W_1 + \tau)/2$. The entropy-gap identity (EG) gives

$$\delta(\psi) := 1 + \tfrac{1}{2} \log_2 3 - E(\psi) \;\geq\; k\mathcal{D}(\psi) \geq kQ(\psi). \tag{3}$$

Indeed the left side of (EG) is $3 \ln 2\, \delta(\psi)$, and its coefficients $\alpha = 15/2 - 6 \ln 3$ of $P - 1$ and $\beta = 3(\ln 3 - 1)/2$ of F satisfy $\alpha > 6\beta$, since $\ln 3 = 1.0986\ldots < 11/10$. As $k = \beta/(3 \ln 2)$, dropping $\Delta_H \geq 0$ gives $\delta \geq (\alpha(P - 1) + \beta F)/(3 \ln 2) \geq k(6(P - 1) + F) = k\mathcal{D}$. Rational series bounds for $\ln 2, \ln 3$ give $k > 16/225$ (GO-4).

Theorem 18 (Sharp HS overlap stability)**.** *For every unit pure four-qubit state,*

$$Q(\psi) \geq 1 - \Phi(\psi) \geq \frac{6 + \sqrt{6(1 + \sqrt{\tau(\psi)})}}{12}\, d(\psi)^2 \geq \frac{6 + \sqrt{6}}{12}\, d(\psi)^2, \qquad \frac{6 + \sqrt{6}}{12} > \frac{7}{10}. \tag{4}$$

Equality in $Q \geq 1 - \Phi$ holds on $\mathcal{M}$ ($Q = 0$, $\Phi = 1$) and on the three Bell-pair-product LU orbits ($Q = \Phi = 1/2$), so the coefficient one is optimal.

[Exact rational certificate, GO-1–GO-3; proof in Section S7.2 of the supplement.]

Theorem 19 (Explicit global LU-orbit stability)**.** *For every unit pure four-qubit state,*

$$d(\psi)^2 \leq \frac{12}{6 + \sqrt{6}}\, Q(\psi), \qquad \boxed{\delta(\psi) \geq k\, \frac{6 + \sqrt{6}}{12}\, d(\psi)^2}, \qquad k\, \frac{6 + \sqrt{6}}{12} > \frac{1}{20}.$$

For every critical state the optimal bound is $\delta \geq C_* d^2$, with $C_* \approx 0.19058$ (Theorem 21). On an explicit neighbourhood of the two orbits the coefficient improves:

$$d(\psi) \leq 1/20 \quad \Longrightarrow \quad \delta(\psi) \geq \frac{3}{10} d(\psi)^2.$$

Its sharp limiting coefficient is $c_* = (2 - \ln 3)/(2 \ln 2)$, in the precise sense of (S7.12). The power two is optimal. No optimal global coefficient is claimed; the equality states of Theorem 21 show that it is at most C_*.

[Uses Theorem 18; proof in Section S7 of the supplement.]

The overlap bound is an exact positivity certificate for $Q - (1 - \Phi)$ on a 19-variable closest-orbit normal slice (Section S7.2 of the supplement): a degree-six polynomial identity writing it as 277 rational Gram forms weighted by the slice's positive constraints, each Gram matrix certified positive by an integer diagonal-dominance test, plus two explicit nonnegative remainders; the entropy bound then follows from (3). The local radius comes from the transverse quadratic forms of Q and of the entropy at the orbit. Brierley and Higuchi [7] previously proved local maximality of entropy. The proofs are in Sections S7 and S8 of the supplement.

D.1 Sharp constants on the critical locus

A state is critical when all four one-qubit marginals are $I/2$. This locus contains M_4, the cluster state and the Bell-pair products, and on it the constants can be made sharp. Both theorems below are proved in the supplement [40] (Sections S8.2, S8.4 and S8.5 there).

Domain	Coefficient in $\delta \geq Cd^2$	Status and proof
All unit pure states	$C = 1/20$	Certified, nonoptimal; Theorem 18 and (3).
$d \leq 1/20$	$C = 3/10$	Explicit local bound; (S7.13) of the supplement.
$d \to 0$	$c_* = (2 - \ln 3)/(2 \ln 2)$	Sharp limiting coefficient; (S7.12) of the supplement.
Critical states	$C = C_*$	Optimal on the whole critical locus; Theorem 21, analytic bounds and interval certificates.
All unit pure states	$C = C_*$	Open; the critical equality states give an upper bound on any universal coefficient.

Table 7: Distance stability: domains, constants and proof methods. The elementary polarisation and spin-flip defect bounds follow directly from (EG); they do not yet control d.

Theorem 20 (Sharp critical-state orbit stability)**.** *For every critical unit pure four-qubit state, with Φ, d, Q, k as in Appendix D,*

$$Q(\psi) \geq \max\{1 - \Phi(\psi),\ 3/2 - 2\Phi(\psi)\}, \tag{5}$$

$$Q(\psi) \geq \frac{2+\sqrt{2}}{4} d(\psi)^2, \tag{6}$$

$$\delta(\psi) \geq k\big(\max\{1 - \Phi, 3/2 - 2\Phi\} + 5\tau/2\big) \geq k\,\frac{2+\sqrt{2}}{4}\, d(\psi)^2. \tag{7}$$

The coefficient one in $Q \geq 1 - \Phi$, and the coefficient $(2+\sqrt{2})/4$ in (6), are both optimal: Bell-pair products attain them. The comparison $Q \geq 1 - \Phi$ holds for all states (Theorem 18). If the critical state also has $\tau = 0$, equivalently if $P = 1$, then

$$\Phi \geq \frac{2+\sqrt{3}}{6}, \qquad Q \geq \frac{3}{4-\sqrt{3}}(1 - \Phi) \geq \frac{3+\sqrt{3}}{4} d^2. \tag{8}$$

These coefficients and the fidelity floor are optimal, attained by cluster states.

[Checks CO-1–CO-3. Uses Theorem 18; proof in Section S8.2 of the supplement.]
The moment route thus gives the entropy coefficient $k(2+\sqrt{2})/4$ on the critical locus. On the slice $P = 1$ a critical normal form is described by two auxiliary Bloch vectors $r, s \in S^2$ (equation (S8.4) of the supplement), in which Section S8.3 of the supplement proves a one-parameter sharpening of (8). The optimal entropy coefficient on the critical locus is larger. Put $p = \cos^2(\pi/8) = (2+\sqrt{2})/4$, $H_{\mathrm{b}}(x) = -x \log_2 x - (1-x)\log_2(1-x)$ and

$$C_* = \frac{\frac{1}{2}\log_2 3 - \frac{1}{3}(1 + 2H_{\mathrm{b}}(p))}{2 - \sqrt{(1 + 1/\sqrt{3})(1 + \sqrt{2/3})}} = 0.190578947\ldots \tag{9}$$

Theorem 21 (Sharp entropy stability on the critical locus)**.** *Every critical unit pure four-qubit state satisfies*

$$\delta(\psi) \ \geq\ C_*\, d(\psi)^2,$$

and equality with $d > 0$ holds exactly for states whose critical normal form has one of the following 144 Bloch-coordinate pairs, all with $P = 1$. In the auxiliary Bloch coordinates $(r, s) \in S^2 \times S^2$ of the null-critical normal form they are: one of r, s is a signed coordinate axis and the other is $(\pm e_b \pm e_c)/\sqrt{2}$, $b \neq c$ ($6 \cdot 12$ pairs, twice for the swap). These pairs attain the bound, so C_* is the optimal coefficient on the critical locus. Consequently every universal coefficient in $\delta \geq Cd^2$ is at most C_*, and the question left open is whether C_* is optimal over all states.

[Interval-certified; proof in Sections S8.4 and S8.5 of the supplement.]

The equality states explain the formula (9). For $r = e_1$, $s = (e_2 + e_3)/\sqrt{2}$ one cut is maximally mixed and the other two have spectrum $(p, p, 1-p, 1-p)/2$ (Lemma S33 of the supplement), so the numerator is the entropy deficit $\delta = 1 + \frac{1}{2}\log_2 3 - \frac{1}{3}(2 + 2(1 + H_b(p)))$; the closest cube pair gives $2\sqrt{\Phi} = \sqrt{(1 + 1/\sqrt{3})(1 + \sqrt{2/3})}$, so the denominator is $d^2 = 2 - 2\sqrt{\Phi}$. The proof represents a critical state by a 3×3 correlation matrix in a six-parameter chart, settles $\tau \geq 0.36$ analytically, and covers the rest by an interval-certified bisection together with a local cone lemma at the equality points.

D.2 Spectral functionals and limitations

Corollary 22 (Spectral-majorant stability)**.** *In Proposition 6, suppose $a > 0$ and $\alpha = -b - 17a/12 > 0$. Then*

$$\mathcal{E}_f(M_4) - \mathcal{E}_f(\psi) \geq \frac{c_O}{3} \min(\alpha, a/12) d(\psi)^2, \qquad c_O = \frac{6 + \sqrt{6}}{12}.$$

For critical states, c_O can be replaced by $(2 + \sqrt{2})/4$. For $d \leq 2^{-10}$ the local bound $Q \geq 3d^2$ replaces $c_O/3$ by 1. In particular this applies to every fixed Tsallis parameter $0 < q < 2$, including its von Neumann limit in nats. At $q = 2$, the enlarged maximiser set prevents such stability relative to $\mathcal{M}$. The same holds for every defect estimate of this shape: the Rényi defect estimate (RE) in the proof of Theorem S18 of the supplement, $3m_q \ln 2\,(\mathcal{E}_q(M_4) - \mathcal{E}_q(\psi)) \geq \alpha_q(P-1) + \frac{a_q}{12}F$ with $m_q = 2^{-q} + 3 \cdot 6^{-q}$ and $a_q, \alpha_q > 0$ the cubic and purity coefficients defined there, gives for $1 < q \leq 5/4$

$$\mathcal{E}_q(M_4) - \mathcal{E}_q(\psi) \geq \frac{c_O \min(\alpha_q, a_q/12)}{3m_q \ln 2}\, d(\psi)^2.$$

[Uses Theorem 18.]

Proof. Drop the scalar remainder in (SF). Since $P \geq 1$, $F \geq 0$ and $Q = F + P - 1$ on unit states, $\alpha(P-1) + \frac{a}{12}F \geq \min(\alpha, a/12)\, Q$; this is also the Rényi case. Then use $Q \geq c_O d^2$ (Theorem 18), (6) for critical states, or, for the local statement, $Q \geq 3d^2$ when $d \leq 2^{-10}$ (Section S7.3 of the supplement). □

Limitations. The global coefficient remains a worst-case bound. Bell-pair products have $F = 0$ but $Q = 1/2$, and the cluster state has $W_1 = \tau = 0$ but $Q = F = 1/2$, so neither term of $\mathcal{D} = F + 3(W_1 + \tau)$ alone controls the distance. The sharp local entropy coefficient is c_*; the quadratic coefficients of Q at the orbit are 6 and 20. The comparison $Q(\psi) \geq 1 - \Phi(\psi)$ is sharp (Theorem 18), but $1 - \Phi$ and d^2 are comparable only up to the factor $(1 + \sqrt{\Phi})/2 \in [\frac{6+\sqrt{6}}{12}, 1]$, and $\mathcal{D} \geq Q$ discards the spin-flip and polarisation terms.

E Proofs for Rényi, Tsallis and mixed-state entropies

Proof of Corollary 7. The limit $q \to 1$ is the von Neumann entropy bound *in nats* (divide by $\ln 2$ for bits). For the proof take $f_q(x) = (x^q - x)/(1-q)$, with $f_q(0) = 0$. On $(0, 1)$,

$$f_q'''(x) = q(2-q)x^{q-3} > 0, \qquad f_q^{(4)}(x) = -q(2-q)(3-q)x^{q-4} < 0.$$

Thus its Hermite cubic has $a_q > 0$ and majorises f_q, including 0 by continuity. Set $g(q) = (15 - 7q)3^{q-1}/(5 + 3q)$. Direct divided differences give

$$b_q + \tfrac{17}{12}a_q = \frac{6^{3-q}}{48(1-q)}\left[5 + 3q - (15 - 7q)3^{q-1}\right] = \frac{6^{3-q}(5+3q)}{48} \cdot \frac{1 - g(q)}{1 - q} < 0. \tag{T}$$

The prefactor is positive, so the sign claim is that $g > 1$ on $(0,1)$ and $g < 1$ on $(1,2)$. Indeed $g(0) = g(1) = 1$ and

$$(\log g)' = \ln 3 - \frac{80}{(15-7q)(5+3q)}.$$

Since $(15-7q)(5+3q) = 75+10q-21q^2$, and $16/15 < \ln 3 < 5/4$, this derivative is positive at $q = 0$, increases up to the vertex $q = 5/21$ of the quadratic and then decreases, so it crosses zero exactly once in $(0,1)$ (it is $\ln 3 - \frac{5}{4} < 0$ at $q = 1$) and remains negative on $[1,2)$. Thus $g > 1$ on $(0,1)$ and $g < 1$ on $(1,2)$, proving the strict sign in (T) on either side of $q = 1$. The logarithm bounds follow, for example, from $\ln 3 = \sum_{k\geq 0} 1/((2k+1)4^k)$: its first two terms give the lower bound $13/12$, and using $2k+1 \geq 3$ in the terms with $k \geq 2$ gives the upper bound $13/12 + \frac{1}{3}\sum_{k\geq 2} 4^{-k} = 10/9$. At $q = 1$, since $g(1) = 1$, the quotient $(1-g(q))/(1-q)$ tends to $g'(1) = (\log g)'(1) = \ln 3 - \frac{5}{4}$, so the continuous limit is $b_1 + 17a_1/12 = \frac{36\cdot 8}{48}(\ln 3 - \frac{5}{4}) = 6\ln 3 - \frac{15}{2} < 0$. As for the von Neumann entropy, $f_q^{(4)} < 0$ makes the Hermite remainder vanish on $[0,1]$ only at the two nodes, so equality can also be identified without the zeros of F: $P = 1$ and $\Delta_f = 0$ give every cut the spectrum $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$, and Proposition 4 applies.

□

Sharpness of the Tsallis range. At $q = 2$ the bound is still valid, but equality is not unique: $S_2(\rho) = 1 - \operatorname{tr}\rho^2$ and Lemma 2 give $\mathcal{E}_{S_2}(\psi) = 1 - P/3 \leq 2/3$, with equality iff all one-qubit marginals are maximally mixed and $\tau = 0$. The cluster state C_4 of the introduction has cut spectra $(1/2, 1/2, 0, 0)$, $(1/4)^4$, $(1/4)^4$, and so also attains $2/3$ at $q = 2$. More strongly, M_4 is not a mean Tsallis-q maximiser for any $q > 2$. The averaged cut moments are

$$m_q = 2^{-q} + 3\,6^{-q}, \qquad c_q = (2^{1-q} + 2\,4^{1-q})/3, \qquad m_q - c_q = \frac{2^{-q}}{3}(1 + 9\,3^{-q} - 8\,2^{-q}).$$

The last parenthesis vanishes at $q = 2$ and has strictly positive derivative $8(\ln 2)2^{-q} - 9(\ln 3)3^{-q}$ for $q \geq 2$: indeed $(3/2)^q \geq 9/4 > 9\ln 3/(8\ln 2)$ follows from $4 > 3$. Therefore $S_q(C_4) > S_q(M_4)$ for all $q > 2$. This explains why the parameter range $0 < q < 2$ for unique M_4 maximisers ends sharply at 2.

Expanded spectral gap. In the setting of Proposition 6, with α and Δ_f as in its proof, the purity identity (Lemma 2) and the remainder identity (S) at $N = 1$, $z = \psi$, turn (SF) into the equivalent formula

$$\begin{aligned} 3(\mathcal{E}_f(M_4) - \mathcal{E}_f(\psi)) = {} & \left(\frac{\alpha}{2} + \frac{a}{16}\right) W_1 + \frac{\alpha}{2}\tau + \frac{a}{32}\sum_i \|V_i - 8U_i\|^2 \\ & + \frac{a}{8}\sum_{i<j} \left\|(R_i^{(i)} - R_j^{(j)})z\right\|^2 + \Delta_f, \end{aligned} \tag{SF+}$$

in which every term is nonnegative. The residual and pair terms sum to $\frac{a}{12}(F - \frac{3}{4}W_1)$, which Theorem 15 bounds below by $\frac{a}{12}(\tau - \tau^{3/2})$.

Tsallis stability. For $0 < q < 2$, with $\delta_q = \mathcal{E}_{S_q}(M_4) - \mathcal{E}_{S_q}(\psi)$, the identity (SF+), with $\alpha_q = -b_q - 17a_q/12 > 0$, and Theorem 15 give

$$3\delta_q \geq \left(\frac{\alpha_q}{2} + \frac{a_q}{16}\right) W_1 + \frac{\alpha_q}{2}\tau + \frac{a_q}{12}(\tau - \tau^{3/2}).$$

Here $\alpha_q, a_q > 0$ for every fixed q in this interval (with the von Neumann limit at $q = 1$), and $\tau - \tau^{3/2} \geq 0$. At $q = 2$ the bound becomes the identity $\delta_2 = (W_1 + \tau)/6$.

Quantitative consequence for von Neumann entropy. With $\delta = E(M_4) - E(\psi)$ in bits, Theorem 15 gives

$$3\ln 2\,\delta \geq \frac{21 - 15\ln 3}{8} W_1 + \Big(\frac{15}{4} - 3\ln 3\Big)\tau + \frac{3}{2}(\ln 3 - 1)(\tau - \tau^{3/2}).$$

Both τ terms are nonnegative, and for small τ the coefficient of τ rises from $\frac{15}{4} - 3\ln 3 \approx 0.454$ to $\frac{9}{4} - \frac{3}{2}\ln 3 \approx 0.602$. Dropping F in (EG) gives the simpler joint estimate

$$W_1 + \tau \leq \frac{6\ln 2}{\frac{15}{2} - 6\ln 3}\,\delta \approx 4.57862\,\delta.$$

Thus near-maximal entropy forces small one-qubit polarisation and spin-flip overlap. These conditions alone do not identify the state: cluster states also have $W_1 = \tau = 0$. Appendix D adds control of the distance to the two HS orbits and transfers it to other strict spectral majorants.

Proof of Corollary 10. The ceiling is proved in the main text after the corollary; it remains to show that no mixed state attains it. Fix one $2|2$ cut. The support of a mixed ρ contains independent vectors with 4×4 flattenings A, B. Their pencil contains a nonzero singular matrix: either a generator is singular, or both are invertible and the positive-degree polynomial $\det(A + tB)$ has a complex root. Let χ be the normalised support vector represented by that matrix. Because ρ is positive definite on its support, for some $p > 0$ the operator $\rho - p|\chi\rangle\langle\chi|$ is positive. Thus an ensemble for ρ can contain χ with positive weight. But χ has deficient Schmidt rank, whereas every pure maximiser has full rank on this cut. Applying the strict pure-state bound to that ensemble makes the mean of the convex roofs strictly smaller than the ceiling.

For Rényi entropies in the stated range, use the pure-state bound of Corollary 9 in place of (SF); the same support argument proves strictness. □

The same exclusion of mixed maximisers works whenever all pure maximisers have full Schmidt rank on one fixed square bipartition.

F Proofs for Rényi orders above one

The proof separates physical compatibility from a classical entropy problem. For a triple of four-point spectra, let p_j be the purity of spectrum j and write

$$\varepsilon = P - 1, \qquad F = 17P - 12T - 12, \qquad V_P = \sum_j (p_j - P/3)^2.$$

Every pure four-qubit state supplies a triple in the relaxation

$$\mathcal{R} = \Big\{\varepsilon \geq 0, \quad F \geq 0, \quad 6V_P \leq F^2 + \tfrac{9}{2}\varepsilon\Big\}.$$

The first two constraints are the purity and sextic bounds; the third is (QK), Corollary 28 of the sharp bound (QV). No spectral cap is imposed.

F.1 High orders: an elementary bound

Every order $q \geq 2$ is settled by purity alone, with no anchor, crossing or certificate; the rest of this appendix is needed only for $1 < q < 2$.

Corollary 23 (Unrestricted high-order Rényi purity deficit)**.** *For every pure four-qubit state and every $q \geq 2$,*

$$\boxed{\mathcal{E}_q(\psi) \leq \frac{5}{3} - \frac{1}{3}\log_2(2P - 1) \leq \frac{5}{3}.} \tag{R2}$$

At $q = 2$, the first bound is sharp at every $P \in [1, 3/2]$ and equality holds exactly when two of the three cut marginals are maximally mixed. For $q > 2$, equality in the first bound requires and is attained by the cluster or Bell-pair-product cut spectra. In particular, for all $q \geq 2$ the global ceiling $5/3$ has equality exactly at the cluster-type cut spectra: $(1/2, 1/2, 0, 0)$, $(1/4)^4$, $(1/4)^4$, up to ordering of cuts. No eigenvalue-cap hypothesis is assumed.

Proof. Put $p_j = \operatorname{tr} \rho_{1j}^2$ and $x_j = 4p_j - 1 \geq 0$. Lemma 2 gives $P = \sum_j p_j \geq 1$. Expanding a product of nonnegative terms yields

$$64 \prod_{j=2}^{4} p_j = \prod_j (1 + x_j) \geq 1 + \sum_j x_j = 4P - 2,$$

so $\prod_j p_j \geq (2P - 1)/32$. Consequently

$$\mathcal{E}_2(\psi) = -\frac{1}{3} \log_2 \prod_j p_j \leq \frac{5}{3} - \frac{1}{3} \log_2 (2P - 1).$$

Rényi entropy is nonincreasing in its order, proving the bound for $q \geq 2$. At $q = 2$ equality in the product inequality holds exactly when at most one x_j is positive, equivalently when at least two cut spectra are uniform-four. The cluster-to-Bell family (QF) of Section F.5, with cut spectra $((1 + r)/2, (1 - r)/2, 0, 0), (1/4)^4, (1/4)^4$ for $0 \leq r \leq 1$, has precisely the equality form and $P = 1 + r^2/2$, proving sharpness for every $P \in [1, 3/2]$.

For $q > 2$, equality also requires equality in Rényi-order monotonicity on every cut, hence each cut spectrum is uniform on its support. The two already-uniform-four spectra contribute total purity $1/2$. Since $P \geq 1$, the last uniform support has rank at most two: rank two gives the cluster pattern, and rank one the Bell-pair-product pattern. Both attain the displayed bound. Finally the weaker ceiling $5/3$ requires $P = 1$, which excludes the Bell pattern and leaves only the cluster-type spectra. □

F.2 Plan of the anchor proof

Two anchors suffice. Set $q_s = 1587/1000 < q_* < q_u = 8/5$. The anchors must bracket q_*, because an HS bound transfers only to lower orders and a cluster bound only to higher ones, and they are rational so that the supporting inequalities are statements with exact rational data. Since $\mathcal{E}_q$ decreases in q, the cluster bound is easier to prove as q increases towards q_*: q_s is the lower end of the certified enclosure $1587/1000 < q_* < 1589/1000$ of Proposition 24, and it exceeds the threshold q_b that the cubic envelope requires (Corollary 33). The upper anchor $q_u = 8/5$, the end of the interval in that proposition, is a simple order at which the HS supporting inequality has explicit constants (Lemma S6 of the supplement). Theorem 36 partitions $\mathcal{R}$ into just two regions for order transfer:

region	anchor bound	anchor equality
$F < 3/20, \quad 20\varepsilon + 2F < 1$	$\mathcal{E}_{q_u} \leq h_{q_u}$	three HS spectra
complement in $\mathcal{R}$	$\mathcal{E}_{q_s} \leq 5/3$	cluster triple.

Here $h_q = H_q(1/2, 1/6, 1/6, 1/6)$ and the cluster triple is $((1/2, 1/2, 0, 0), (1/4)^4, (1/4)^4)$, up to cut order. Single crossing (Corollary 26) transfers an HS bound to every lower order above one; ordinary order monotonicity transfers a constant bound to every higher order. Since $h_{q_s} > 5/3 > h_{q_u}$, the two anchors give $\mathcal{E}_q \leq \max\{h_q, 5/3\}$ for all $q > 1$. With the anchor equality cases, and single crossing strict off HS spectra, equality is HS below q_*, cluster above it, and both at q_*. The proof of Corollary 9 below gives the order transfer and physical rigidity explicitly. Orders $0 < q \leq 1$ were settled in Section 3.

How the anchor proof is organised. The subsections below give single crossing, (QV) and (QK), determinant reduction, the cubic envelope and the Lagrangian dual. Determinant replacement keeps each cut's second and third moments and does not decrease its entropy, so the supporting bounds need only two double-root charts, which include spectra with an eigenvalue above 1/2. The supplement proves the anchors through the disjoint partition of Table S3 of the supplement, then the supporting inequalities in order of use.

The physical input (QK) follows from (QV), which has an exact-rational certificate and an independent analytic proof on critical states (Section S6.17 of the supplement). The entropy estimates use calculus and explicit polynomial arithmetic, with exact polynomial sign checks. An independent near-one defect estimate, critical-locus variance refinements and a shorter proof for states with a rank-two cut, with further purity-controlled bounds, are in Sections S6.14–S6.21 of the supplement; the proof here does not use them.

F.3 The HS–cluster transition

Put $m_q = 2^{-q} + 3\,6^{-q}$, so the HS entropy is $h_q = \log_2 m_q/(1-q)$.

Proposition 24 (HS–cluster crossing)**.** *There is a unique $q_* \in (3/2, 8/5)$ satisfying*

$$\frac{\log_2 m_{q_*}}{1-q_*} = \frac{5}{3}.$$

For every $q > q_*$, the cluster state has strictly larger mean Rényi entropy than M_4; at q_* the two values coincide. Since $1587/1000 < q_* < 1589/1000$ (RA-4), q_* is the threshold $\alpha_0 \approx 1.59$ of the Rényi conjecture of Gour and Wallach [8, Fig. 2]. That the two states are the global maximisers below and above q_* is Corollary 9.

Proof. The cluster spectra recalled above give $(1+2+2)/3 = 5/3$ for every $q > 0$. The Rényi entropy of the nonuniform full-support HS spectrum is strictly decreasing: $g(q) = \ln m_q$ is strictly convex, $g(1) = 0$, and its secant slope $g(q)/(q-1)$ is strictly increasing. Since $1-q < 0$ for $q > 1$, $h_q > 5/3$ if and only if $m_q < 2^{-5(q-1)/3}$. As $m_q = 2^{-q}(1+3^{1-q})$, we have $m_{3/2} < 2^{-5/6}$, as $(1+1/\sqrt{3})^3 = 2+10/(3\sqrt{3}) < 4$ reduces to $100 < 108$; and $m_{8/5} > 1/2$, as $2^{3/5} < 47/31 < 1+3^{-3/5}$ reduces to $8 \cdot 31^5 < 47^5$ and $27 \cdot 16^5 < 31^5$. So its Rényi value crosses 5/3 strictly between those parameters. Continuity and strict monotonicity finish. The same monotonicity and the exact enclosures RA-4 of $\log_2 m_q$ at $q = 1587/1000$ and $q = 1589/1000$ give $h_{1587/1000} > 5/3 > h_{1589/1000}$, hence $1587/1000 < q_* < 1589/1000$. □

F.4 Single crossing in the order

The orders above one are reduced to two anchor orders by a property of single spectra: as the order increases, the Rényi entropy of any four-point spectrum can overtake that of the HS spectrum at most once, and never falls back below it. For a probability vector λ and $q > 1$ put

$$u_q(\lambda) = \ln \sum_i \lambda_i^q, \qquad v_q(\lambda) = \partial_q u_q(\lambda) = \frac{\sum_i \lambda_i^q \ln \lambda_i}{\sum_i \lambda_i^q}, \qquad \kappa_q = \frac{1}{q} + \frac{\ln 3}{\sinh((q-1)\ln 3)},$$

with $0^q \ln 0 = 0$, so that $H_q(\lambda)\ln 2 = u_q(\lambda)/(1-q)$. Write $\sigma = (\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ for the HS spectrum.

Lemma 25 (Single crossing)**.** *For every $q > 1$ and every four-point probability vector λ,*

$$v_q(\lambda) - v_q(\sigma) \le \kappa_q\big(u_q(\lambda) - u_q(\sigma)\big), \tag{SX}$$

with equality only when λ is a permutation of σ.

Corollary 26 (Order transfer)**.** *For every four-point spectrum λ the function*

$$G_\lambda(q) = \frac{(q-1)\big(H_q(\lambda) - H_q(\sigma)\big)}{q\tanh\big((q-1)\ln 3/2\big)}$$

is nondecreasing on $(1,\infty)$, strictly unless λ is a permutation of σ. Consequently, for every triple of four-point spectra and $1 < q < q_1$: if $\mathcal{E}_{q_1} \le h_{q_1}$ then $\mathcal{E}_q \le h_q$, with equality only when all three spectra are HS spectra; and if $\mathcal{E}_{q_1} < h_{q_1}$ then $\mathcal{E}_q < h_q$.

Proof. Put $U(q) = u_q(\lambda) - u_q(\sigma)$ and $K(q) = \ln q + \ln\tanh((q-1)\ln 3/2)$, so that $K' = \kappa_q$ and $G_\lambda = -e^{-K}U/\ln 2$. By (SX), $(e^{-K}U)' = e^{-K}(U' - \kappa_q U) \le 0$, with equality at an order only for HS. For a triple, the average of $G_{\lambda^{(j)}}(q)$ is a positive multiple of $\mathcal{E}_q - h_q$. It increases strictly unless all three spectra are HS. Its sign at q_1 therefore gives the two transfers and the equality statement. □

Proof of Lemma 25. Fix $q > 1$, and put $s = q-1$, $L = \ln 3$, $\beta = q\kappa_q - 1 = qL/\sinh(sL) > 0$ and $\varphi = v_q - \kappa_q u_q$. We maximise this continuous function on the simplex.

A maximiser has full support. On the positive coordinates, with $Z = \sum_i \lambda_i^q$,

$$\partial_i\varphi = \frac{g(\lambda_i)}{Z}, \qquad g(x) = x^{q-1}(q\ln x + C), \qquad C = 1 - qv_q - q\kappa_q.$$

The derivative in a zero coordinate is zero, since $x^{q-1}\ln x \to 0$. At a maximiser on a face, the positive-coordinate derivatives equal a multiplier μ. But $\sum_i \lambda_i\partial_i\varphi = 1 - q\kappa_q = -\beta$, so $\mu = -\beta$. Moving mass from a positive coordinate to a zero one therefore has derivative $\beta > 0$, contradicting maximality.

Only one coordinate can be larger. Since

$$g'(x) = x^{q-2}\big[(q-1)(q\ln x + C) + q\big],$$

the bracket is strictly increasing, so g decreases then increases and takes each value at most twice. The Lagrange condition thus gives at most two coordinate values, $a > b > 0$. If both occur, $g(a) = g(b)$ implies $g'(a) > 0$. Two coordinates equal to a would admit the trace-preserving variation $(a+t, a-t)$. Its first variations of Z and v_q vanish, and its second variation of φ is $2g'(a)/Z > 0$, again contradicting maximality. Thus every maximiser is uniform or a permutation of (a,b,b,b).

The one-large family. Parametrise it by $a = (1+3e^{-z})^{-1}$, $b = ae^{-z}$, $z \ge 0$, and put $w = 3e^{-qz}/(1+3e^{-qz})$, $p = 3e^{-z}/(1+3e^{-z})$. Direct differentiation gives, for $z > 0$,

$$\begin{aligned}\varphi(z) &= \beta\ln(1+3e^{-z}) - \kappa_q\ln(1+3e^{-qz}) - zw,\\ \varphi'(z) &= \beta(w-p) + qzw(1-w) = zw(1-w)\big(q - \beta R(z)\big),\end{aligned}$$

where the elementary hyperbolic identity

$$R(z) = \frac{(e^{sz}-1)(1+3e^{-qz})}{z(1+3e^{-z})} = \frac{\sinh(sz)}{z} + \tanh\left(\frac{z-L}{2}\right)\frac{\cosh(sz)-1}{z} \tag{SR}$$

settles the sign. The function $\sinh(sz)/z$ is strictly increasing for $z > 0$, and the second summand has the sign of $z - L$. Consequently

$$R(z) < R(L) = \frac{\sinh(sL)}{L} = \frac{q}{\beta} \quad (0 < z < L), \qquad R(z) > R(L) \quad (z > L).$$

Hence φ strictly increases up to $z = L$ and strictly decreases afterwards. This also excludes the uniform endpoint $z = 0$. Its unique maximum is $z = L$, namely σ, proving both the inequality and its equality case. □

F.5 The universal quadratic variance bound

For $0 \le r \le 1$ the critical normal-form vector $v = (\sqrt{(1+r)/2},\, i\sqrt{(1-r)/2},\, 0,\, 0)$ (Appendix B) has cut spectra

$$((1+r)/2, (1-r)/2, 0, 0), \qquad (1/4)^4, \qquad (1/4)^4, \tag{QF}$$

since its two nonzero components are in quadrature. This *cluster-to-Bell family* joins the cluster state ($r = 0$) to a product of two Bell pairs ($r = 1$).

Theorem 27 (Sharp quadratic purity variance)**.** *Every unit pure four-qubit state satisfies*

$$\boxed{F^2 + 3P(P-1) \ge 6V_P, \qquad V_P = \sum_{j=2}^{4} (p_j - P/3)^2.} \tag{QV}$$

Equality holds along the whole cluster-to-Bell family (QF)*, so the coefficient* 3 *cannot be decreased.*

[Exact-rational certificate, QVA.]

Corollary 28 (Quadratic purity variance)**.** *Every unit pure four-qubit state satisfies*

$$\boxed{F^2 + \frac{9}{2}(P-1) \ge 6V_P.} \tag{QK}$$

Equality holds exactly for HS cut spectra, cluster cut spectra, or products of two Bell pairs, up to ordering of cuts and eigenvalues. The coefficient $9/2$ *cannot be decreased, and the coefficient* 6 *cannot be increased while retaining* $9/2$*.*

Proofs in outline. On critical states Proposition S22 of the supplement proves (QV) by a joint-cut analytic argument, and an elementary box bound on the purities proves it whenever $F \ge 3/2 - P$. The certificate proves it for all states at once. Homogenised by $N = \|z\|^2$, the defect of (QV) is a qubit-symmetric, conjugation-even local-unitary invariant of bidegree $(6, 6)$ that vanishes at HS and cluster; such invariants form an 83-dimensional space, and 83 explicit Gaussian-integer states determine its elements (Section S2 of the supplement). Ninety-five six-copy covariants, built from z and its spin flip and vanishing at HS and cluster, have squared norms in that space. With positive integers q_i and D, the certificate reads $2D\,(F^2 + 3P(P-1) - 6V_P) = \sum_i q_i \|\Phi_i\|^2$ for unit states; its homogeneous form is (S6.1) of the supplement. That homogeneous identity holds at the 83 states, hence everywhere. Every term on the right is nonnegative. For the corollary, the defect of (QK) exceeds that of (QV) by $3(P-1)(3/2-P) \ge 0$ on $1 \le P \le 3/2$, and the purity box makes (QK) strict above $3/2$; equality is then analysed at $P = 1$ by Proposition S21 of the supplement and at $P = 3/2$ by the purity box, which leaves purities $(1, \frac{1}{4}, \frac{1}{4})$ and hence products of two Bell pairs. The family (QF) fixes both coefficients. The complete proofs are in Section S6.1 of the supplement [40] (exact rational certificate; checks QVA-1–QVA-3, QG-1, QG-3, SD-4). □

F.6 Determinant reduction for four-point spectra

Lemma 29 (Determinant monotonicity at fixed second and third moments)**.** *For* $1 < q < 2$*, let* λ, λ' *be four-point spectra with the same second moment* p *and third moment* t*. If* $\prod_i \lambda_i < \prod_i \lambda'_i$*, then* $H_q(\lambda) < H_q(\lambda')$*. Consequently, at fixed* p *and* t*, the maximum of* H_q *is attained at a spectrum of one of the two types*

$$(a, b, c, c) \quad \text{or} \quad (a, a, b, c), \qquad a \ge b \ge c \ge 0, \tag{DM}$$

including their degenerations.

Proof idea. Write e_k for the elementary symmetric polynomials of the eigenvalues and $d = \prod_i \lambda_i$. Newton's identities fix e_1, e_2, e_3, so $\prod_i (\lambda_i + s)$ depends on the spectrum only through d; a Stieltjes representation of x^q then shows that $\sum_i \lambda_i^q$ decreases in d, and a spectrum of maximal determinant has a repeated outer pair. The complete proof is in Section S6.2 of the supplement [40]. □

Corollary 30 (Six-variable classical reduction)**.** *Let three four-component probability vectors be constrained by any conditions involving only their total second and third moments P, T and the sum of their squared second moments $U = \sum_j p_j^2$. For every $1 < q < 2$, the supremum of their mean Rényi entropy is unchanged if each cut is required to have one of the spectral forms* (DM)*. Up to permutation of cuts, only four patterns need be considered: $k = 0, 1, 2, 3$ cuts with two largest equal, the others with two smallest equal. Each cut has two free real spectral parameters. In particular, this reduction is exact for the classical relaxation $P \geq 1$, $F = 17P - 12T - 12 \geq 0$, and $F^2 + \frac{9}{2}(P-1) \geq 6(U - P^2/3)$. Theorem 36 bounds this relaxation at the two anchor orders; with Corollary 26 and order monotonicity this gives the envelope on it for $q > 1$.*

Proof. At fixed p_j, t_j for each cut, replacing that spectrum by one of maximum determinant preserves P, T, U and thus all the stated constraints. Lemma 29 shows that this replacement cannot decrease any cut entropy. The two forms in (DM) have two free coordinates after the trace constraint. The cut permutation symmetry reduces the eight form choices to four. □

F.7 A sharp cubic Rényi bound from binary grouping

Binary grouping gives an independent entropy bound that will control the region of large moment slack. Set

$$q_b = \frac{4\ln 2}{4\ln 2 - 1}, \qquad r_b = q_b - 1 = \frac{1}{4\ln 2 - 1}, \qquad \frac{3}{2} < q_b < \frac{313}{200} < \frac{1587}{1000} = q_s. \tag{BT}$$

Thus $q_b/(r_b \ln 2) = 4$ and $1/2 < r_b < 1$. The bounds on q_b follow from three terms of the atanh series: $56/81 + 2/1215 < \ln 2 < 1123/1620 < 3/4$.

Lemma 31 (Binary grouping)**.** *For any finite probability vectors u, v and $0 \leq a \leq 1$, set $w = a^2(3 - 2a)$. Then*

$$H_{q_b}(au, (1-a)v) \leq 4a(1-a) + wH_{q_b}(u) + (1-w)H_{q_b}(v). \tag{BG}$$

For $0 < a < 1$, equality holds exactly when $a = 1/2$ and $H_{q_b}(u) = H_{q_b}(v)$. In particular $H_q(a, 1-a) \leq 4a(1-a)$ holds for every a if and only if $q \geq q_b$; its equality cases are $a = 0, 1/2, 1$.

Proof. Write $q = q_b$, $r = r_b$, $L = \ln 2$ and $h(w) = -w\ln w - (1-w)\ln(1-w)$. The log-sum inequality gives, for positive A, B, $\ln(A + B) \geq w\ln A + (1-w)\ln B + h(w)$. Apply it with $A = a^q 2^{-rH_q(u)}$ and $B = (1-a)^q 2^{-rH_q(v)}$. The required assertion follows if

$$J(a) := q\{w\ln a + (1-w)\ln(1-a)\} + h(w) + 4rLa(1-a) \geq 0.$$

The function J is symmetric about $1/2$, and $J(1/2) = J(1) = 0$, with the latter understood by continuity. Since $w - a = a(1-a)(2a-1)$ and $4rL = q$, differentiation cancels the terms outside the derivative of w:

$$J'(a) = 6a(1-a)\left(q\ln\frac{a}{1-a} - \ln\frac{w}{1-w}\right).$$

For $a > 1/2$, put $t = a/(1-a) > 1$. Then

$$\frac{w}{1-w} = t\,T_2(t), \qquad T_2(t) = \frac{t(t+3)}{3t+1}, \qquad \operatorname{sgn} J'(a) = \operatorname{sgn}\{r\ln t - \ln T_2(t)\}.$$

The function $s \mapsto \ln T_2(e^s)$ is strictly convex for $s > 0$, since

$$\left(t\frac{d}{dt}\right)^2 \ln T_2(t) = \frac{24t(t^2-1)}{(t+3)^2(3t+1)^2} > 0.$$

Its derivative starts at $1/2$ and tends to 1. Consequently $\ln T_2(e^s) - rs$ starts at zero with negative derivative, is strictly convex, tends to $+\infty$, and has exactly one further zero. Thus J increases

and then decreases between its zero endpoints $1/2, 1$; it is strictly positive in between. Symmetry proves $J \geq 0$ throughout. Equality in the log-sum inequality at $a = 1/2$ requires $A = B$, which is precisely the stated entropy condition.

Taking both conditional vectors pure proves the binary bound at q_b; order monotonicity proves it for every $q \geq q_b$ with the same equality cases. Necessity follows as $a = \epsilon \downarrow 0$: $H_q(\epsilon, 1-\epsilon) = q\epsilon/((q-1)\ln 2) + o(\epsilon)$, whereas $4\epsilon(1-\epsilon) = 4\epsilon + O(\epsilon^2)$. Orders $q \leq 1$ are excluded by monotonicity from any $1 < q < q_b$. □

Proposition 32 (Sharp cubic entropy bound)**.** *For every four-point probability vector λ and every $q \geq q_b$,*

$$H_q(\lambda) \leq \frac{10}{3} - 6p + \frac{8}{3}t = 4e_2(\lambda) + 8e_3(\lambda), \qquad p = \sum_i \lambda_i^2, \quad t = \sum_i \lambda_i^3. \tag{CB}$$

The threshold q_b is necessary for a bound valid on the entire simplex. Equality occurs exactly at pure, uniform-two and uniform-four spectra.

Proof. By order monotonicity it suffices to take $q = q_b$. Group the entries as $\lambda = (au, a(1-u), (1-a)v, (1-a)(1-v))$. For $0 < a < 1$, Lemma 31 and its binary consequence give

$$H_q(\lambda) \leq 4a(1-a) + 4a^2(3-2a)u(1-u) + 4(1-a)^2(1+2a)v(1-v).$$

The right side is exactly $4e_2 + 8e_3$, by expanding across the two groups. If $a = 0$ or 1 the assertion is the binary bound. The identity with (p, t) follows from Newton's identities and $\sum \lambda_i = 1$. For equality with $0 < a < 1$, grouping forces $a = 1/2$ and equal conditional entropies; binary equality then makes both conditional spectra pure or both uniform-two. This gives uniform-two or uniform-four. The boundary case also permits a pure spectrum. Each such spectrum stays an equality at every order, and all other spectra are strict already at q_b. The binary restriction proves sharpness of the threshold. □

Corollary 33 (Analytic cubic envelope)**.** *Every triple of four-point probability vectors satisfies, for every $q \geq q_b$,*

$$\mathcal{E}_q \leq \frac{5}{3} - \frac{20}{27}(P-1) - \frac{2}{27}(F - 1/2). \tag{HC}$$

In particular $\mathcal{E}_q \leq 5/3$ whenever $20(P-1) + 2F \geq 1$. Equality in the $5/3$ ceiling holds exactly at a permutation of $((1/2, 1/2, 0, 0), (1/4)^4, (1/4)^4)$. No quadratic variance bound or physical compatibility is required.

Proof. Sum (CB) over the three cuts, substitute $T = (17P - 12 - F)/12$, and divide by three. Equality in the ceiling requires $20(P-1) + 2F = 1$ and equality in all three scalar bounds. Each cut is then pure, uniform-two or uniform-four, with entropy 0, 1 or 2 at every order, and a mean of $5/3$ forces exactly one uniform-two and two uniform-four spectra; these have $P = 1$, $F = 1/2$ and attain $5/3$ at every order. □

F.8 An exact Lagrangian dual for the classical relaxation

We now prove the anchor-order bounds used with Corollary 26. The argument uses a classical relaxation of the cut-spectrum constraints: three four-point probability vectors $\lambda^{(2)}, \lambda^{(3)}, \lambda^{(4)}$ with $p_j = \sum_i (\lambda_i^{(j)})^2$, $t_j = \sum_i (\lambda_i^{(j)})^3$, $P = \sum p_j \geq 1$, $F = 17P - 12T - 12 \geq 0$ ($T = \sum t_j$), and the quadratic bound (QK). The cut spectra of every physical state form such a triple (Lemma 2, Theorem 3, Corollary 28). Write $\varepsilon = P - 1$, $U = \sum p_j^2$, $f_j = 17p_j - 12t_j$, and $\mathcal{E}_q = \frac{1}{3}\sum_j H_q(\lambda^{(j)})$ in bits.

Lemma 34 (Lagrangian dual bound)**.** *For $\theta = (\alpha, \beta, \gamma) \in \mathbb{R}^2 \times [0, \infty)$ put*

$$\Lambda_\theta(\lambda) = H_q(\lambda) + \alpha(p - \tfrac{1}{3}) + \beta(f - 4) - \gamma(p^2 - \tfrac{1}{9}), \qquad \ell_\theta(\varepsilon, F) = \alpha\varepsilon + \beta F - \frac{\gamma}{6}\Big(F^2 + \frac{17}{2}\varepsilon + 2\varepsilon^2\Big).$$

For every triple of the relaxation, $\sum_j H_q(\lambda^{(j)}) \leq \sum_j \Lambda_\theta(\lambda^{(j)}) - \ell_\theta(\varepsilon, F)$. The function ℓ_θ is concave, so on a rectangle its minimum is attained at a corner.

Proof. $\sum_j (p_j - \frac{1}{3}) = \varepsilon$, $\sum_j (f_j - 4) = F$ and $\sum_j (p_j^2 - \frac{1}{9}) = U - \frac{1}{3}$, hence $\sum_j H_q = \sum_j \Lambda_\theta - \alpha\varepsilon - \beta F + \gamma(U - \frac{1}{3})$. The quadratic bound reads $6U \le F^2 + \frac{9}{2}\varepsilon + 2P^2$, i.e. $U - \frac{1}{3} \le \frac{1}{6}(F^2 + \frac{17}{2}\varepsilon + 2\varepsilon^2)$, and $\gamma \ge 0$. □

Lemma 35 (Sharp third-moment bound at fixed purity)**.** *If a four-point probability vector has purity $p = \frac{1}{4} + x$, then $0 \le x \le \frac{3}{4}$ and $|t - \frac{1}{16} - \frac{3}{4}x| \le x^{3/2}/\sqrt{3}$. The upper bound is attained only at the one-large spectrum $((1+3s)/4, (1-s)/4, (1-s)/4, (1-s)/4)$, up to permutation, where $s = \sqrt{4x/3}$. The lower bound is attained precisely at its one-small counterpart when $x \le 1/12$; at $x = 0$ both are uniform.*

Proof. The vectors with given p form a round two-sphere $\{\lambda = \frac{1}{4}\mathbf{1} + w : \sum w_i = 0,\ |w|^2 = x\}$ of radius $\sqrt{x}$. Optimising on the whole sphere gives a valid bound even when it extends outside the simplex. On it $t = \frac{1}{16} + \frac{3}{4}x + \sum w_i^3$; critical points of the cubic $\sum w_i^3$ have at most two distinct coordinates, namely the patterns $(3,1)$ and $(2,2)$. The pattern $(2,2)$ has $\sum w^3 = 0$, and the patterns $(z,z,z,-3z)$ with $12z^2 = x$ give $\sum w^3 = -24z^3 = \mp x^{3/2}/\sqrt{3}$. The upper extremiser has three entries $(1-s)/4 \ge 0$ for every $x \le 3/4$; the lower has smallest entry $(1-3s)/4 \ge 0$ exactly when $x \le 1/12$. □

Exact contact identities. Write $c_1 = (1/2, 1/2, 0, 0)$ and $c_2 = (1/4)^4$. Their (H_q, p, f, p^2) values, independent of q, are $(1, 1/2, 11/2, 1/4)$ and $(2, 1/4, 7/2, 1/16)$. Consequently, with $\Lambda_i^\star = \Lambda_\theta(c_i)$,

$$\Lambda_1^\star = 1 + \frac{\alpha}{6} + \frac{3}{2}\beta - \frac{5}{36}\gamma, \qquad \Lambda_2^\star = 2 - \frac{\alpha}{12} - \frac{\beta}{2} + \frac{7}{144}\gamma, \qquad \Lambda_1^\star + 2\Lambda_2^\star - \ell_\theta(0, 1/2) = 5, \quad \text{(DC)}$$

since $\ell_\theta(0, 1/2) = \beta/2 - \gamma/24$.

Theorem 36 (Two Rényi anchors)**.** *Let $q_s = 1587/1000$ and $q_u = 8/5$. Every triple satisfying $P \ge 1$, $F \ge 0$ and (QK) obeys one of the following.*

1. *If $F \ge 3/20$ or $20\varepsilon + 2F \ge 1$, then $\mathcal{E}_{q_s} \le 5/3$, with equality only at a permutation of (c_1, c_2, c_2).*
2. *Otherwise $\mathcal{E}_{q_u} \le h_{q_u}$, with equality only when all three spectra are HS spectra.*

Proof outline. The cubic envelope first removes $20\varepsilon + 2F \ge 1$. On the residual, $\varepsilon < 1/20$ and every $p_j < 11/20$. For $F < 3/20$ use the HS supporting inequality at q_u. For $3/20 \le F \le 11/25$ use the strict dual band at q_s. The remaining region lies near the cluster triple: the capped cubic bound handles cuts with $p_j \le 47/100$, and the sliver bound handles the unique cut above that threshold. Table S3 and Section S6.4 of the supplement [40] give the complete partition, summation and equality argument. □

Proof of Corollary 9. For $0 < q < 1$ use Corollary 8, and for $q = 1$ use Theorem 1. For $q > 1$, the three cut spectra of ψ form a triple of the relaxation (Lemma 2, Theorem 3 and Corollary 28); recall $q_s < q_* < q_u$ and that $\mathcal{E}_q$ and h_q decrease in q.

In case (1) of Theorem 36, $\mathcal{E}_{q_s} \le 5/3 < h_{q_s}$. Corollary 26 gives $\mathcal{E}_q < h_q$ for $1 < q \le q_s$, and $\mathcal{E}_q \le \mathcal{E}_{q_s} \le 5/3$ for $q \ge q_s$. So $\mathcal{E}_q \le \max\{h_q, 5/3\}$, strictly for $q < q_*$; for $q \ge q_*$ equality forces $\mathcal{E}_{q_s} = 5/3$, hence the cluster-type triple.

In case (2), Corollary 26 gives $\mathcal{E}_q \le h_q$ for $1 < q \le q_u$, with equality only for three HS spectra, and $\mathcal{E}_q \le \mathcal{E}_{q_u} \le h_{q_u} < 5/3$ for $q \ge q_u$. As $h_q < 5/3$ for $q > q_*$, equality occurs only for $q \le q_*$ and three HS spectra.

Three HS spectra imply $P = 1, F = 0$, so Proposition 4 gives the two HS LU orbits. A physical state with triple (c_1, c_2, c_2) has two maximally mixed cut marginals; Proposition 40 identifies it with the cluster endpoint of (QF). □

Remark 37 (Scope of the relaxation). The classical relaxation uses only $P \geq 1$, $F \geq 0$ and (QK); no joint purity bound, spectrum cap or four-tangle information. It is needed only at the two anchor orders: Corollary 26 acts cut by cut and needs no compatibility at all. The multiplier β may have either sign; the dual has no gap on the sliver only after the third-moment link of Lemma 35 (a constant θ on a fixed (ε, F) cell fails there). The equality classification uses the same strict local bounds as the inequality; it requires no additional covering.

F.9 A spectral cap from spin flip

The following bound controls the part of a cut spectrum above $1/2$ directly by two-copy quantities. It holds for every pure state, not only on the critical locus, and needs neither a four-qubit normal form nor a polynomial certificate. Write $x_+ = \max\{x, 0\}$ and use the Hilbert–Schmidt norm throughout.

Proposition 38 (Spectral spin-flip bound). *For every unit pure four-qubit state and every balanced cut $A|A^c$,*

$$\boxed{(2\lambda_{\max}(\rho_A) - 1)_+^2 \leq \tau + \frac{1}{2}W_1 = 2(P-1) - \frac{1}{2}W_1 \leq 2(P-1).} \tag{10}$$

Both coefficients on the first right-hand side are optimal. In particular, $P = 1$ forces every balanced eigenvalue to be at most $1/2$.

The matrix inequality behind the proposition holds in every bipartite dimension. The two vectors below need only have the same Schmidt coefficients; their Schmidt bases may be unrelated.

Lemma 39 (Overlap and marginal mismatch). *Let ψ, ϕ be unit bipartite pure states with common squared Schmidt coefficients $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Let ρ_A, ρ_B and σ_A, σ_B be their marginals. Then*

$$|\langle \psi, \phi \rangle|^2 + \frac{1}{2}\big(\|\rho_A - \sigma_A\|_F^2 + \|\rho_B - \sigma_B\|_F^2\big) \geq (2\lambda_1 - 1)_+^2. \tag{11}$$

For each common Schmidt spectrum with $\lambda_1 \geq 1/2$, the bound is attained by changing the sign of every Schmidt term except the first.

Proof. The case $\lambda_1 \leq 1/2$ is immediate. Assume $\lambda_1 > 1/2$. Pad the coefficient matrices with zeros if the local dimensions differ, and choose Schmidt bases for ψ. Its coefficient matrix is $D = \operatorname{diag}(\sqrt{\lambda_i})$; write W for that of ϕ. Thus $WW^\dagger$ and $W^\dagger W$ have eigenvalues λ_i. Put

$$\Lambda = D^2, \qquad r = 1 - \lambda_1, \qquad d = 2\lambda_1 - 1, \qquad S = \operatorname{diag}(0, r - \lambda_2, \ldots, r - \lambda_n).$$

The entries of S are nonnegative and increasing. Trace rearrangement therefore gives, for $R = WW^\dagger$ and for $R = W^\dagger W$,

$$\operatorname{tr}(SR) \geq \sum_{i=2}^{n} (r - \lambda_i)\lambda_i = r^2 - \sum_{i=2}^{n} \lambda_i^2. \tag{12}$$

Indeed, diagonalising R expresses its diagonal as a doubly stochastic average of permutations of the decreasing vector (λ_i); pairing it with the increasing diagonal of S is minimised in the displayed order.

Define the quadratic form

$$\mathcal{B}_D(W) = \|W\|_F^2 + |\operatorname{tr}(DW)|^2 - \operatorname{tr}\big(\Lambda(WW^\dagger + W^\dagger W)\big).$$

Its useful lower bound has an explicit positive remainder. Let $W_{\mathrm{t}} = (W_{ij})_{i,j \geq 2}$ and $t = \sum_{i=2}^{n} \sqrt{\lambda_i} W_{ii}$, so $\operatorname{tr}(DW) = \sqrt{\lambda_1} W_{11} + t$. Since $\Lambda + S = rI + dE_{11}$, with E_{11} the first matrix unit,

and $1-2r=d$, the left side below is $d(\|W_{\mathrm{t}}\|_F^2-|W_{11}|^2)+|\sqrt{\lambda_1}W_{11}+t|^2$; for $r>0$, completing the square in W_{11} with $\lambda_1-d=r$ gives

$$\mathcal{B}_D(W)-\operatorname{tr}(S(WW^\dagger+W^\dagger W))=d\left(\|W_{\mathrm{t}}\|_F^2-\frac{|t|^2}{r}\right)+\frac{|rW_{11}+\sqrt{\lambda_1}t|^2}{r}\geq 0. \tag{13}$$

Cauchy–Schwarz gives $|t|^2\leq r\sum_{i\geq 2}|W_{ii}|^2\leq r\|W_{\mathrm{t}}\|_F^2$. This proves positivity without any condition on W. When $r=0$, the remainder is simply $\|W_{\mathrm{t}}\|_F^2\geq 0$.

Apply (12) twice to obtain

$$\mathcal{B}_D(W)\geq 2\left(r^2-\sum_{i=2}^{n}\lambda_i^2\right).$$

Set $c_2=\sum_i\lambda_i^2$. In these coordinates $\langle\psi,\phi\rangle=\operatorname{tr}(DW)$, $\rho_A=\Lambda$, $\sigma_A=WW^\dagger$, and ρ_B,σ_B are the transposes of $\Lambda, W^\dagger W$. Since $\operatorname{tr}(WW^\dagger)^2=\operatorname{tr}(W^\dagger W)^2=c_2$, the mismatch terms are $2c_2-2\operatorname{tr}(\Lambda WW^\dagger)$ and $2c_2-2\operatorname{tr}(\Lambda W^\dagger W)$. With $\|W\|_F^2=1$, the left-hand side of (11) is therefore $\mathcal{B}_D(W)-1+2c_2$. Its lower bound is $2r^2+2\lambda_1^2-1=d^2$, as required. Taking $W=\operatorname{diag}(\sqrt{\lambda_1},-\sqrt{\lambda_2},\dots,-\sqrt{\lambda_n})$ proves sharpness. □

Proof of Proposition 38. Apply Lemma 39 to $\phi=\widetilde{\psi}=\sigma_y^{\otimes 4}\overline{\psi}$. Spin flip preserves every Schmidt spectrum and $|\langle\psi,\widetilde{\psi}\rangle|^2=\tau$. For a two-qubit marginal on sites i,j, Pauli expansion gives

$$\rho_{ij}-\widetilde{\rho}_{ij}=\frac{1}{2}\left(\mathbf{r}_i\cdot\boldsymbol{\sigma}\otimes I+I\otimes\mathbf{r}_j\cdot\boldsymbol{\sigma}\right),\qquad \|\rho_{ij}-\widetilde{\rho}_{ij}\|_F^2=\|\mathbf{r}_i\|^2+\|\mathbf{r}_j\|^2.$$

Thus the two marginal mismatch terms add to $W_1/2$. The purity identity gives the remaining equalities and inequality in (10).

For sharpness take $\psi=(a|00\rangle+b|11\rangle)_{12}\otimes(|00\rangle+|11\rangle)_{34}/\sqrt{2}$, with real $a,b\geq 0$ and $a^2+b^2=1$. On $12|34$, the left side is one, while $\tau=4a^2b^2$ and $W_1=2(a^2-b^2)^2$, so equality holds throughout. The choices $a=b$ and $b=0$ force respectively the coefficients of τ and W_1 to be at least 1 and $1/2$. □

Equation (10) supplies a fully analytic quantum compatibility constraint. The separate-cut bounds do not by themselves replace the quadratic variance certificate: SCB-5 gives exact classical spectra satisfying all three caps and even $F\geq\tau$ (hence the critical envelope (CE)) but violating (QK), so compatibility between the three cut spectra still matters.

F.10 Quantitative rigidity of two maximally mixed cuts

Proposition 40 (Crossed purity costs and two-flat rigidity)**.** *Let ψ be any unit pure four-qubit state. Fix a balanced cut k, write its ordered spectrum as $\lambda_1\geq\lambda_2\geq\lambda_3\geq\lambda_4\geq 0$, and put*

$$\Delta_k=\sum_{j\neq k}(p_j-1/4),\qquad p_j=\operatorname{tr}\rho_{1j}^2.$$

If $\lambda_1\geq 1/2$, then

$$\boxed{\Delta_k\geq\frac{1}{4}W_1+2(\lambda_2\lambda_3+\lambda_2\lambda_4+\lambda_3\lambda_4)\geq\frac{1}{4}W_1+(\lambda_3+\lambda_4)^2.} \tag{TF}$$

The first inequality is sharp for every chosen spectrum with $\lambda_1\geq 1/2$. On the critical locus, equality holds exactly when all nonzero squares of the other normal-form amplitudes have a phase opposite to the square of the largest amplitude. The noncritical sharpness family in Proposition 38 also attains the first inequality.

Consequently, whenever two *balanced cuts of a pure four-qubit state have maximally mixed marginals $(1/4)^4$, the third has Schmidt rank at most two. Up to local unitaries and a permutation of the qubits, every such state belongs to the attained cluster-to-Bell family* (QF)*. In particular the exact Rényi envelope of Proposition S23 of the supplement applies for all $q>0$.*

Proof. The purity identity and Proposition 38 give

$$2\Delta_k = 2(P - p_k - 1/2) = W_1 + \tau - (2p_k - 1) \geq \frac{1}{2}W_1 + (2\lambda_1 - 1)^2 - (2p_k - 1).$$

Using $\sum_i \lambda_i = 1$ gives

$$(2\lambda_1 - 1)^2 - (2p_k - 1) = 4(\lambda_2\lambda_3 + \lambda_2\lambda_4 + \lambda_3\lambda_4).$$

This proves the first bound. Since $\lambda_2 \geq (\lambda_3+\lambda_4)/2$, the second follows by dropping the nonnegative $\lambda_3\lambda_4$ term. On the critical locus, write the chosen spectrum as $\lambda_i = |w_i|^2$ in the normal form, so $\tau = |\sum_i w_i^2|^2$. The bound is then the reverse triangle inequality, with exactly the stated opposite-phase equality condition. Choosing those phases realises every ordered spectrum with $\lambda_1 \geq 1/2$, proving sharpness.

Suppose now two cuts have purity 1/4. Already one maximally mixed two-qubit cut forces all four one-qubit marginals to be $I/2$ by purity of complementary reductions. If k is the remaining cut, then $p_k = P - 1/2 = 1/2 + \tau/2 \geq 1/2$. Since $p_k = \sum_i \lambda_i^2 \leq \lambda_1$, the hypothesis $\lambda_1 \geq 1/2$ holds. The first bound and $\Delta_k = 0$ force at most one among $\lambda_2, \lambda_3, \lambda_4$ to be nonzero. The opposite-phase equality condition means that, up to overall phase and a signed permutation, w has the form

$$w = (\sqrt{a},\ i\sqrt{1-a},\ 0,\ 0), \qquad 1/2 \leq a \leq 1.$$

The magic-basis action is $\mathrm{diag}(w) \mapsto O_1\,\mathrm{diag}(w)O_2^T$, $O_1, O_2 \in SO(4)$: the same permutation on both sides permutes the entries of w, a sign on one side changes the sign of one entry, and extra signs on the zero coordinates make both factors proper rotations. This action is induced by local one-qubit unitaries on each side of the chosen cut, so this is exactly the normal form (QF) up to LU and cut relabelling. □

G Seven qubits: copy moments and the five-copy proofs

We define the copy hierarchy used in Section 5, locate the four-qubit inequalities within it, and record the ingredients of both seven-qubit proofs. The complete proofs are in the supplement [40]; Appendix H constructs the icosahedral family. For a pure state of n qubits put

$$\pi_k(\psi) = \binom{n}{k}^{-1} \sum_{|S|=k} \mathrm{tr}\,\rho_S^2.$$

Pauli expansion. Let A_j be the sum of the squared expectation values of the $3^j\binom{n}{j}$ Pauli strings of weight j (acting nontrivially on exactly j qubits), so $A_0 = 1$ and $A_1 = \sum_i \|\mathbf{r}_i\|^2$ (the W_1 of the four-qubit sections). Expanding each marginal in Pauli strings gives $\pi_k = 2^{-k}\binom{n}{k}^{-1}\sum_{j\leq k}\binom{n-j}{k-j}A_j$ [21]; at $n = 7$, $k = 3$ this is $280\pi_3 = 35 + 15A_1 + 5A_2 + A_3$. An AME state has $A_1 = \cdots = A_k = 0$, which makes it a pure quantum code of distance $k + 1$ [29]. For stabilizer states the A_j and the corresponding shadow coefficients are integers; with $A_1 = A_2 = 0$ the shadow identity $4S_1 = 7 - A_3$ forces $A_3 \geq 3$, and if $A_1 + A_2 \geq 1$ instead, integrality gives $15A_1 + 5A_2 + A_3 \geq 5$. Thus every seven-qubit stabilizer state has $\pi_3 \geq 19/140$ (Proposition 43, which defines the shadow coefficients S_w).

Copy moments. Purities are linear observables of the two-copy density operator: if SW_S swaps the two copies of the qubits in S, then $\mathrm{tr}\,\rho_S^2 = \mathrm{tr}[\mathrm{SW}_S\,(|\psi\rangle\langle\psi|)^{\otimes 2}]$. A lower bound on π_k therefore follows from any convex set of operators that contains the twirled m-copy moments

$$X_m = \mathrm{Av}_{\sigma,K}\int (U|\psi\rangle\langle\psi|U^\dagger)^{\otimes m}\,dU, \qquad m \geq 2,$$

where $U = u_1 \otimes \cdots \otimes u_n$ ranges over local unitaries (LU), and $\mathrm{Av}_{\sigma,K}$ also averages over qubit permutations and complex conjugation (the supplement writes X_m^+). These averages preserve the mean purity objective, and the partial trace over one copy maps X_{m+1} to X_m.

Spin flip supplies further positivity constraints. With $\epsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $\tilde{\psi} = \epsilon^{\otimes n}\overline{\psi}$ the spin-flipped state, the twirl of $|\psi^{\otimes(m-j)}\tilde{\psi}^{\otimes j}\rangle\langle\psi^{\otimes(m-j)}\tilde{\psi}^{\otimes j}|$ is an average of rank-one projectors, $1 \le j \le m/2$; we denote it by $\Lambda_j(X_m)$ when the flipped copies are the last j, and set $\Lambda_0(X_m) = X_m$. Since $\epsilon\overline{u}\epsilon^{-1} = u$ for $u \in SU(2)$, the flip commutes with the local twirl; the extra phase for $U(2)$ cancels in projectors. It also commutes with qubit permutations and complex conjugation. Thus Λ_j is a fixed linear image of X_m: partial transpose on the j flipped copies followed by conjugation by the corresponding ϵ factors. Minimising π_k over trace-one, twirl-invariant operators X_m supported on the symmetric subspace, subject to $X_m \succeq 0$ and $\Lambda_j(X_m) \succeq 0$, is a semidefinite programme. Its optimum L_m bounds $\min \pi_k$ from below, $L_m \le L_{m+1}$, and $L_m \to \min \pi_k$; the level $m = 2$ is the shadow-enumerator linear programme of Rains [21] (Proposition S43 and Theorem S44 of the supplement [40]).

The four-qubit inequalities in this hierarchy. A certificate at level m is an identity

$$\text{objective} - \text{bound} = \sum_j \mathrm{tr}\big[\Xi_j\, \Lambda_j(X_m)\big], \qquad \Xi_j \succeq 0,$$

valid for all X_m; on a genuine state each term is an average of squared norms $\|\Xi_j^{1/2}(\psi^{\otimes(m-j)} \otimes \tilde{\psi}^{\otimes j})\|^2$. At $n = 4$ the purity bound $P \ge 1$ of Lemma 2 is the level-two linear-programming (LP) minimum $\pi_2 = 1/3$ (Theorem S101 of the supplement). Its proof evaluates $\tau = \mathrm{tr}(\rho\tilde{\rho})$, the shadow term. The sextic identity (S) is a level-three certificate (it holds as a linear identity in the three-copy moment) with one flipped copy: every term is $\langle w, Aw\rangle$ with $w = z \otimes z \otimes \tilde{z}$ and $A \succeq 0$ (Appendix A). Its objective F differs from the purity, so it uses the same positivity conditions with a different objective.

At $n = 7$ the levels separate (Table 8). Five copies are the first level that gives the sharp purity bound (levels two to four are provably not sharp). With a purity band and an eigenvalue cap as extra constraints, they also give the sharp bound for the entropy majorant of Section 5.

m	L_m at $n = 7$	status
2	$L_2 = 1/8$	exact (LP-2; a putative AME weight distribution)
3	$L_3 = \frac{125}{978} = 0.1278118\ldots$	proved (Thm. S45)
4	$\frac{211}{1600} \le L_4 < \frac{6597}{50000} < \frac{121}{917}$	bounds proved (Thm. S47, Prop. S49); $L_4 \approx 0.131877$ numerical (FS)
5	$L_5 = \frac{121}{917} = 0.131952\ldots$	verified-arithmetic certificate (Thm. 11)

Table 8: The seven-qubit copy hierarchy for π_3. Strictness $L_2 = 1/8 < L_3$ is read off the table and $L_3 < L_4$ is proved (Corollary S46 of the supplement), and $L_4 < L_5$ by the exact feasible pseudo-moment of Proposition S49 (exact certificate; checks NS-0–NS-4).

Ingredients of the five-copy proofs. Section 5 outlines the proofs of Theorems 11 and 12 in three steps (value, equality moment, orbit), taken at a point ψ_y of the icosahedral family: $y = 37/524$ for the purity and $y = x_*$ for the entropy. The complete proofs are Sections S11 and S12 of the supplement. Here we define the objects used in those outlines and record the sizes of the computations.

Faces. By representation theory the positivity conditions at level five decompose into blocks. The certificates use some of them, $\Gamma_b(X_5)$, each a compression (followed by a fixed congruence) of $\Lambda_{j_b}(X_5)$ with $j_b \in \{0, 1, 2\}$ flipped copies. Exact matrices W_b (rational for the purity; for the entropy defined by exact formulas at x_*, with enclosures used only to check the identity) have columns in the kernels of the blocks at ψ_y, and the *faces* are $W_b^{\mathsf{T}}\Gamma_b(X_5)W_b \succeq 0$. With positive definite C_b, the objective minus its value at ψ_y (for the entropy, the value minus the objective)

equals $\sum_b \operatorname{tr}[C_b W_b^{\mathsf{T}} \Gamma_b(X_5) W_b]$ plus, for the entropy, further nonnegative terms. For the purity there are twelve faces:

$$\pi_3 - \frac{121}{917} = \sum_{b=1}^{12} \operatorname{tr}[C_b W_b^{\mathsf{T}} \Gamma_b(X_5) W_b]$$

for every normalised X_5, in the 3285-dimensional space of conjugation-even five-copy moments (Proposition S60 of the supplement). For the entropy the objective is the mean of the joint majorant of Section 5; there are seventeen faces, and the extra terms are the purity-band constraints times nonnegative cubic polynomials and matrix squares weighted by the eigenvalue cap.

Probes and verified arithmetic. In both cases the target functional and the face entries vanish at $X_5(\psi_y)$ and at $X_5' = \frac{d}{dx} X_5(\psi_x)|_{x=y}$. They therefore lie in the 3283-dimensional *critical space* of linear functionals annihilating these two moments. A computer evaluates the faces at 3424 rational probe moments, and a rank computation with rigorous error bounds shows that the probes separate the critical space; solving the probe equations gives each C_b (for the entropy, the first twelve) as a verified small correction of the identity matrix. In practice numerical linear algebra proposes an approximate inverse, and rigorous rounding-error bounds validate it; Section S11.4 of the supplement gives the rank test, the error model and the margins. The arithmetic is floating point with rigorous error bounds, so the proof uses a verified-arithmetic certificate rather than an exact-rational computation. The same rank computation shows that the face entries span the critical space, which is the input of the equality step. The purity certificate also gives a quantitative bound: $\pi_3 - 121/917$ is at least the least eigenvalue of the C_b times the total face energy $\sum_b \operatorname{tr}[W_b^{\mathsf{T}} \Gamma_b(X_5) W_b]$.

Equality moment and orbit. At equality every face vanishes. Since the face entries span the critical space, the moment then lies on the tangent line $X_5(\psi_y) + tX_5'$; this is Lemma S63 of the supplement, which holds for any number of qubits and copies. The invariant that fixes the tangent parameter is $H = 104S_1 - 25A_2$, with S_1 the weight-one shadow coefficient (Proposition 43) and A_2 the weight-two Pauli sum of the Pauli expansion above (the invariant $\mathcal{E}_2$ of Theorem S64); its family value is $h = 155 + 96y$. Both H and H^2 are linear in X_5, so at $X_5(\psi_y) + tX_5'$ they force $(h + 96t)^2 = h^2 + 192ht$, that is $t = 0$; equivalently, the centred square $(H - h)^2$ is linear in the moment and lies in the critical space, so it vanishes at equality, yet on the actual state it equals $(96t)^2$. Hence $X_5(\psi) = X_5(\psi_y)$. The orbit step is Theorem S64 of the supplement, valid for every $y \in (1/32, 1/12)$: its thirteen invariants, of bidegree at most $(4, 4)$ in $\psi, \bar{\psi}$, are all determined by the four-copy marginal of X_5, and force a local frame in which ψ is fixed by an antilinear operator built from the spin flip, that of Proposition S55; the remaining matrix Q is complex symmetric of order six with zero diagonal, and the real amplitude is fixed by y; the conference-matrix conclusion, unique up to diagonal unitaries and permutations, is Theorem S53 (the identity (1)) with Proposition S54; and Corollary S68 counts the 42 labelled LU orbits. Steps 2 and 3 together are Theorem S66 of the supplement, stated for every $y \in (1/32, 1/12)$.

H Seven qubits: the icosahedral state

Label the qubits $s, 0, 1, \ldots, 5$. Let $s_{ij} = -1$ for $\{i, j\}$ an edge of the pentagon $1{-}3{-}2{-}4{-}5{-}1$ and $s_{ij} = +1$ otherwise. These signs represent the regular two-graph of the six icosahedral axes: a triangle is negative when its three signs multiply to -1, and every pair lies in exactly two negative triangles. The two-graph has symmetry group $A_5 \cong PSL(2, 5)$. For $0 \le x \le \frac{1}{2}$ put $\alpha = \sqrt{(1 - 2x)/30}$, $\beta = \sqrt{x}$ and

$$\begin{aligned} \psi_x = |0\rangle_s \Big(\alpha \sum_{i<j} s_{ij} |e_i + e_j\rangle + \beta |111111\rangle \Big) \\ + |1\rangle_s \Big(\beta |000000\rangle - \alpha \sum_{i<j} s_{ij} |111111 - e_i - e_j\rangle \Big). \end{aligned} \tag{14}$$

Theorem 41 (the icosahedral family)**.**

1. *Every one-qubit marginal of every ψ_x is $I/2$.*

2. *$\pi_3(\psi_x)$ is a quadratic polynomial in x, minimised within the family at $x = 37/524$. The* icosahedral state $\Psi_{\text{ico}} = \psi_{37/524}$ *has* $\pi_3(\Psi_{\text{ico}}) = 121/917 = 0.131952\ldots$

3. *Its* 20 *three-body marginals avoiding s have purity* $4381/34322$ *and its* 15 *containing s have purity* $2363/17161$. *Its pair purities are* $8641/34322$ *(*15 *pairs avoiding s) and* $8761/34322$ *(*6 *pairs containing s).*

4. *Throughout the family, writing $u = \alpha^2 = (1-2x)/30$, its triple spectra are $(4u)^6, (3u+x)^2$ on cuts avoiding s, and $(8u), (4u)^5, (u+x)^2$ on cuts containing s.*

Proof. Let C be the 6×6 zero-diagonal symmetric conference matrix with entries $C_{ij} = s_{ij}$ $(i \neq j)$. Every pair of vertices lies in two negative and two positive triangles, so $C^2 = 5I_6$. In (14) the $|0\rangle_s$ and $|1\rangle_s$ branches are supported on ordinary Hamming weights $\{2,6\}$ and $\{0,4\}$, respectively. Disjointness of these weight sets makes ρ_s diagonal, with each branch of weight $15u + x = 1/2$. Within each branch the ordinary weights have one parity, so every ρ_i is diagonal, and $P(b_i = 1) = 5u + x + 10u = 1/2$. This proves item (1).

For an ordinary triple $I = \{i,j,k\}$ with complement $J = \{0,\ldots,5\} \setminus I$, group the partial trace over $\{s\} \cup J$ by Hamming weight on J. This block-diagonalises ρ_I by Hamming weight on I. On $|000\rangle$ and $|111\rangle$ the diagonal entries are $3u + x$ (three pairs inside J plus the endpoint term). On the weight-1 subspace $\text{span}\{|100\rangle, |010\rangle, |001\rangle\}$, the $|0\rangle_s$ branch (weight 1 on J) contributes $u\, B_{IJ} B_{IJ}^{\mathsf{T}}$, where B_{IJ} is the 3×3 block of C between I and J, while the $|1\rangle_s$ branch (weight 3 on J) contributes $u\,|v_1\rangle\langle v_1|$ with $|v_1\rangle = (s_{jk}, s_{ik}, s_{ij})^{\mathsf{T}}$. For any zero-diagonal $\{\pm 1\}$ matrix C_I of order three, $C_I^2 = I_3 + |v_1\rangle\langle v_1|$, while the I-block of $C^2 = 5I_6$ reads $C_I^2 + B_{IJ}B_{IJ}^{\mathsf{T}} = 5I_3$. Their sum is therefore

$$u(B_{IJ}B_{IJ}^{\mathsf{T}} + |v_1\rangle\langle v_1|) = u((5I_3 - C_I^2) + (C_I^2 - I_3)) = 4u\, I_3.$$

Since $X^{\otimes 7}\psi_x$ is ψ_x with $\alpha \mapsto -\alpha$, and the weight-1 block depends only on α^2, the weight-2 block is also $4u\, I_3$, so $\rho_I = \text{diag}(3u + x, (4u)^6, 3u + x)$ in the computational basis.

For a cut $\{s,i,j\}$ containing s, with complement $K = \{0,\ldots,5\} \setminus \{i,j\} = \{k,l,m,n\}$, the five Hamming-weight sectors on K map to four orthogonal 2×2 blocks of ρ_{sij}:

- weights 0 and 4 on K span $\{|011\rangle, |100\rangle\}$ with orthogonal conditional vectors $\alpha s_{ij}|011\rangle + \beta|100\rangle$ and $\beta|011\rangle - \alpha s_{ij}|100\rangle$, giving $(u+x)I_2$;
- weight 1 on K spans $\{|010\rangle, |001\rangle\}$ and weight 3 spans $\{|101\rangle, |110\rangle\}$, each giving $4u\, I_2$ because $\sum_{r\in K} s_{ir}s_{jr} = (C^2)_{ij} = 0$;
- weight 2 on K spans $\{|000\rangle, |111\rangle\}$ with matrix $\left(\begin{smallmatrix} 6u & -2u\,\eta_K \\ -2u\,\eta_K & 6u \end{smallmatrix}\right)$, where $\eta_K = s_{kl}s_{mn} + s_{km}s_{ln} + s_{kn}s_{lm} \in \{\pm 1\}$, giving eigenvalues $6u \pm 2u = (8u, 4u)$.

Here the sign magnitude follows intrinsically from the conference equation: $C^{-1} = C/5$ and $\det C = -125$, so Jacobi's complementary-minor identity gives $\det C_K = (-125)(-1/25) = 5$. Expanding a symmetric zero-diagonal 4×4 determinant gives $\det C_K + \eta_K^2 = 6$, since all off-diagonal entries are signs. Thus $\eta_K^2 = 1$, without a case split on pentagon subgraphs. This proves item (4). Tracing out one further site gives $\rho_{ij} = \text{diag}(\frac{1}{2} - 8u, 8u, 8u, \frac{1}{2} - 8u)$ and $\rho_{si} = \text{diag}(10u, 5u+x, 5u+x, 10u)$. Summing the squared eigenvalues gives (15) and $\pi_3(\psi_x) = \frac{121}{917} + \frac{(524x-37)^2}{137550}$, proving (2) and (3). IC-1–IC-6 independently check all marginals, polynomial identities and derivatives. □

In particular the 20 cuts avoiding s at Ψ_{ico} have spectrum $(41/262, 41/262, (15/131)^6)$ and the 15 containing s have $(30/131, (15/131)^5, (13/131)^2)$. Throughout ψ_x the mean two- and three-body purities and the mean three-body third moment $b_3 = \frac{1}{35}\sum_{|S|=3} \text{tr}\,\rho_S^3$ are

$$\pi_2(\psi_x) = \frac{832x^2 - 112x + 163}{630}, \qquad \pi_3(\psi_x) = \frac{2096x^2 - 296x + 149}{1050}, \tag{15}$$

$$b_3(\psi_x) = \frac{35008x^3 + 17328x^2 - 3084x + 709}{31500},$$

with $b_3'(37/524) = -432/120127 < 0$ (IC-6).

Corollary 42 (purity and entropy have different stationary points)**.** *Let $f(t) = -t\log_2 t$, $f(0) = 0$. The mean balanced entropy $E_3 = \binom{7}{3}^{-1}\sum_{|S|=3} S(\rho_S)$, in bits, on the icosahedral family is the strictly concave function*

$$E_3(\psi_x) = \frac{195f(4u) + 40f(3u + x) + 15f(8u) + 30f(u + x)}{35}, \qquad u = (1 - 2x)/30.$$

It has a unique maximum on $[0, 1/2]$, lying strictly between 0 and $37/524$. In particular Ψ_{ico} is not an entropy maximiser, even locally in the full pure-state space, since decreasing x already increases E_3.

[Check IC-6.]

Proof. The spectra give the formula. Every eigenvalue is affine in x, and the nonconstant eigenvalue $4u$ makes the sum strictly concave. At the purity minimum,

$$E_3'(37/524) = \frac{4}{35\ln 2}\ln\frac{30^{10}15^5}{13^7 41^8} < 0,$$

where the sign is an exact integer comparison. At $x = 0$ the derivative is $32\ln(16/3)/(35\ln 2) > 0$. Strict concavity proves the asserted location and uniqueness. This locates the maximum within the family; it is the global maximum over all pure seven-qubit states by Theorem 12. □

Comparison. Di Martino, Facchi and Florio [10] found numerically a seven-qubit state with $\pi_3 = 0.131952$, given by coefficients to six digits in a basis of products of four-qubit and three-qubit states, and did not claim it optimal. Computed from their coefficients, its marginal spectra and pair purities agree with those of Ψ_{ico} to that precision, which suggests that the two states are LU-equivalent; Theorem 41 gives an explicit closed-form state with the exact value $121/917$. This value is below the Fano/wheel graph state ($19/140 = 0.1357\ldots$, Zha et al. [27], also [16]; the best stabilizer value, Proposition 43) and the earlier numerical 0.13387 of Facchi et al. [28]. We know of no state in the literature with smaller π_3.

Proposition 43 (stabilizer states)**.** *Every seven-qubit stabilizer state has $\pi_3 \geq 19/140$. Hence the seven-qubit minimum is not attained by a stabilizer state.*

Proof. For any pure seven-qubit state with Pauli weight enumerator $A_w = \sum_{\mathrm{wt}(P)=w}\langle\psi|P|\psi\rangle^2$ ($A_0 = 1$) and shadow enumerator $S_w = \sum_{\mathrm{wt}(P)=w}|\langle\psi|P|\tilde{\psi}\rangle|^2$, expanding π_3 in subset purities gives the first identity below; the second follows from complementarity of the mean s-qubit purities, $\pi_s = \pi_{7-s}$ (the pure-state MacWilliams identity), and Rains' shadow transform $S(x, y) = A(\frac{x+3y}{2}, \frac{y-x}{2})$ [21]:

$$280\pi_3 = 35 + 15A_1 + 5A_2 + A_3, \qquad 4S_1 = 7 + A_1 + A_2 - A_3.$$

For a seven-qubit stabilizer state $|\psi\rangle$ with stabilizer group $\mathcal{S}$ ($|\mathcal{S}| = 2^7$), every Pauli string P has $\langle\psi|P|\psi\rangle^2 \in \{0, 1\}$. Moreover, since $\tilde{\psi} = \epsilon^{\otimes 7}\bar{\psi}$ satisfies $M\tilde{\psi} = (-1)^{\mathrm{wt}(M)}\tilde{\psi}$ for every $M \in \mathcal{S}$, the state $P\tilde{\psi}$ is a joint eigenvector of $\mathcal{S}$, so $|\langle\psi|P|\tilde{\psi}\rangle|^2 \in \{0, 1\}$ as well. Hence the A_w and S_w are nonnegative integers. If $A_1 + A_2 \geq 1$, then $15A_1 + 5A_2 + A_3 \geq 5 > 3$. If $A_1 = A_2 = 0$, then $A_3 = 7 - 4S_1 \geq 0$ with $S_1 \in \mathbb{Z}_{\geq 0}$ forces $S_1 \in \{0, 1\}$ and $A_3 \in \{3, 7\}$, so $A_3 \geq 3$. In both cases $15A_1 + 5A_2 + A_3 \geq 3$ and $\pi_3 \geq 38/280 = 19/140$, with equality iff $(A_1, A_2, A_3) = (0, 0, 3)$ ($S_1 = 1$), attained by an explicit seven-vertex graph state. Then $121/917 < 19/140$ by Theorem 41; GS-4 checks both identities and the equality witness. □

I Exact finite checks

Table 9 lists, for each result, the labels of the computer checks its proof uses and their class. *Exact* checks are symbolic or rational-arithmetic computations: polynomial identities, ranks, dimensions and Bernstein coefficient signs. *Verified arithmetic* uses explicit rounding-error bounds, including outward-rounded interval or ball arithmetic, so that every sign decision is rigorously certified. Where the class is *analytic*, the proof is complete in the text and the checks only confirm its algebra. Section S20 of the supplement describes each label. The scripts, data and captured outputs are in the companion repository [40]: `verifiers/MAP.md` gives the script for each label, and the results table of its `README.md` gives the labels for each theorem. Checks marked numerical there (TCN, FS, NV, TA) are evidence only, and no proof uses them.

Result	Checks	Class
Four-qubit maximum, moment inequality and equality (Theorems 1, 3; Appendices A–C)	ME, SP, SI, CE, PZ, ER, CN, RG	analytic
General envelope (Theorem 15)	FI, FG-6–FG-8, ESH; direct proof FG-1–FG-5	exact (point evaluation on a 19-dimensional invariant space)
Tsallis, Rényi $q < 1$, mixed states (Section 3)	EF	analytic
Rényi $1 < q < 2$ (Corollary 9)	QVA, QG, SD, CV; SHD, LHD; OT, BC, DA, UR, SCB, RA; HD, LT, CG, NU, SL, QV	exact six-copy certificate for (QV); exact Bernstein sign checks; otherwise analytic
Orbit stability (Section 4, Appendix D)	GO; OS, BF, EH, NSL, NC, RP, CO, FP, PM, PS, TC, MS, LCC, CCV, EX	exact integer Gram certificate; verified arithmetic
Icosahedral family and stabilizer bound (Appendix H)	IC, EG, GS-4	analytic; EG verified arithmetic
Seven-qubit purity minimum (Theorem 11)	S5, Y5, IK5, PK5, DP5, FP5, TT, MO, FO, CC, CP; LH, QB	verified arithmetic (five-copy certificate FP5-1)
Seven-qubit entropy maximum (Theorem 12)	ES, SEH, EB, SC-2, SP7, AH, E7, FO, EO	verified arithmetic (five-copy certificate E7-1)
Copy hierarchy at seven qubits (Table 8) and its covariant structure	TCB, FC, NS, SF, SDS, HI	exact
Other qubit counts (Table 5)	LP, GS, GL, E18, LP/QP-13, NG, SW, E13L, EU	exact rational; E13L verified arithmetic; EU high precision, rounded up
Purity–time-reversal tradeoffs (Section 6)	ST, TY	exact

Table 9: Computer checks used by each result, by label, with their class. Section S20 of the supplement describes each label.

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